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(54) **ZOOM OPTICAL SYSTEM, OPTICAL APPARATUS AND METHOD FOR MANUFACTURING THE ZOOM OPTICAL SYSTEM**

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G02B 27/00	(2006.01)

(52) **U.S. Cl.**

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(57) **ABSTRACT**

A zoom optical system (ZL), comprises, in order from an object: a first lens group (G1) having a positive refractive power; a second lens group (G2) having a negative refractive power; a third lens group (G3) having a positive refractive power; a fourth lens group (G4) having a positive refractive power; a fifth lens group (G5); and a sixth lens group (G6). In the zoom optical system, upon zooming, a distance between the adjacent lens groups changes. The zoom optical system satisfies the following conditional expression.

$$1.00<Mv4/Mv3<3.00$$

where Mv3: an amount of movement of the third lens group upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

Mv4: an amount of movement of the fourth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

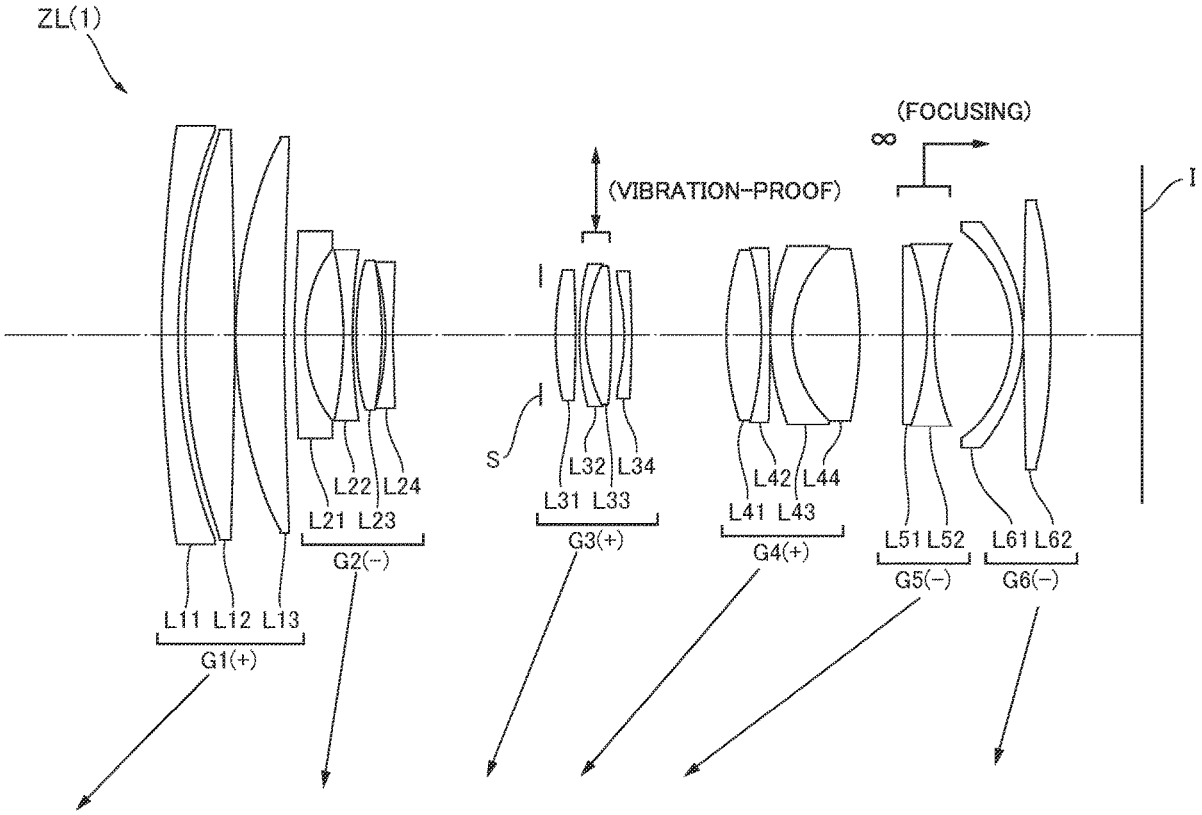


FIG.1

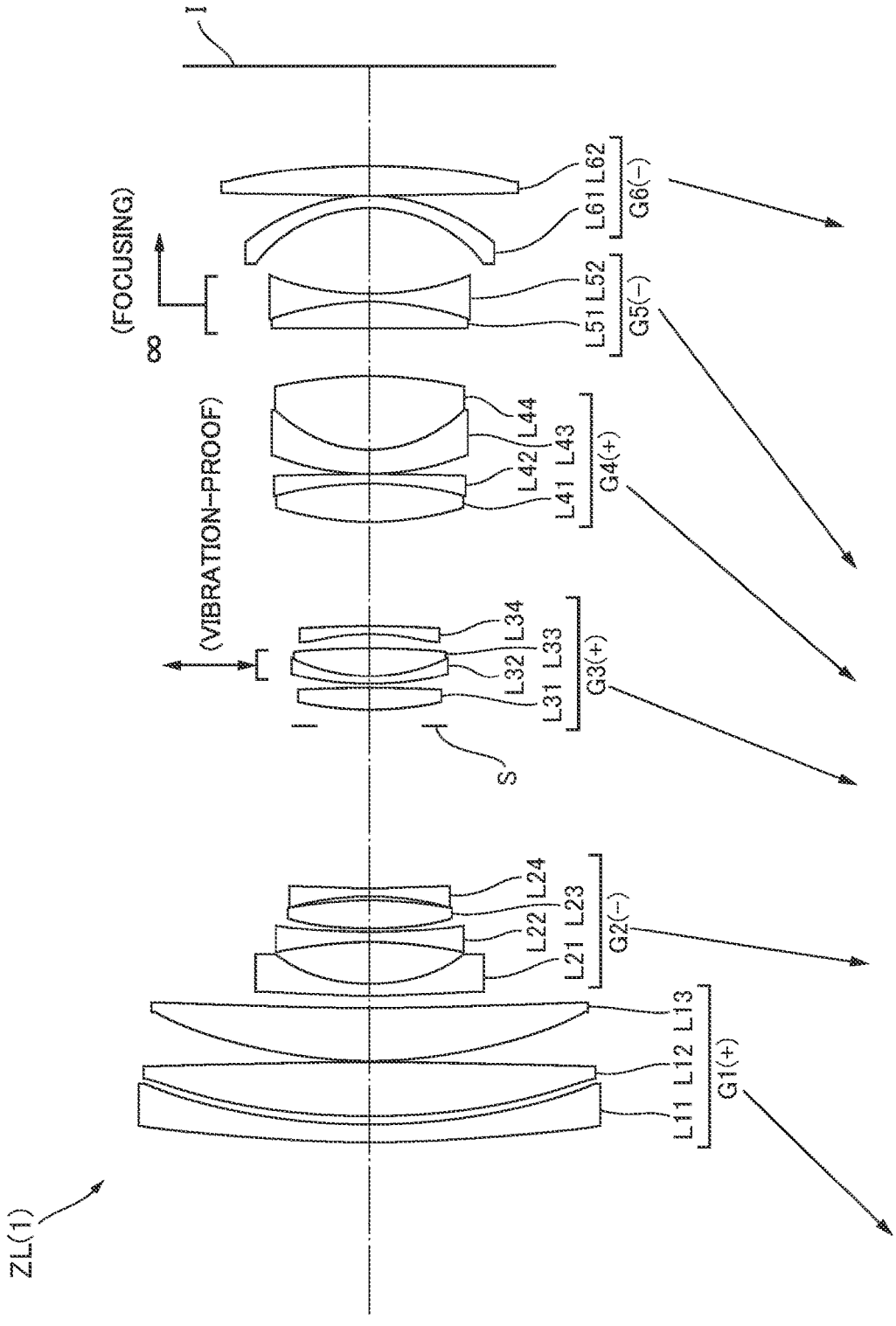

ZL(1)
I
(FOCUSING)
∞
(VIBRATION-PROOF)
S
L11 L12 L13
G1(+)
L21 L22 L23 L24
G2(−)
L31 L32 L33 L34
G3(+)
L41 L42 L43 L44
G4(+)
L51 L52
G5(−)
L61 L62
G6(−)

FIG. 2A

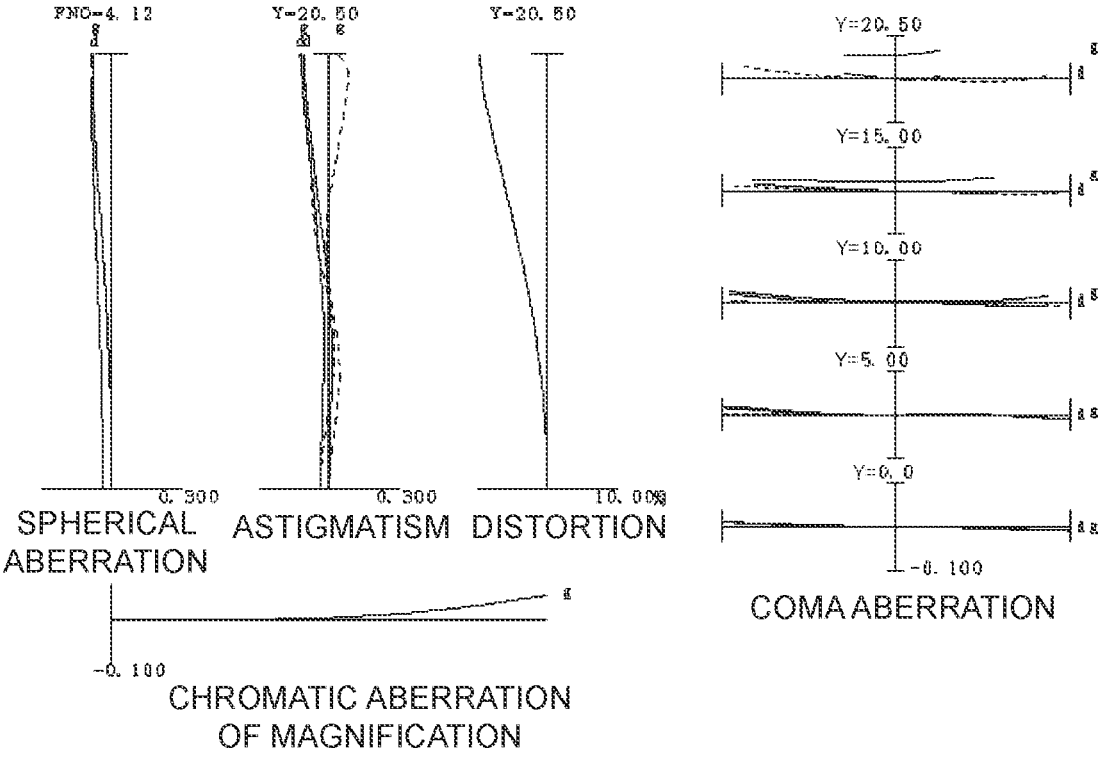

FNO=4.12
Y=20.50
Y=20.50
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y=20.50
Y=15.00
Y=10.00
Y=5.00
Y=0.0
-0.100
COMA ABERRATION

FIG. 2B

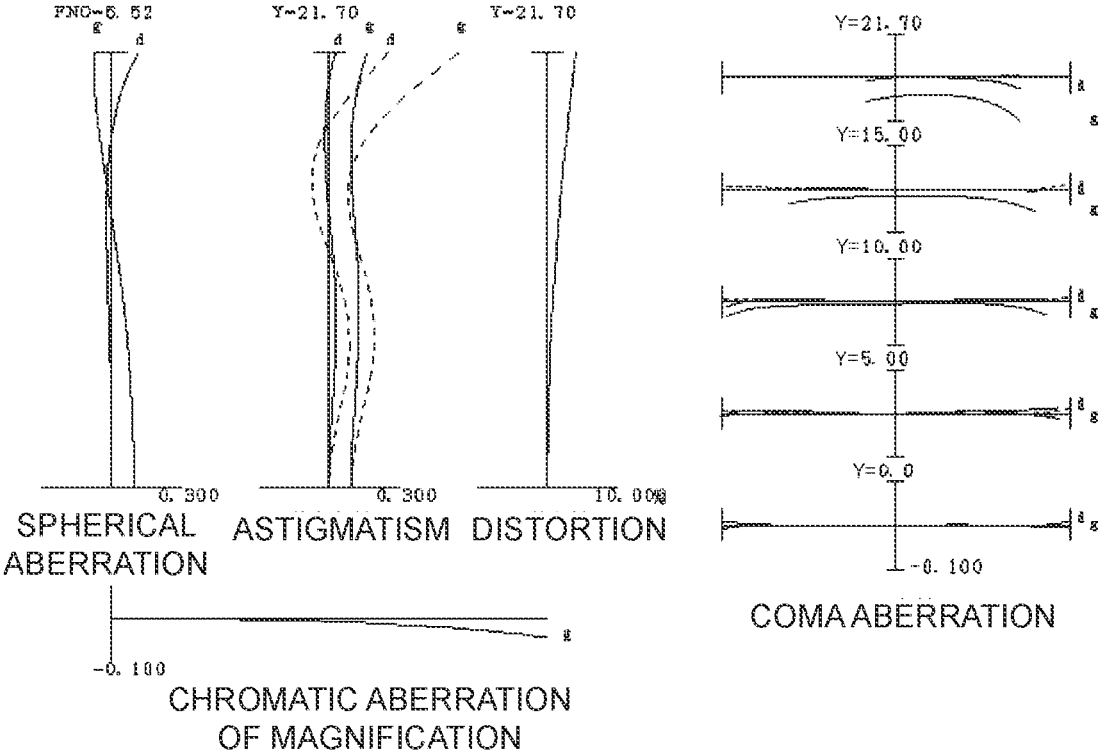
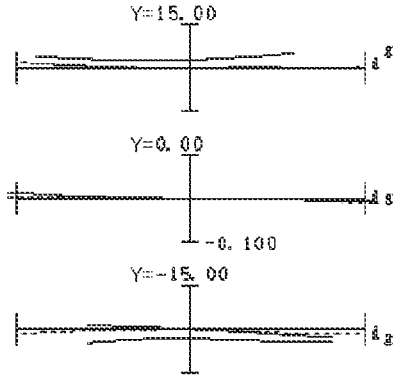
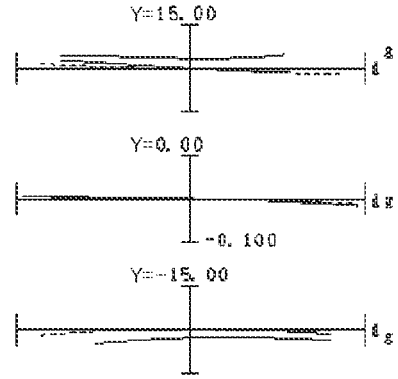

FNO=5.52
Y=21.70
Y=21.70
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y=21.70
Y=15.00
Y=10.00
Y=5.00
Y=0.0
-0.100
COMA ABERRATION

FIG. 3A



COMA ABERRATION

FIG. 3B



COMA ABERRATION

FIG.4

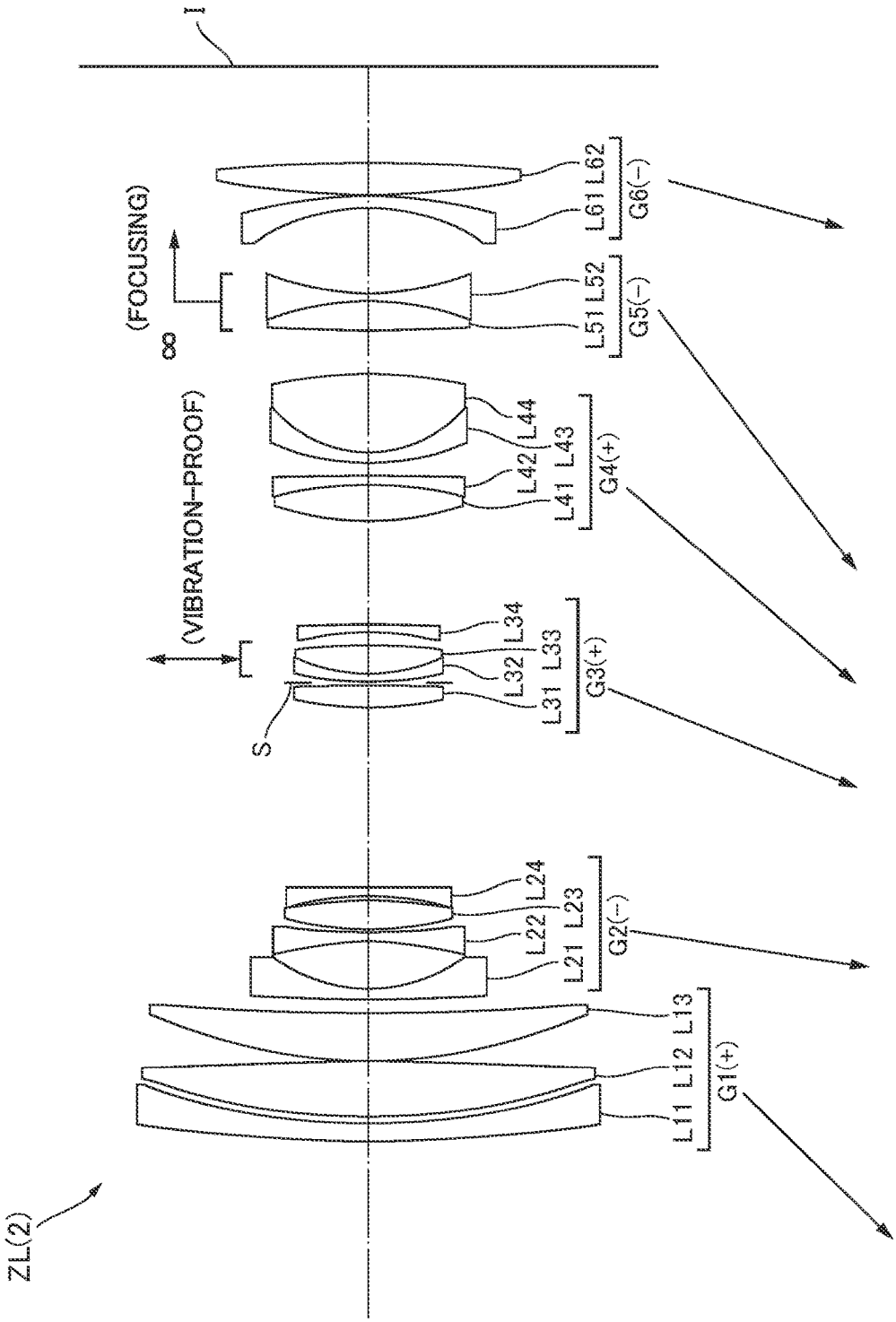

I
(FOCUSING)
∞
(VIBRATION-PROOF)
S
ZL(2)
L11 L12 L13
G1(+)
L21 L22 L23 L24
G2(−)
L31 L32 L33 L34
G3(+)
L41 L42 L43 L44
G4(+)
L51 L52
G5(−)
L61 L62
G6(−)

FIG. 5A

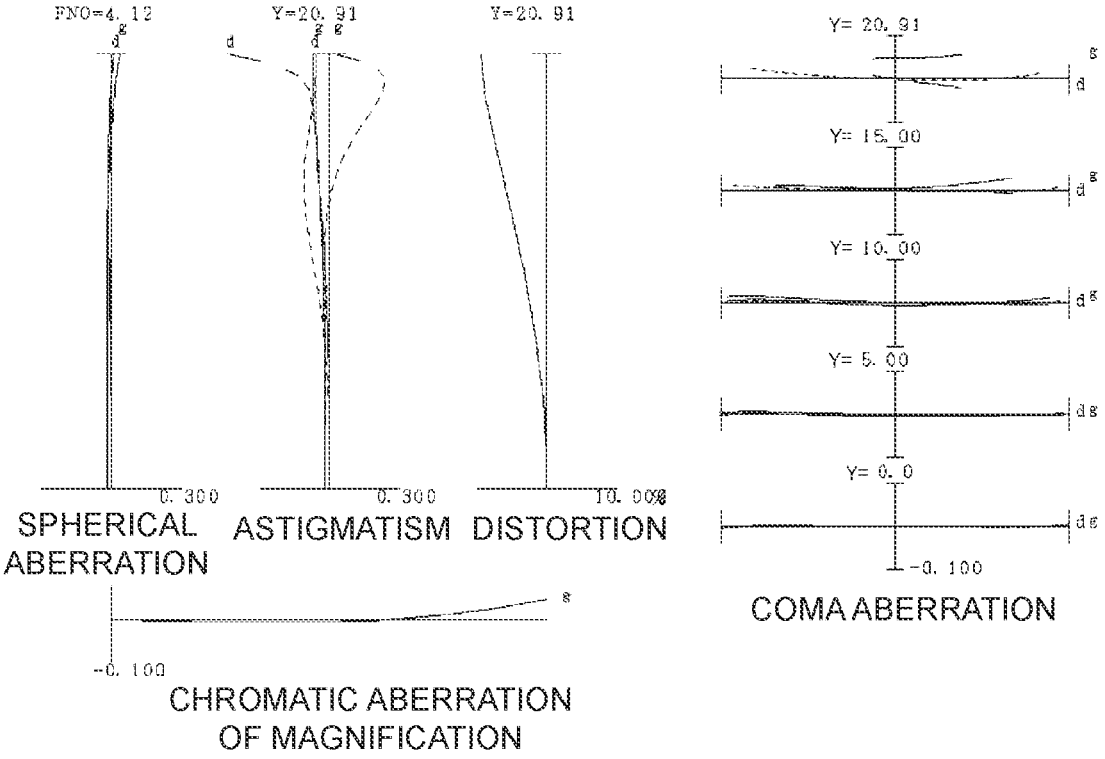

FNO=4.12
Y=20.91
Y=20.91
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y= 20.91
Y= 15.00
Y= 10.00
Y= 5.00
Y= 0.0
-0.100
COMA ABERRATION

FIG. 5B

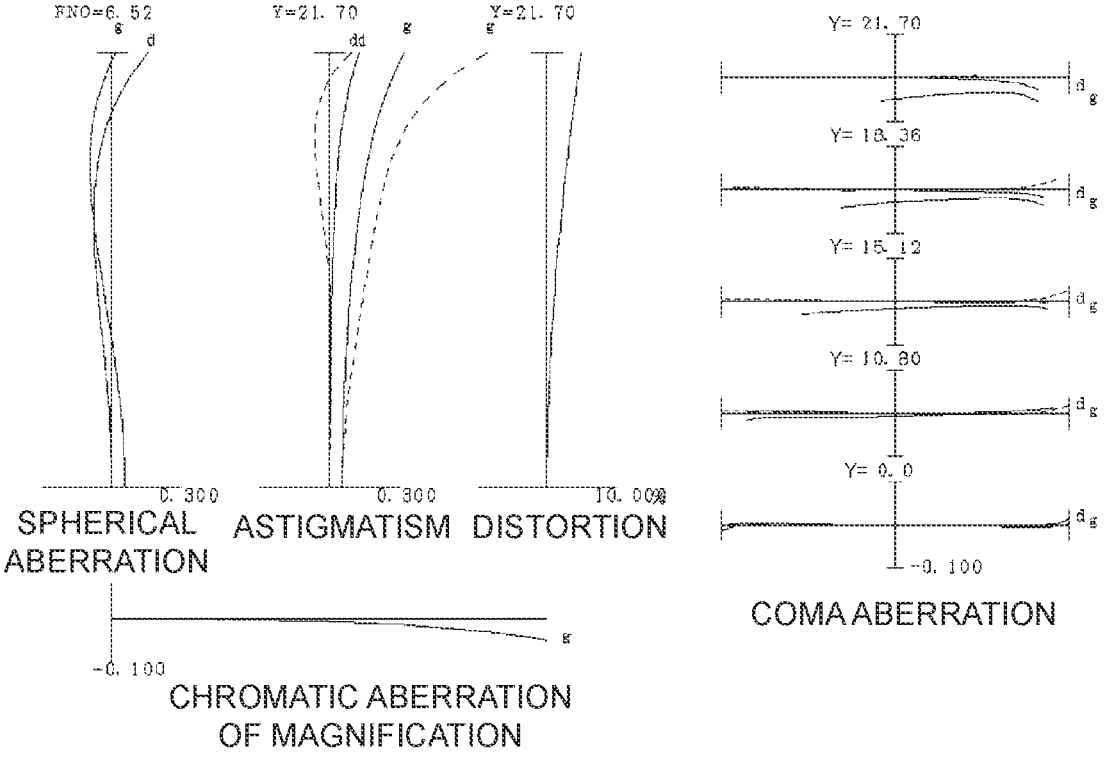
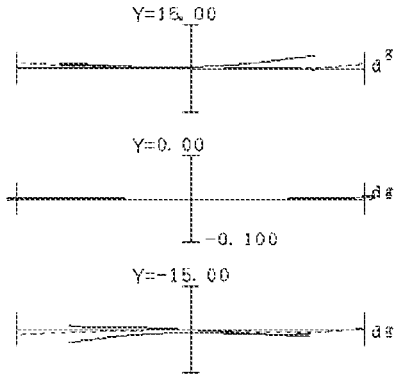
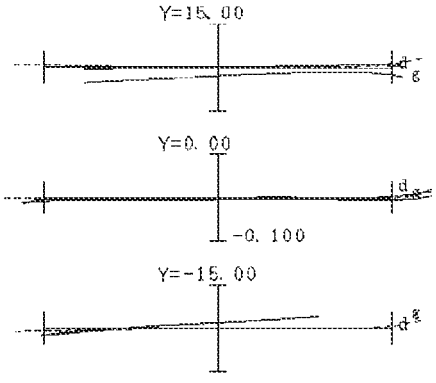

FNO=6.52
Y=21.70
Y=21.70
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y= 21.70
Y= 18.36
Y= 15.12
Y= 10.80
Y= 0.0
-0.100
COMA ABERRATION

FIG. 6A



COMA ABERRATION

FIG. 6B



COMA ABERRATION

FIG.7

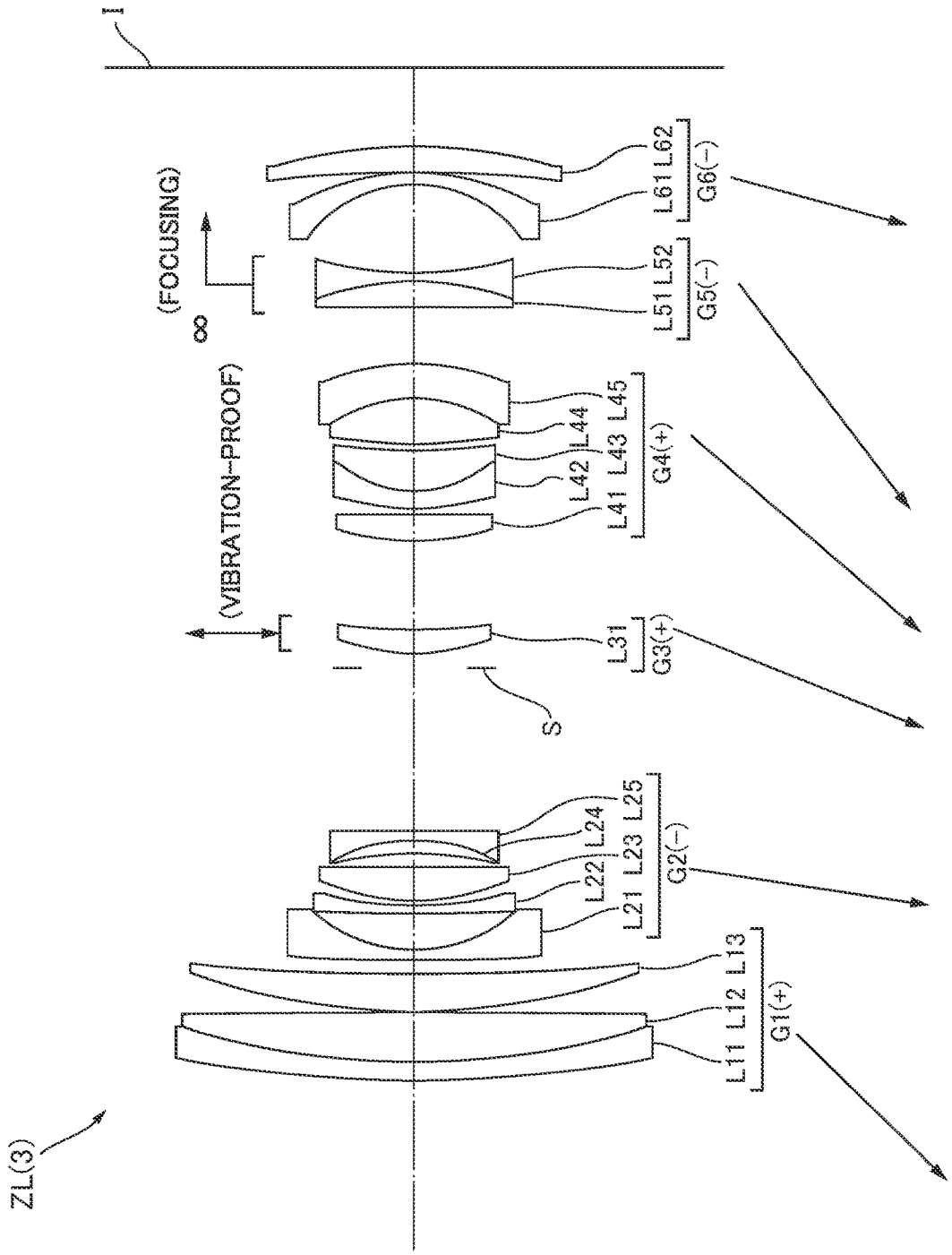

ZL(3)
I
(FOCUSING)
∞
(VIBRATION-PROOF)
L11 L12 L13
G1(+)
L21 L22 L23 L24 L25
G2(−)
L31
G3(+)
S
L41 L42 L43 L44 L45
G4(+)
L51 L52
G5(−)
L61 L62
G6(−)

FIG. 8A

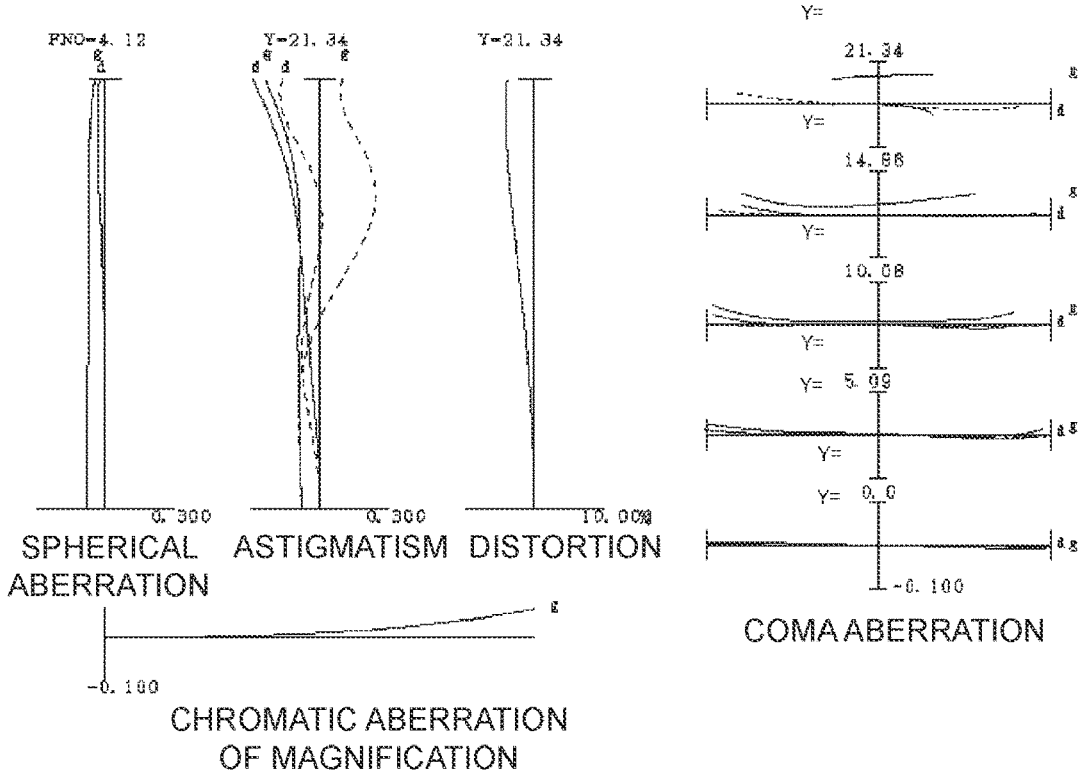

FNO=4.12
Y=21.34
Y=21.34
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y=
21.34
14.96
10.68
5.09
0.0
-0.100
COMA ABERRATION

FIG. 8B

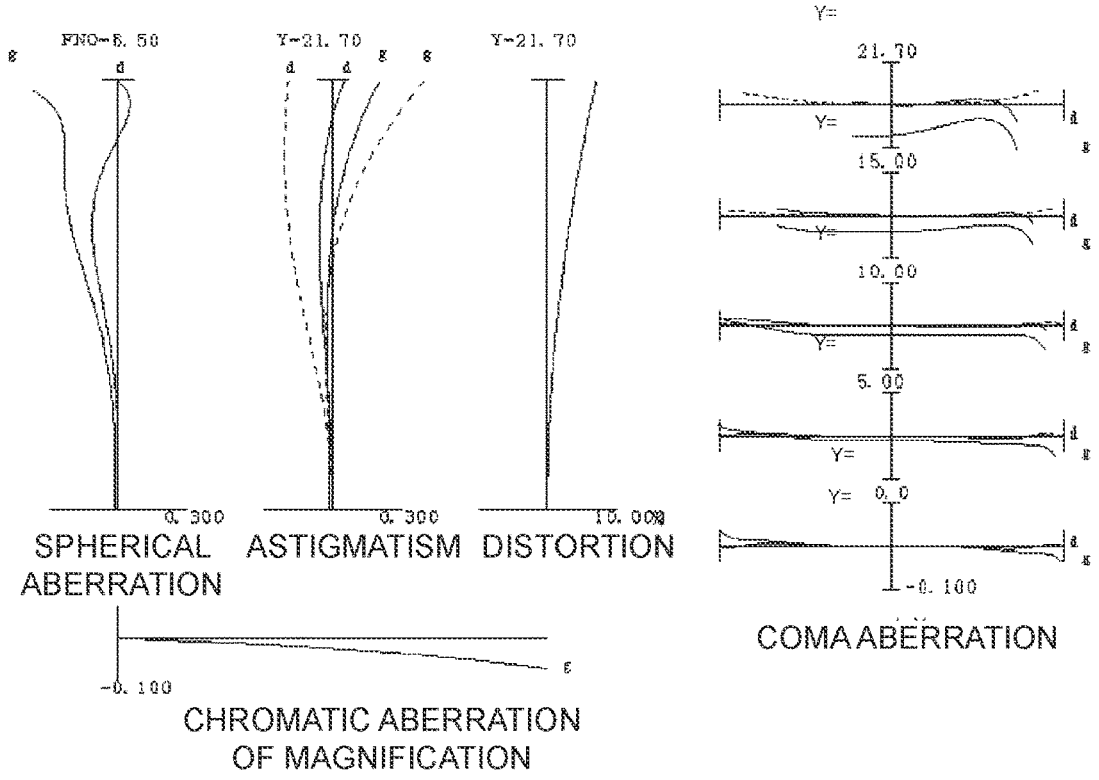
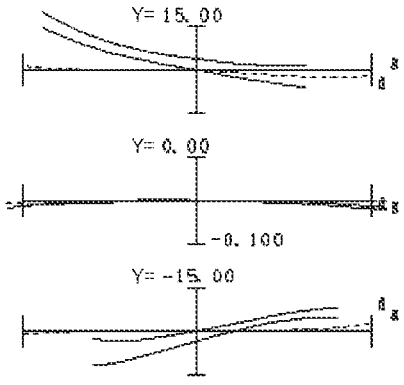
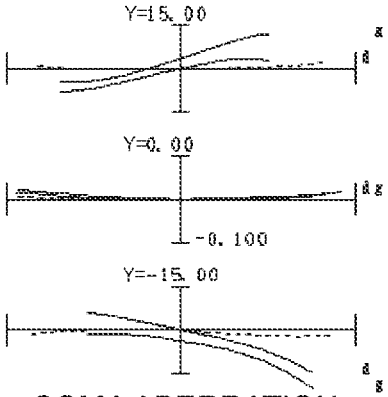

FNO=5.50
Y=21.70
Y=21.70
0.300
0.300
10.00%
SPHERICAL ABERRATION
ASTIGMATISM
DISTORTION
-0.100
CHROMATIC ABERRATION OF MAGNIFICATION
Y=
21.70
15.00
10.00
5.00
0.0
-0.100
COMA ABERRATION

FIG. 9A



COMA ABERRATION

FIG. 9B



COMA ABERRATION

FIG.10

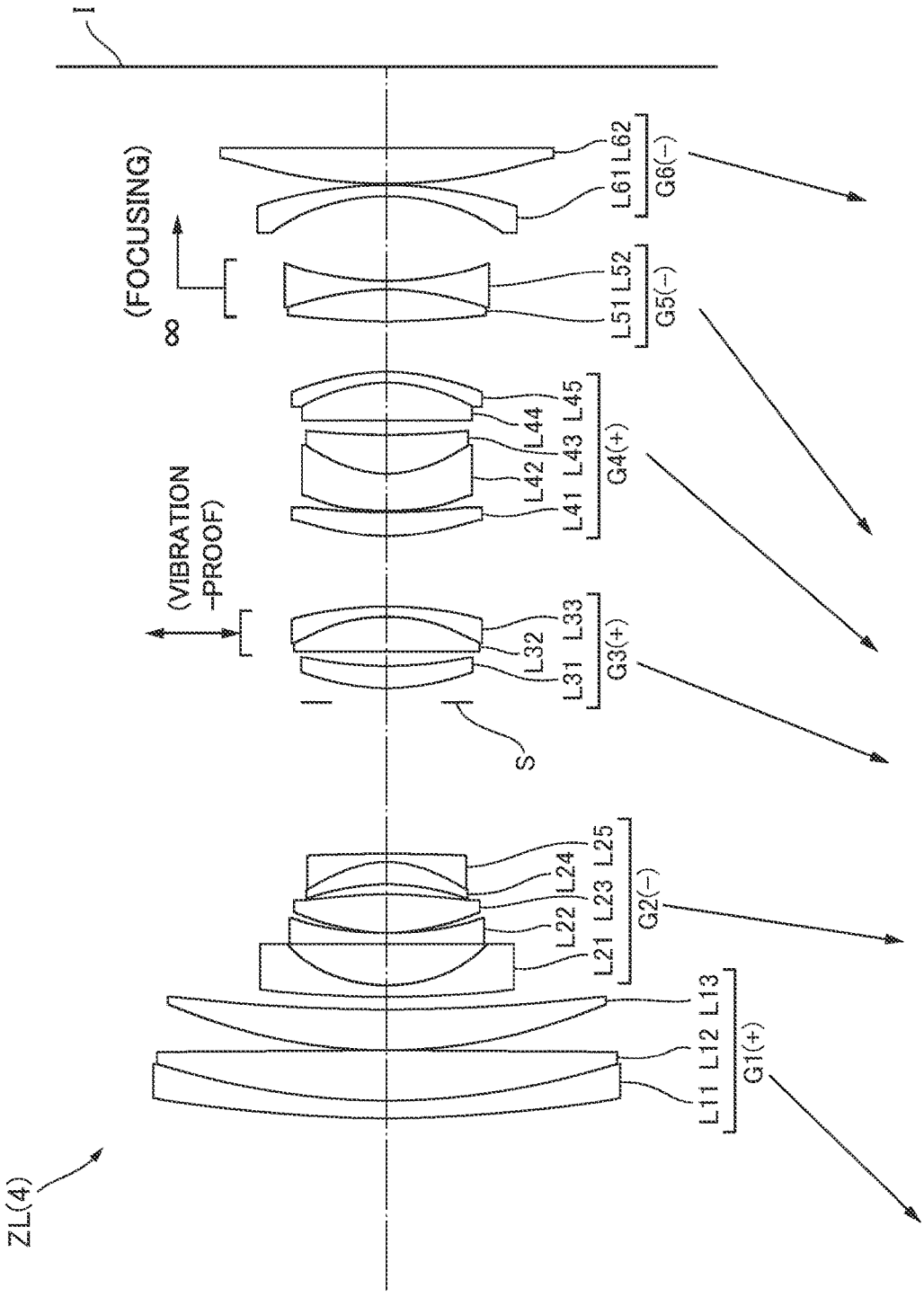

I
(FOCUSING)
∞
(VIBRATION -PROOF)
S
ZL(4)
L61 L62
G6(−)
L51 L52
G5(−)
L41 L42 L43 L44 L45
G4(+)
L31 L32 L33
G3(+)
L21 L22 L23 L24 L25
G2(−)
L11 L12 L13
G1(+)

FIG. 11A

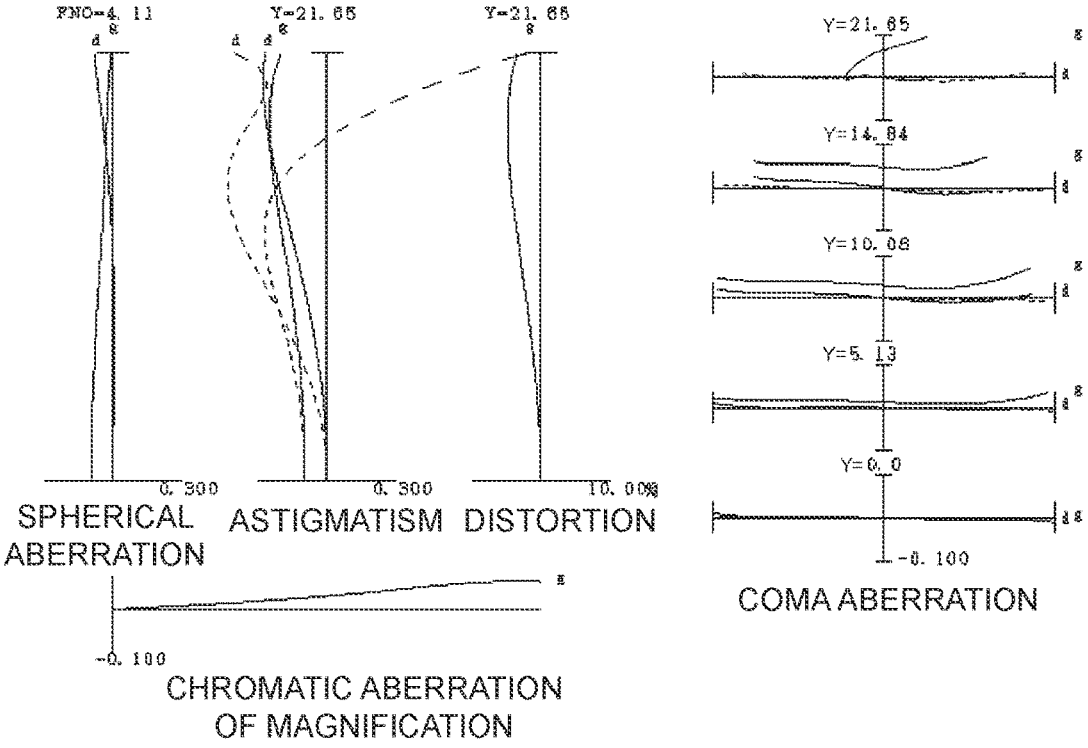


FIG. 11B

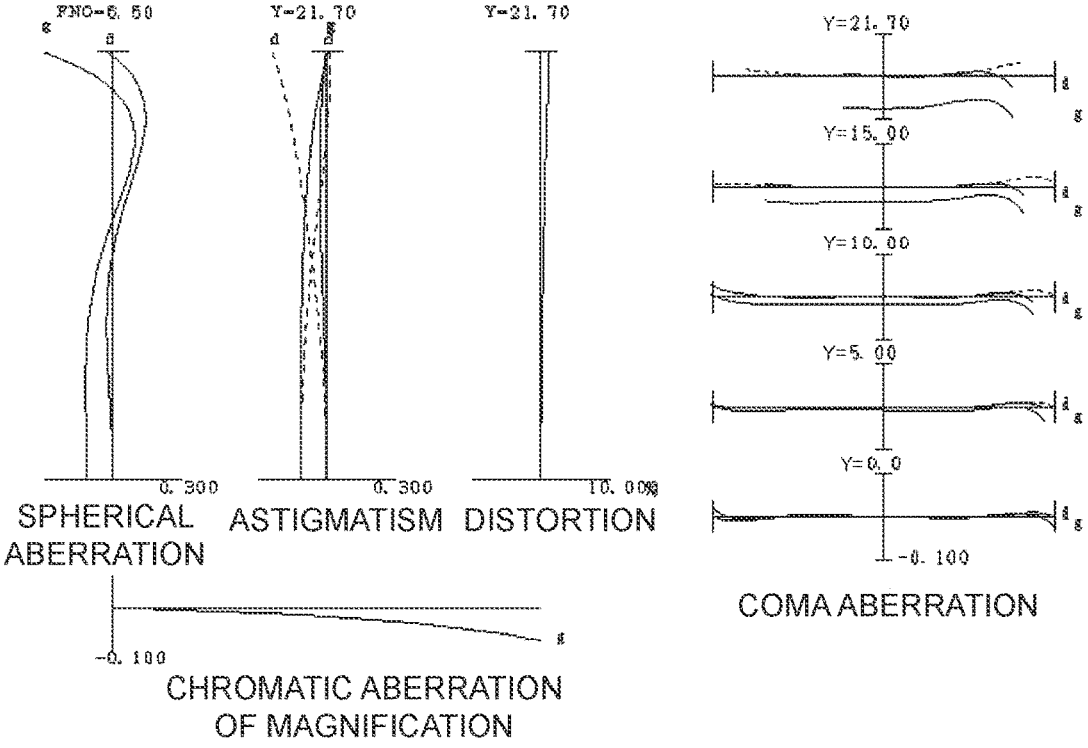
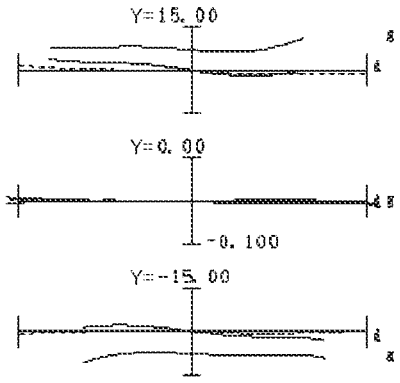
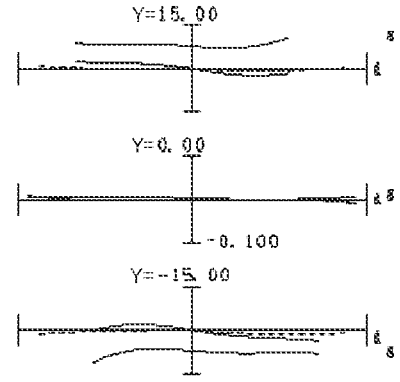


FIG. 12A



COMA ABERRATION

FIG. 12B



COMA ABERRATION

FIG.13

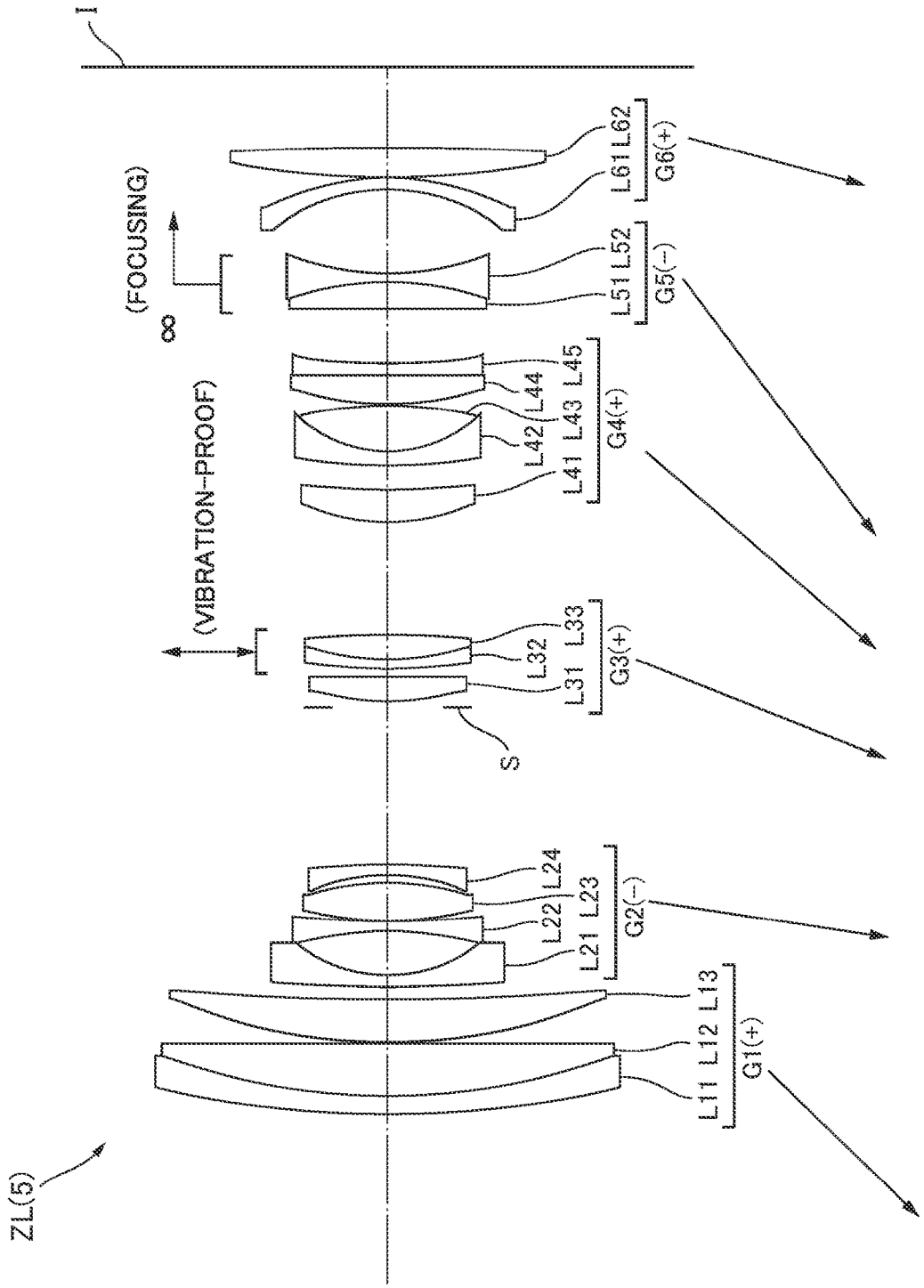

I
(FOCUSING)
∞
(VIBRATION-PROOF)
L61 L62
G6(+)
L51 L52
G5(−)
L41 L43 L45
L42 L44
G4(+)
L31 L33
L32
G3(+)
S
L21 L23
L22 L24
G2(−)
L11 L12 L13
G1(+)
ZL(5)

FIG. 14A

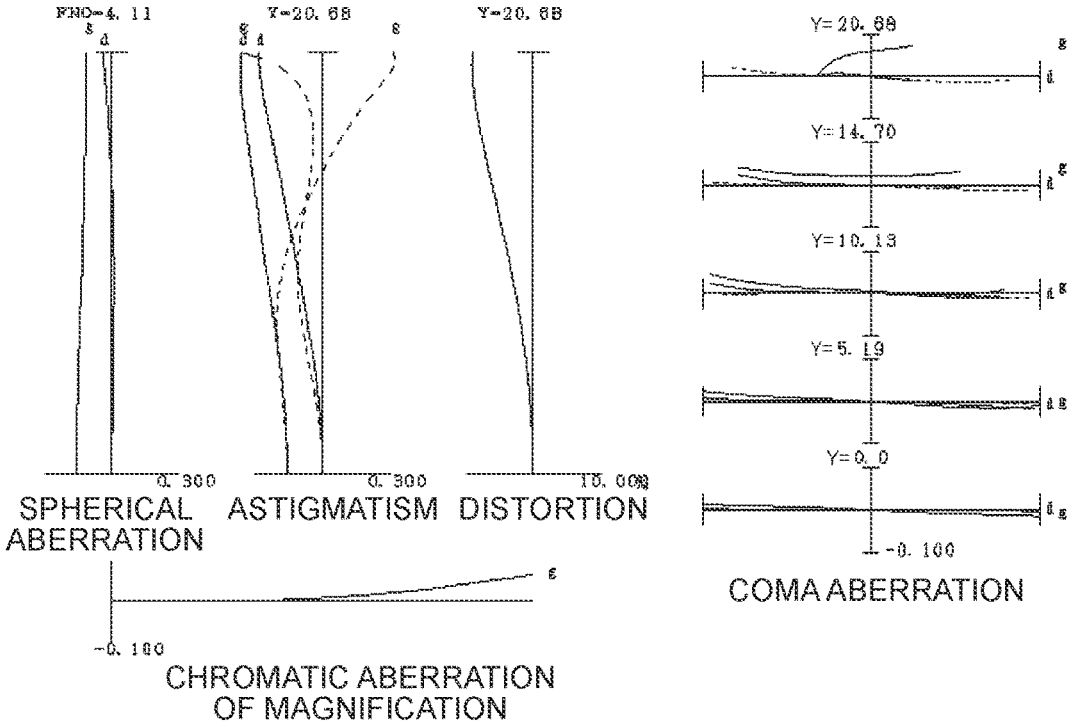


FIG. 14B

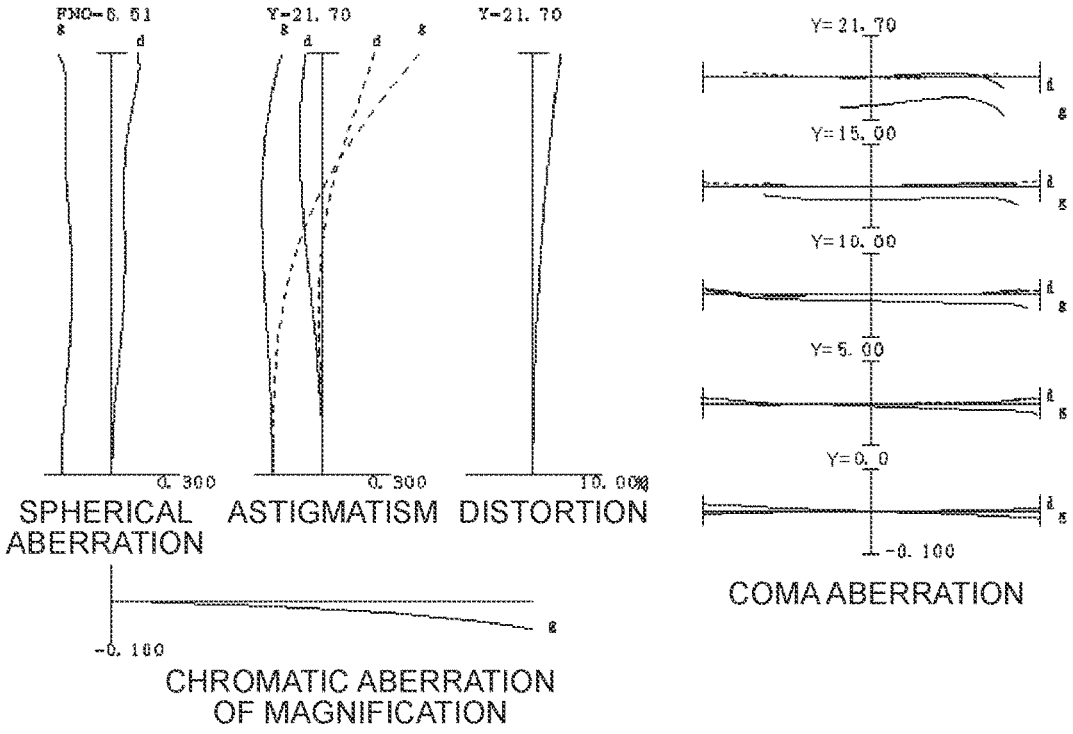
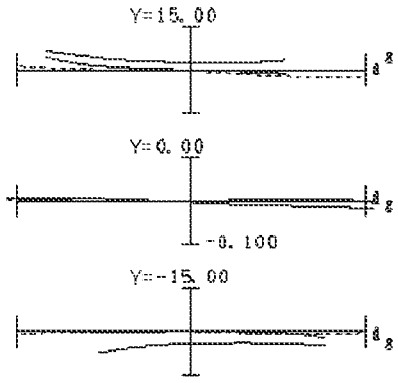
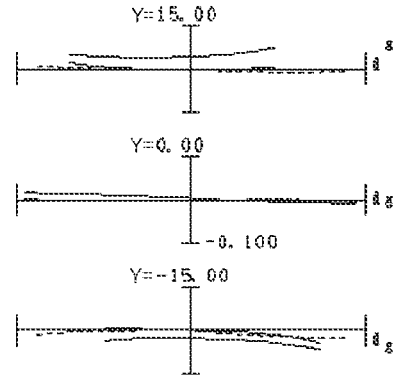


FIG. 15A



COMA ABERRATION

FIG. 15B



COMA ABERRATION

FIG.16

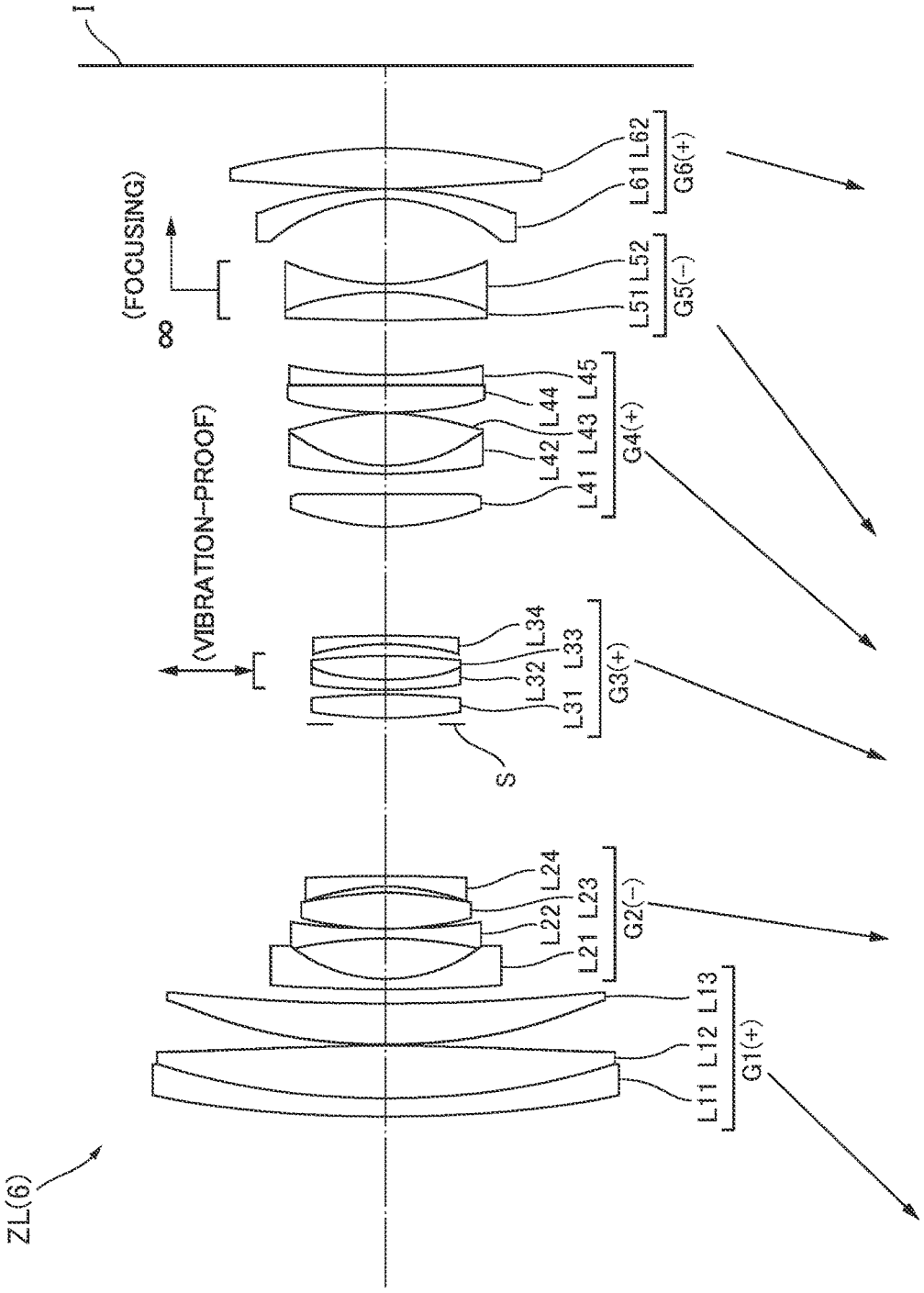

I
(FOCUSING)
∞
(VIBRATION-PROOF)
L61 L62
G6(+)
L51 L52
G5(−)
L41 L43 L45
L42 L44
G4(+)
L31 L33 L34
L32
G3(+)
S
L21 L23 L24
L22
G2(−)
L11 L12 L13
G1(+)
ZL(6)

FIG. 17A

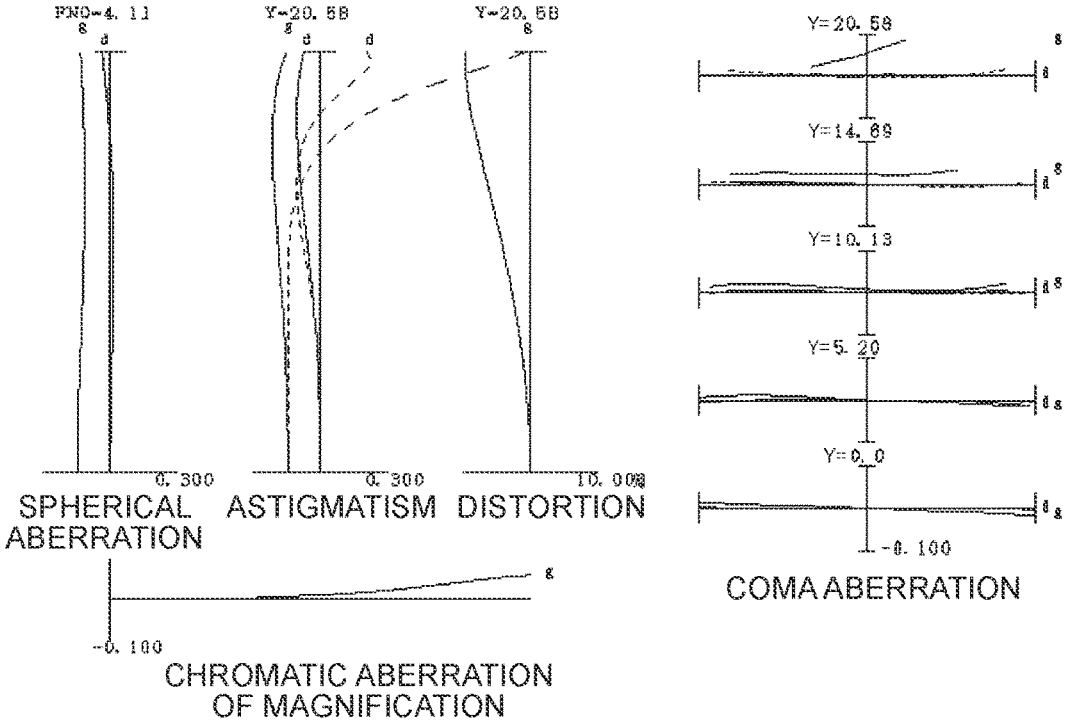


FIG. 17B

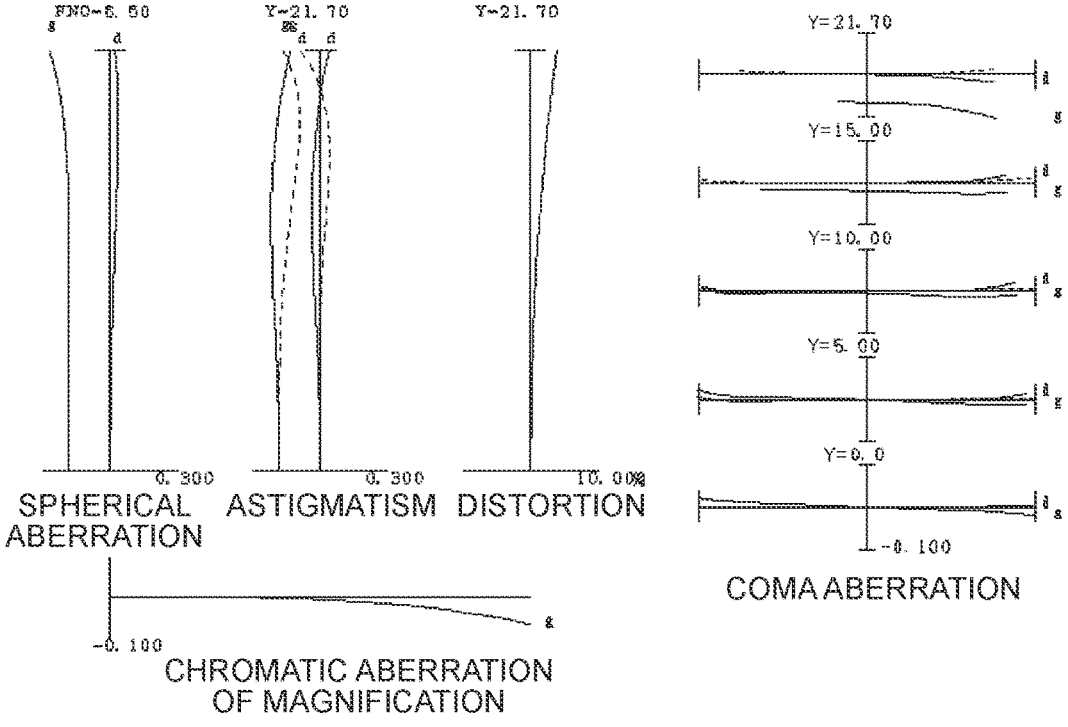
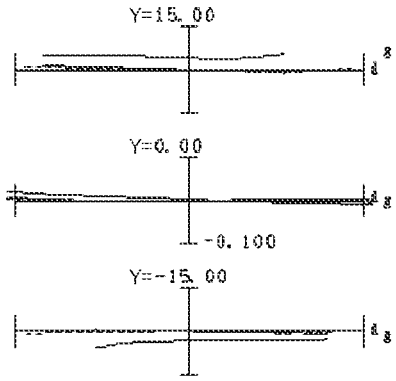
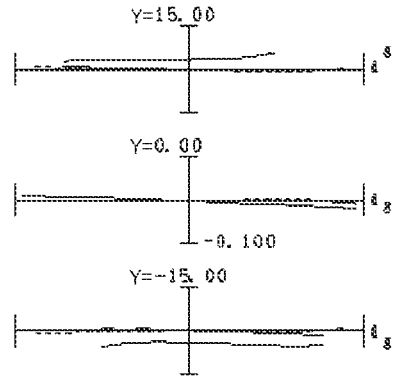


FIG. 18A



COMA ABERRATION

FIG. 18B



COMA ABERRATION

FIG.19

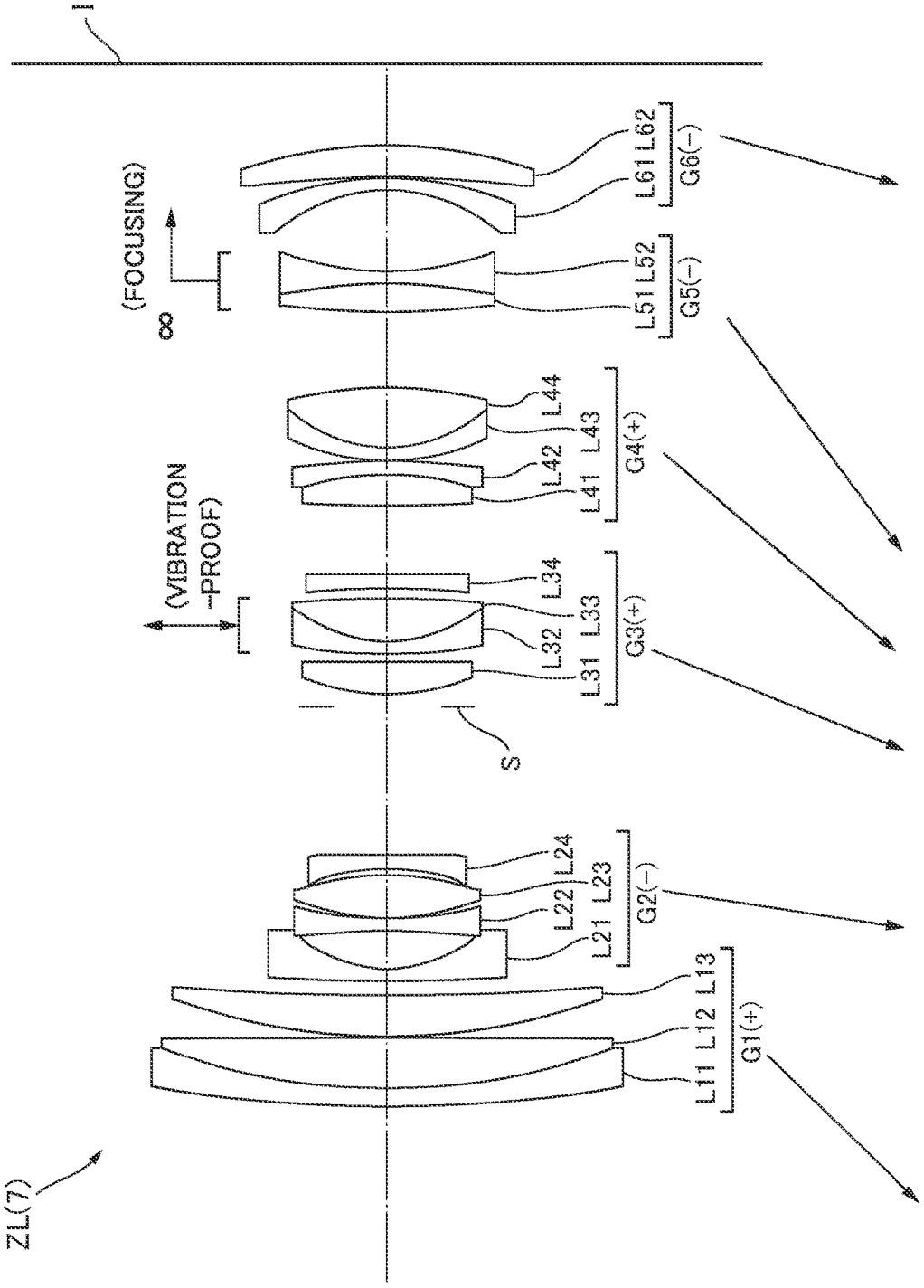

I
(FOCUSING)
∞
L61 L62
G6(−)
L51 L52
G5(−)
L44
L42
L41 L43
G4(+)
(VIBRATION
−PROOF)
L34
L32
L31 L33
G3(+)
S
L24
L22
L21 L23
G2(−)
L13
L11 L12
G1(+)
ZL(7)

FIG. 20A

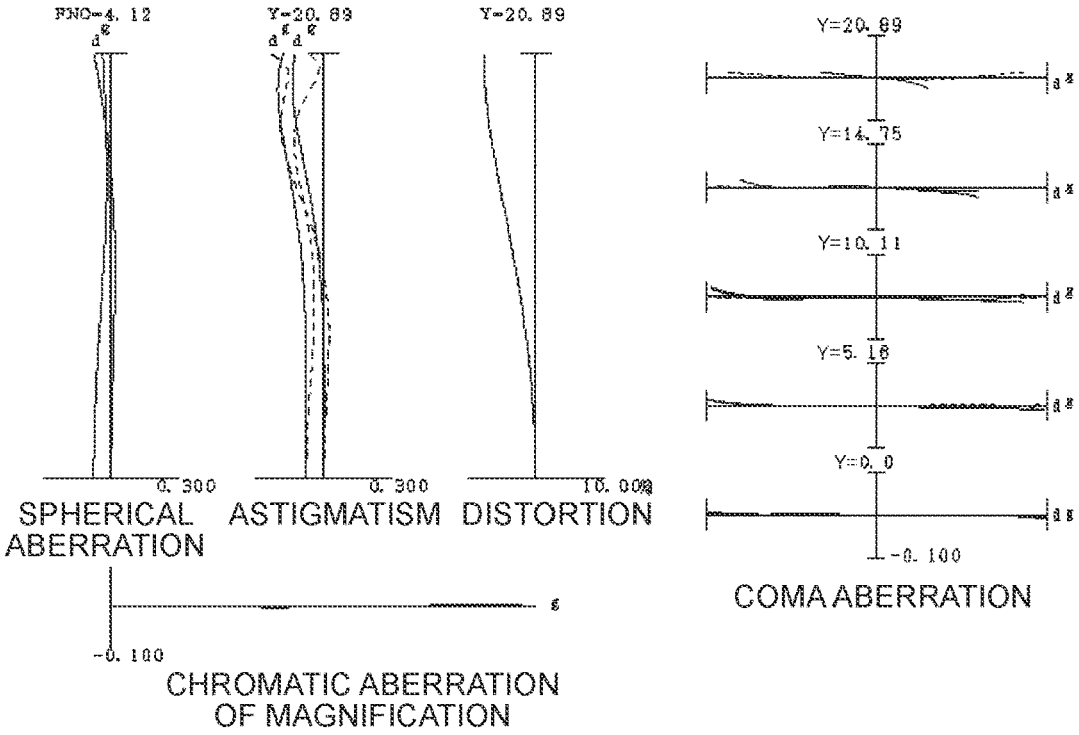


FIG. 20B

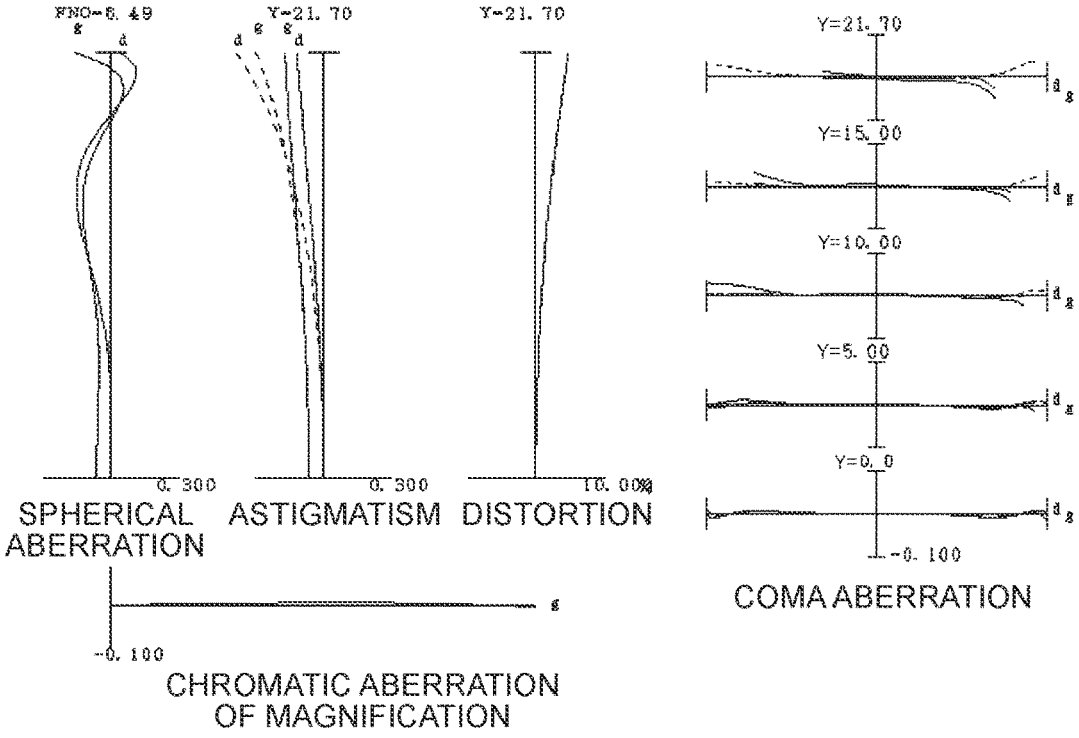
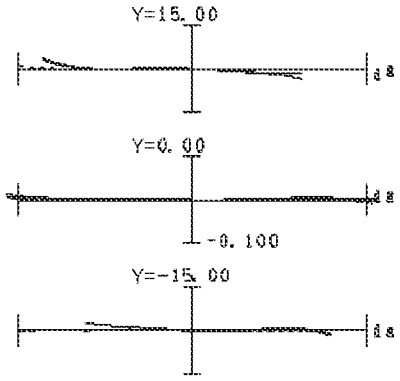
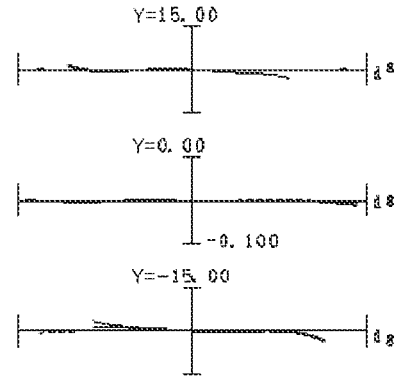


FIG. 21A



COMA ABERRATION

FIG. 21B



COMA ABERRATION

FIG.22

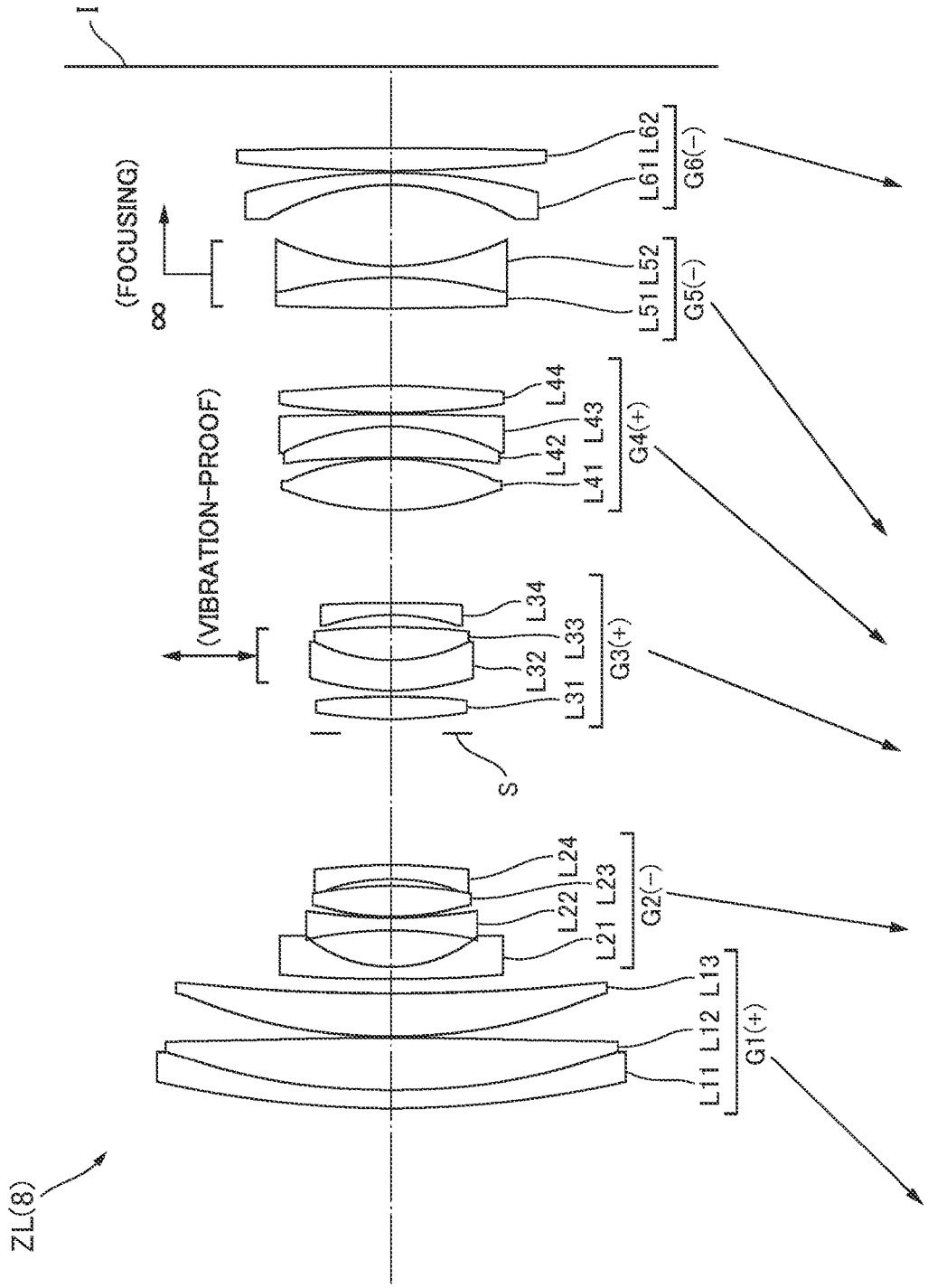

I
(FOCUSING)
∞
(VIBRATION-PROOF)
L61 L62
G6(−)
L51 L52
G5(−)
L44
L42
L41 L43
G4(+)
L34
L32
L31 L33
G3(+)
S
L24
L22
L21 L23
G2(−)
L13
L11 L12
G1(+)
ZL(8)

FIG. 23A

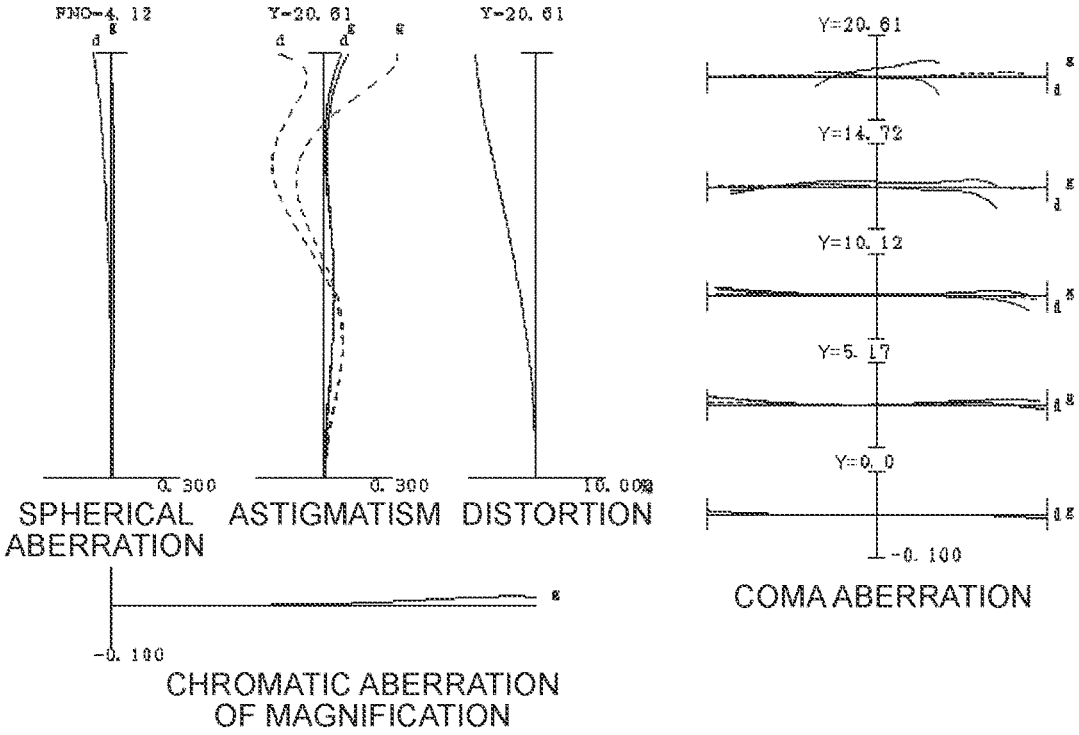


FIG. 23B

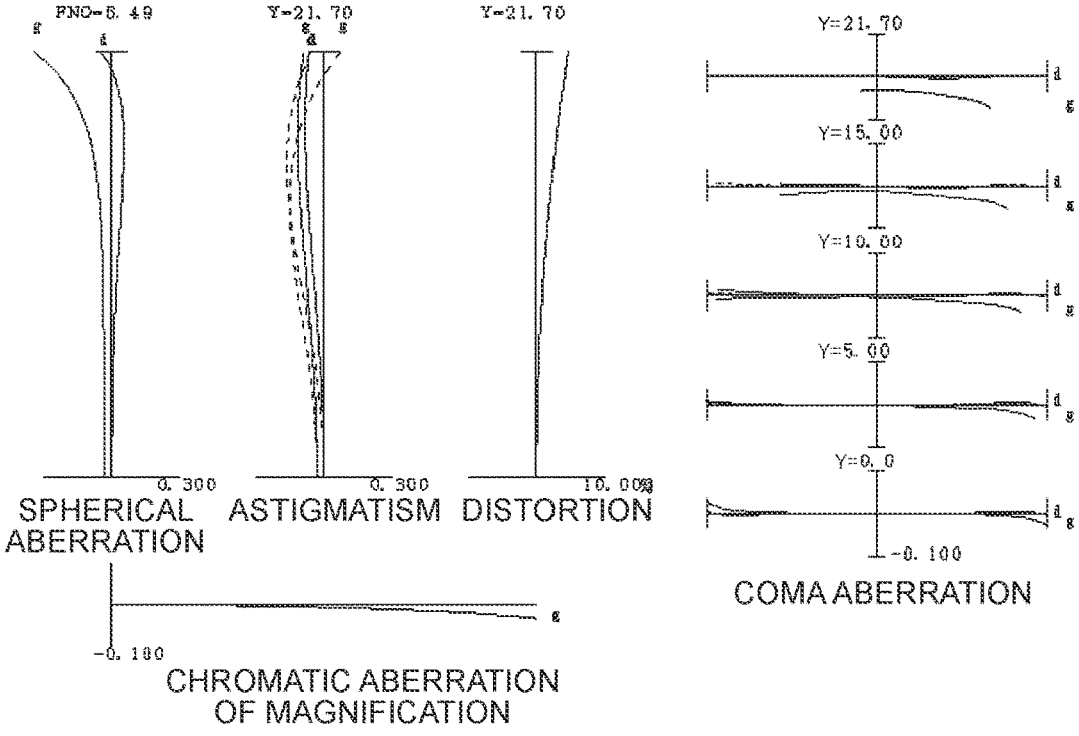
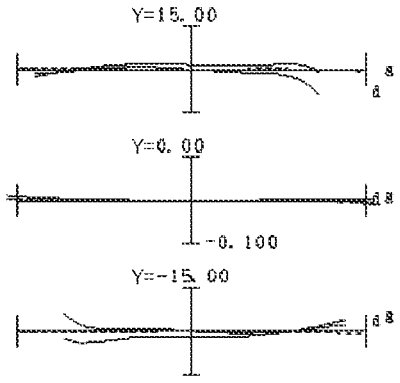
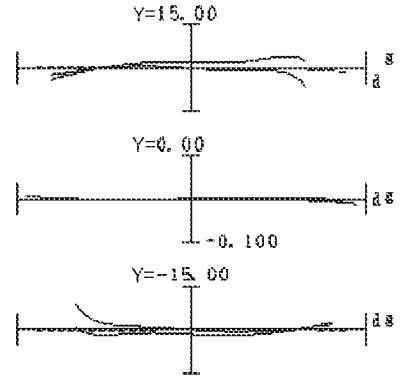


FIG. 24A



COMA ABERRATION

FIG. 24B



COMA ABERRATION

FIG.25

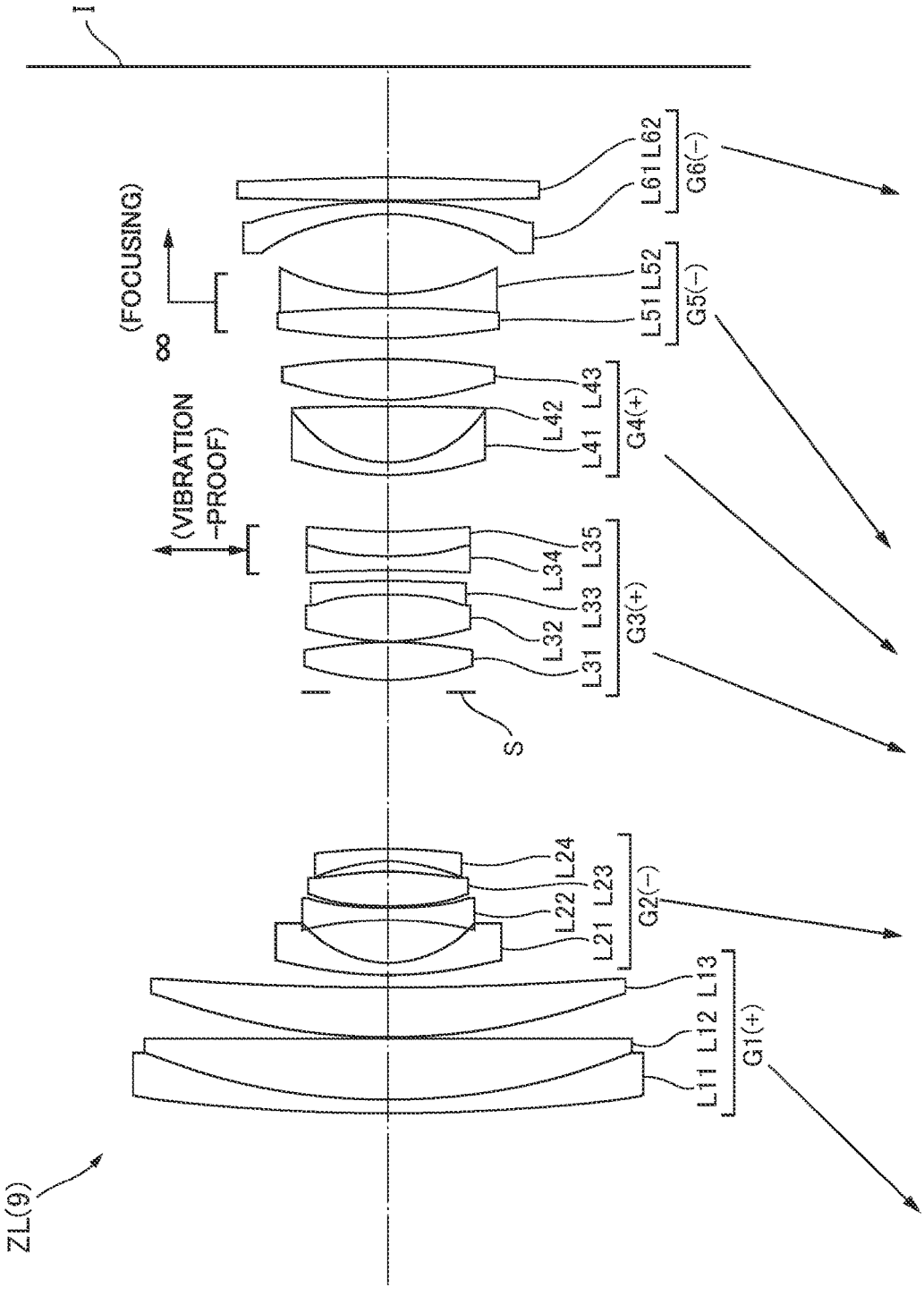

I
(FOCUSING)
∞
(VIBRATION -PROOF)
L61 L62
G6(−)
L51 L52
G5(−)
L42
L41 L43
G4(+)
L34
L35
L32
L33
L31
G3(+)
S
L24
L22
L23
L21
G2(−)
L13
L11 L12
G1(+)
ZL(9)

FIG. 26A

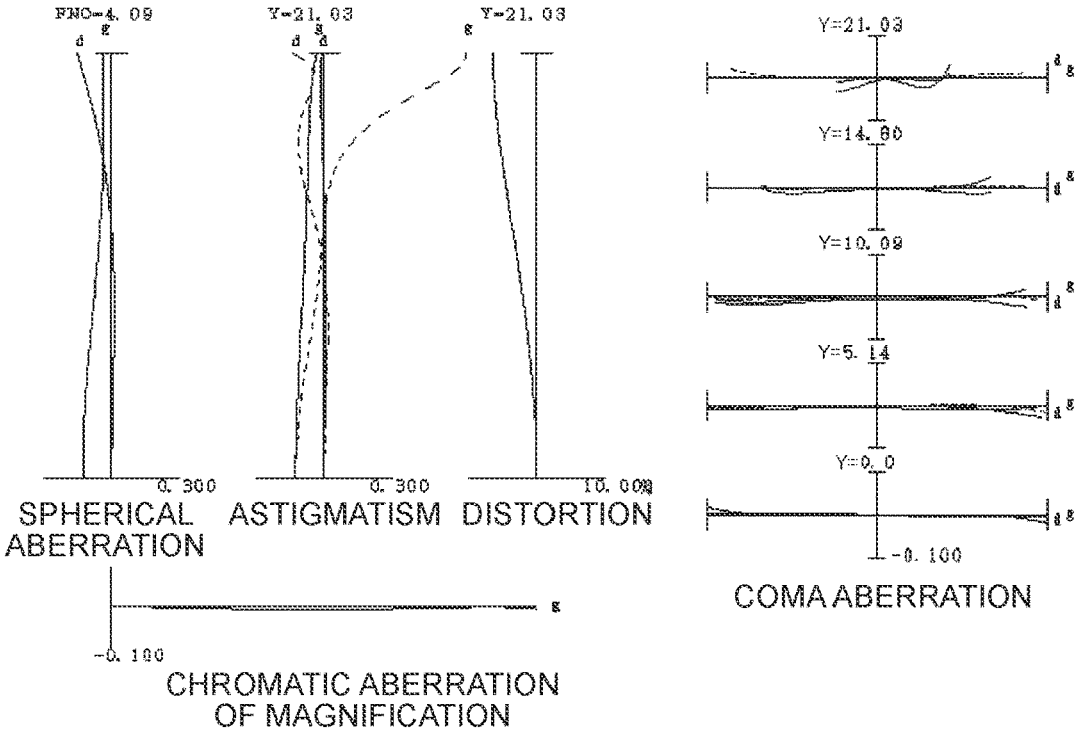


FIG. 26B

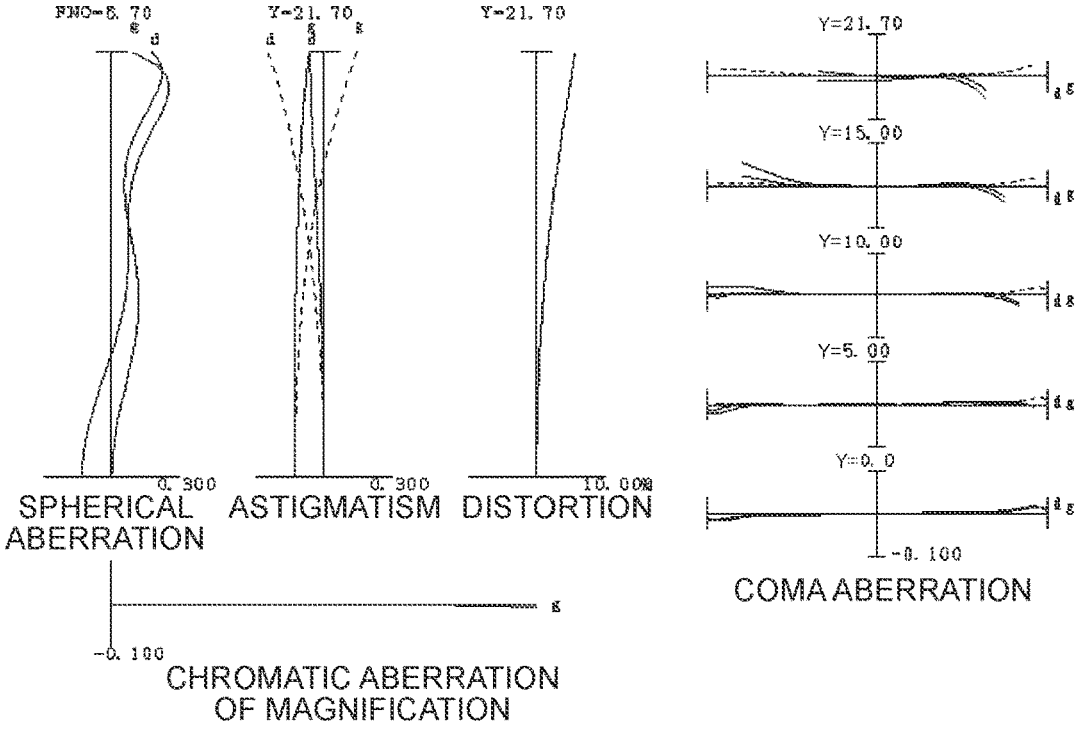
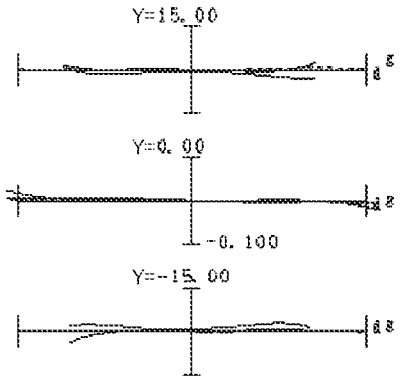
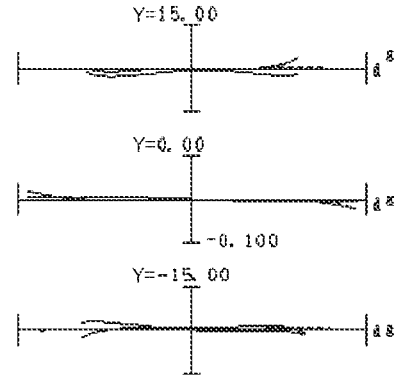


FIG. 27A



COMA ABERRATION

FIG. 27B



COMA ABERRATION

FIG.28

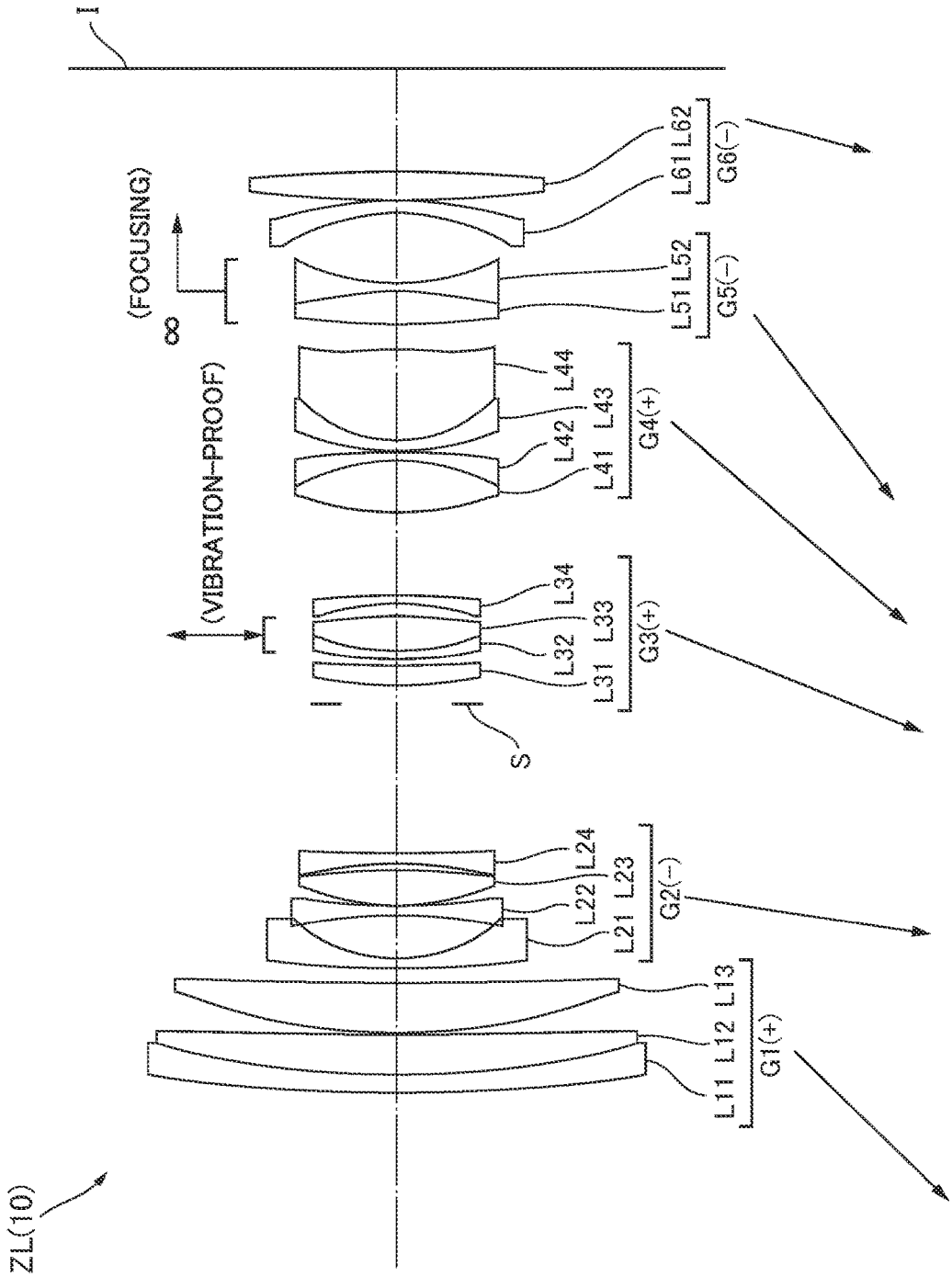

I
(FOCUSING)
∞
(VIBRATION-PROOF)
L61 L62
G6(−)
L51 L52
G5(−)
L44
L42
L41 L43
G4(+)
L34
L32
L31 L33
G3(+)
S
L24
L22
L21 L23
G2(−)
L13
L12
L11
G1(+)
ZL(10)

FIG. 29A

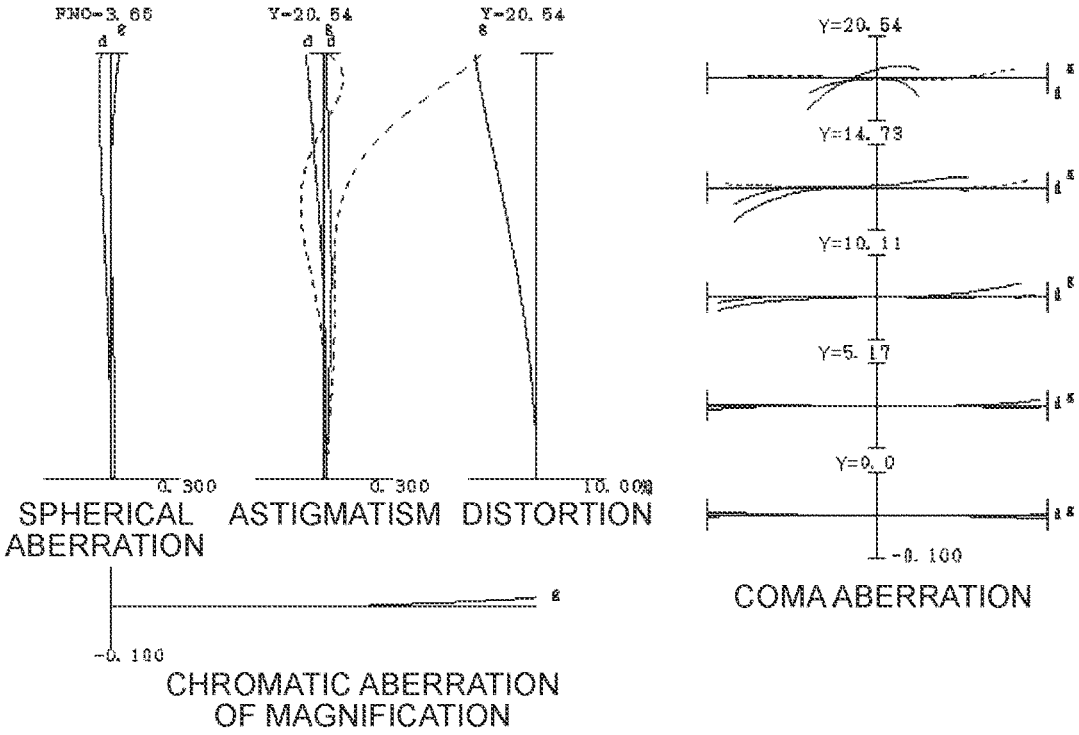


FIG. 29B

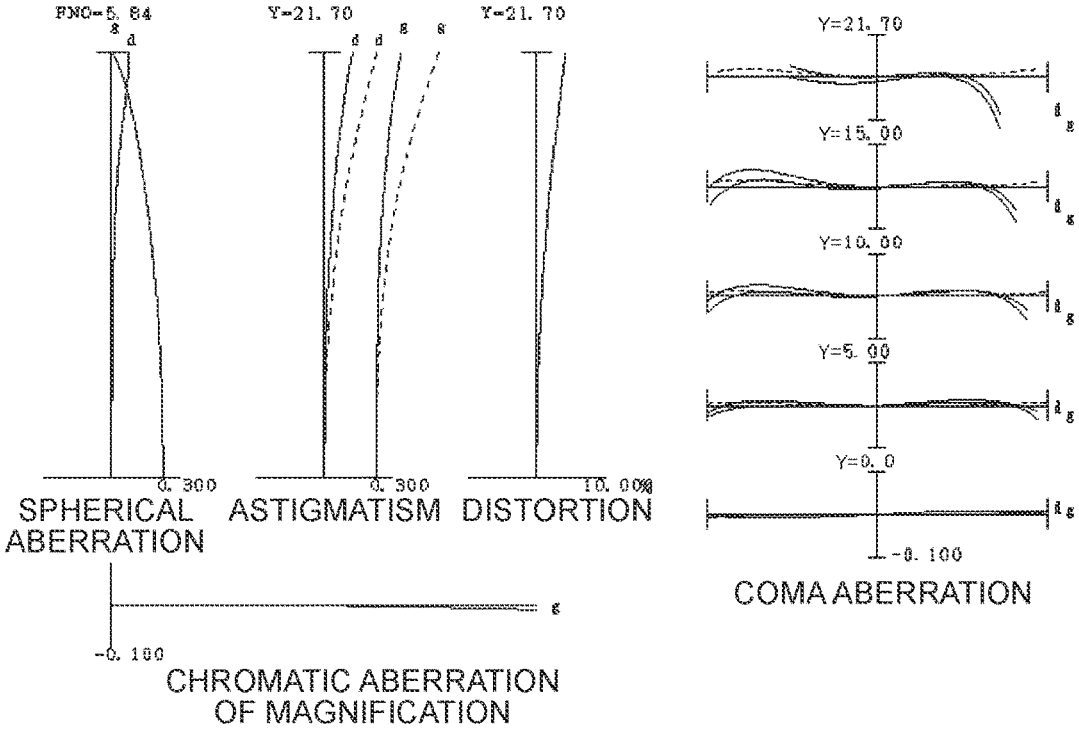
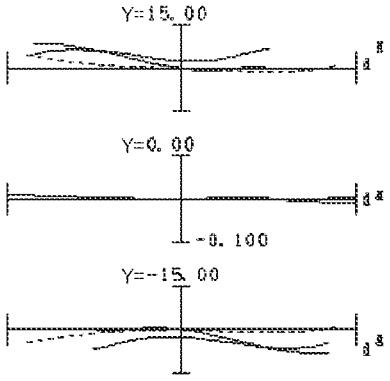
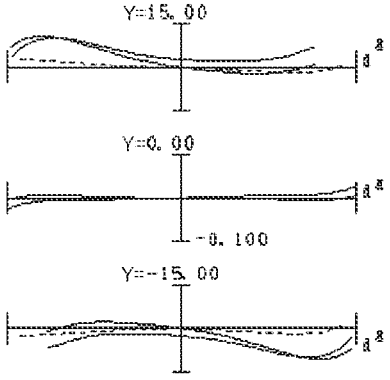


FIG. 30A



COMA ABERRATION

FIG. 30B



COMA ABERRATION

FIG.31

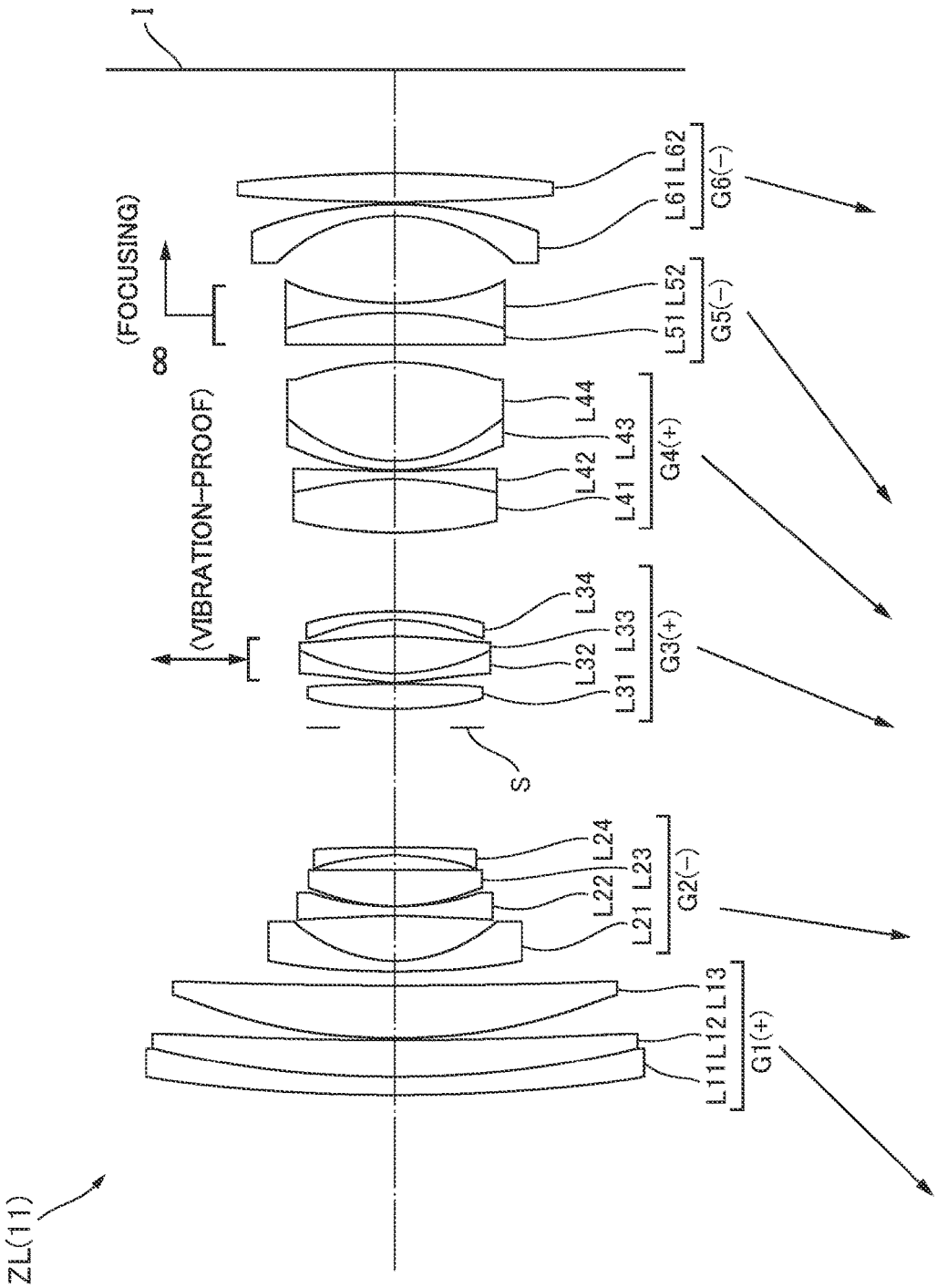

ZL(11)
I
(FOCUSING)
∞
(VIBRATION-PROOF)
L11 L12 L13
G1(+)
L21 L22 L23 L24
G2(−)
L31 L32 L33 L34
G3(+)
S
L41 L42 L43 L44
G4(+)
L51 L52
G5(−)
L61 L62
G6(−)

FIG. 32A

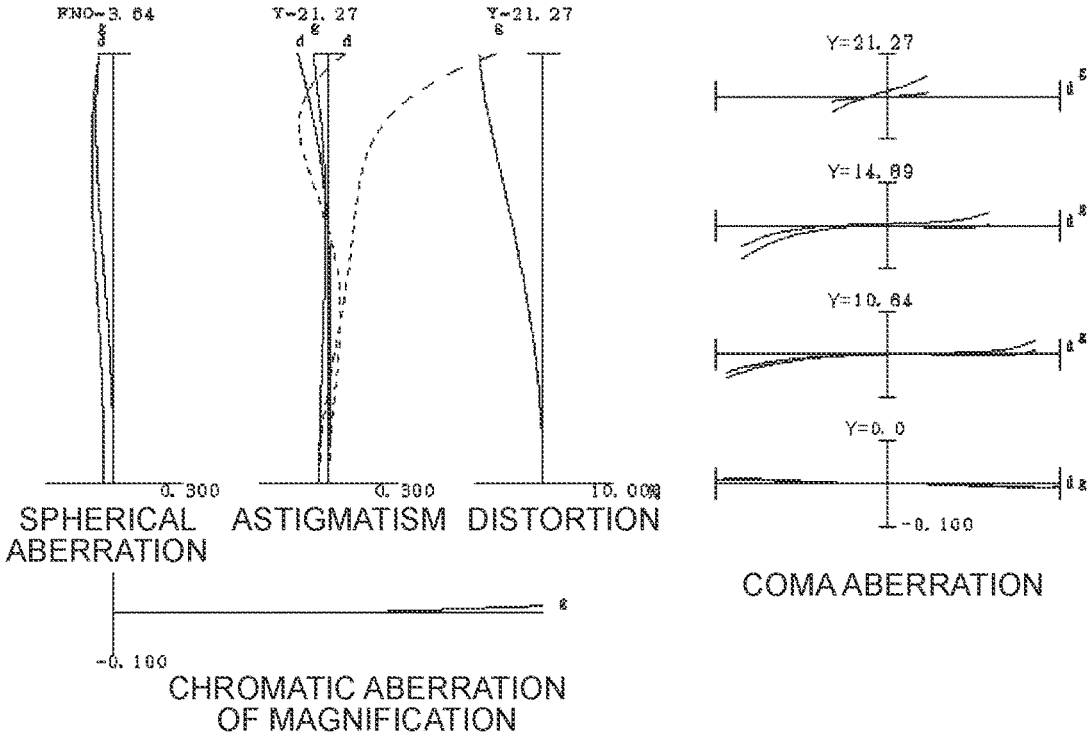


FIG. 32B

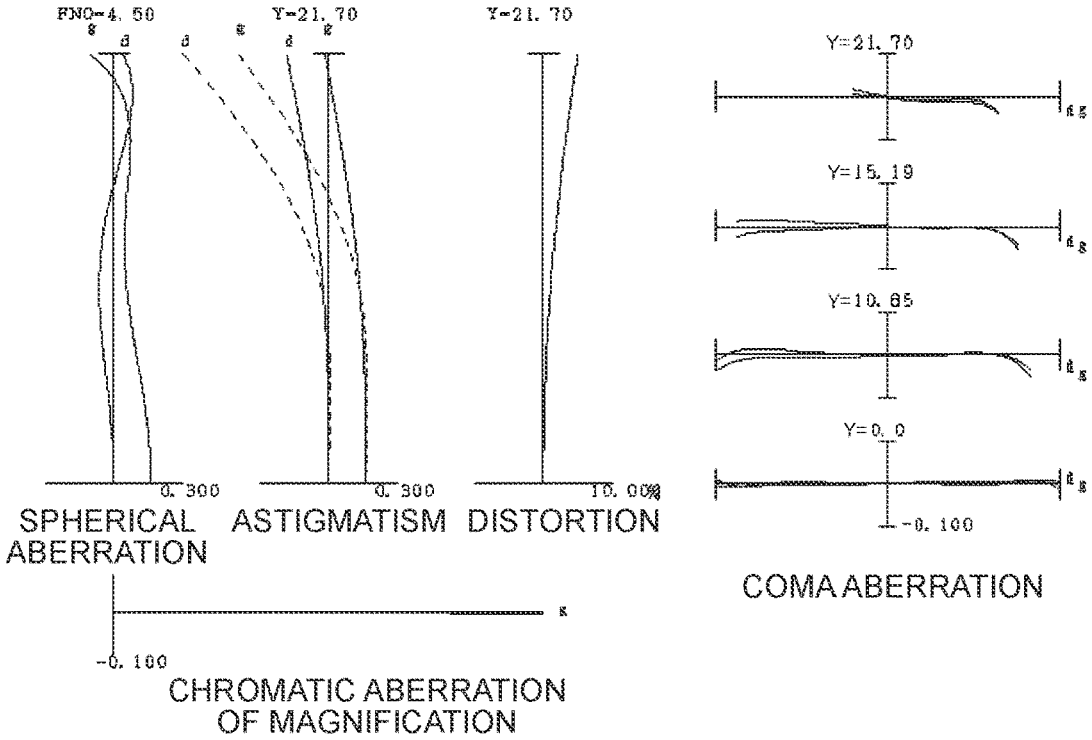
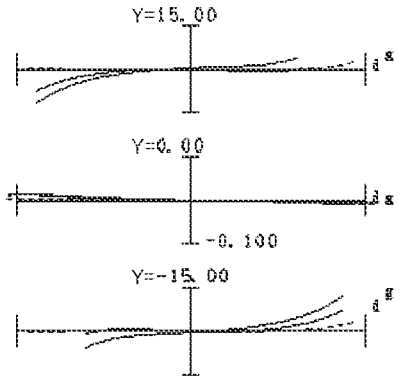
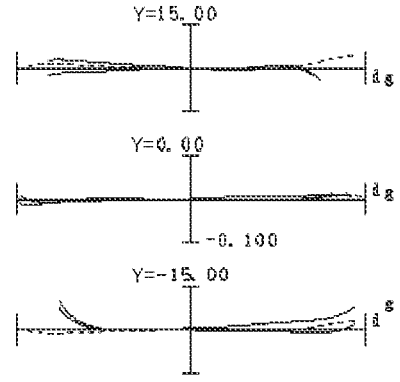


FIG.33A



COMA ABERRATION

FIG.33B



COMA ABERRATION

FIG.34

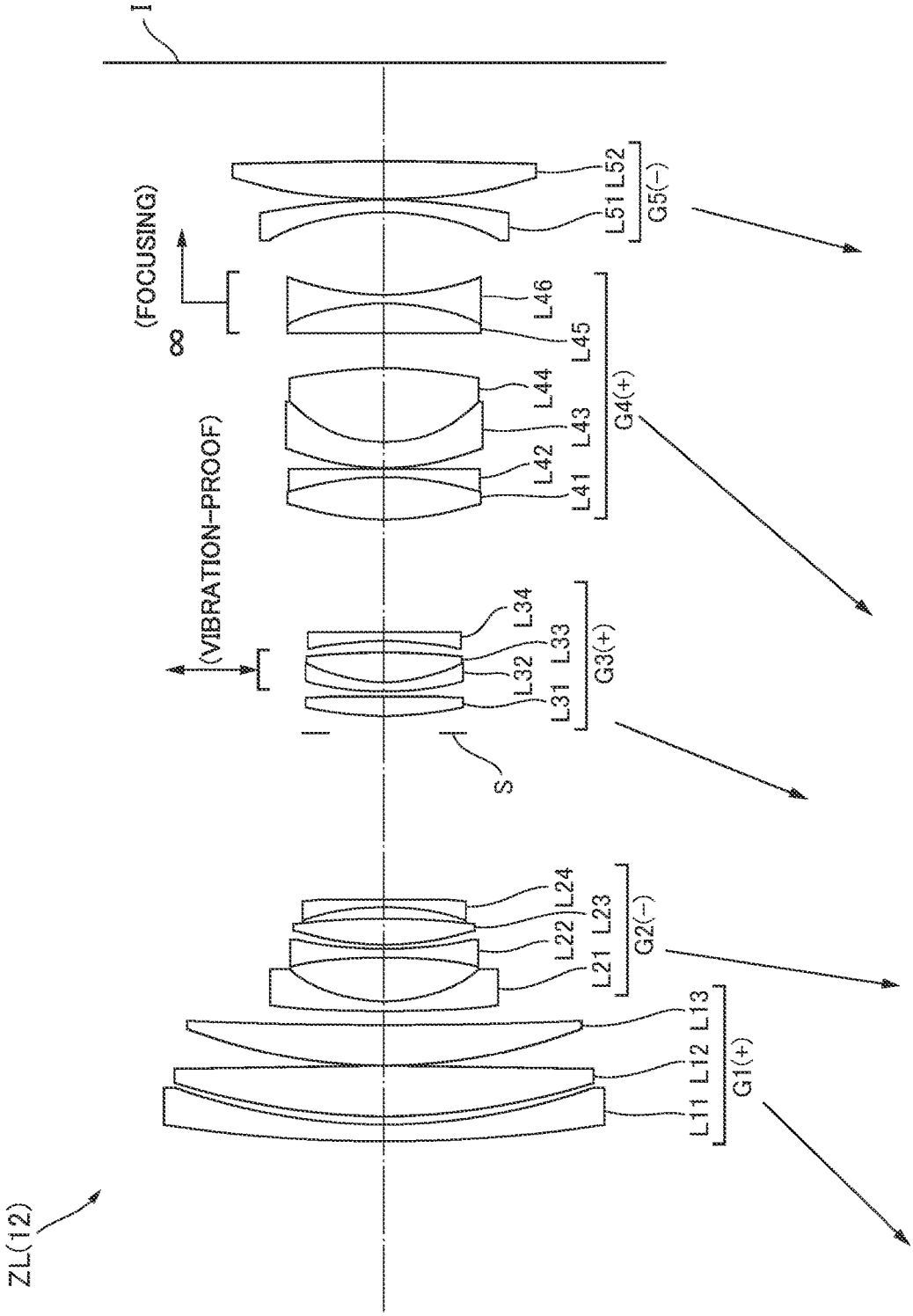

ZL(12)
(FOCUSING)
∞
(VIBRATION-PROOF)
S
L11 L12 L13
G1(+)
L21 L22 L23 L24
G2(−)
L31 L32 L33 L34
G3(+)
L41 L42 L43 L44 L45 L46
G4(+)
L51 L52
G5(−)

FIG. 35A

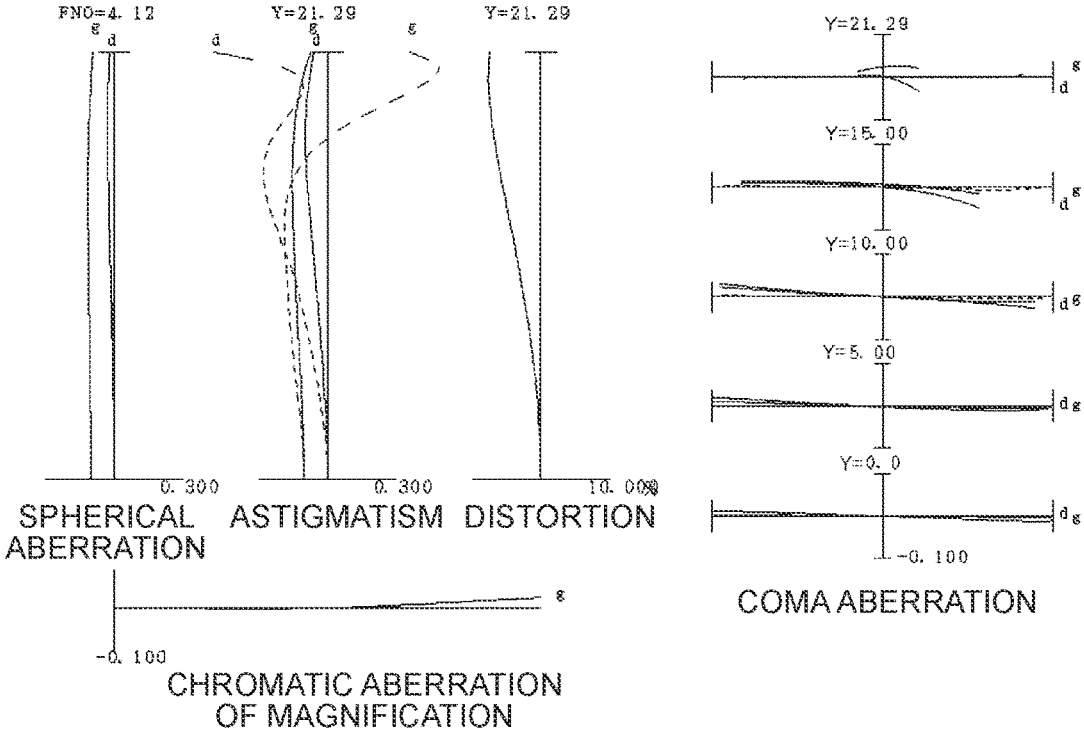


FIG. 35B

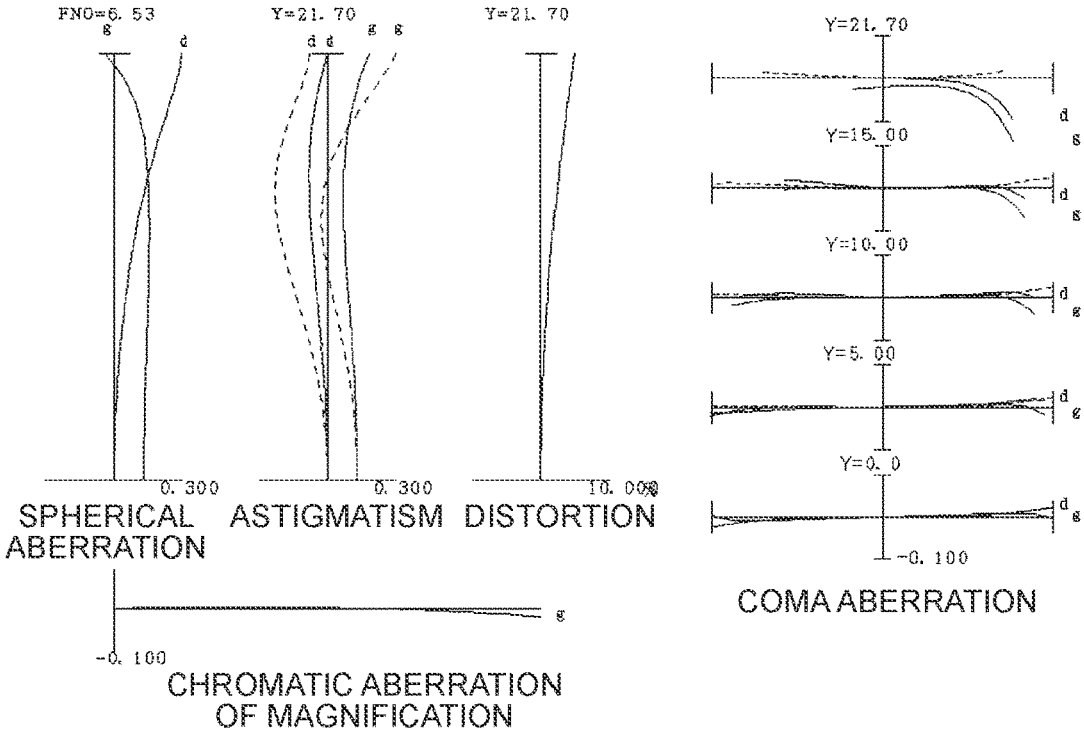
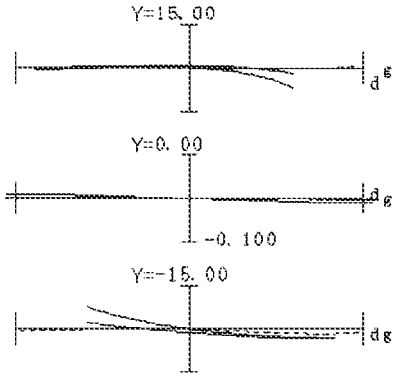
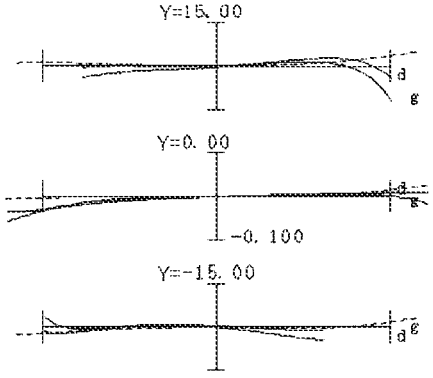


FIG. 36A



COMA ABERRATION

FIG. 36B



COMA ABERRATION

FIG.37

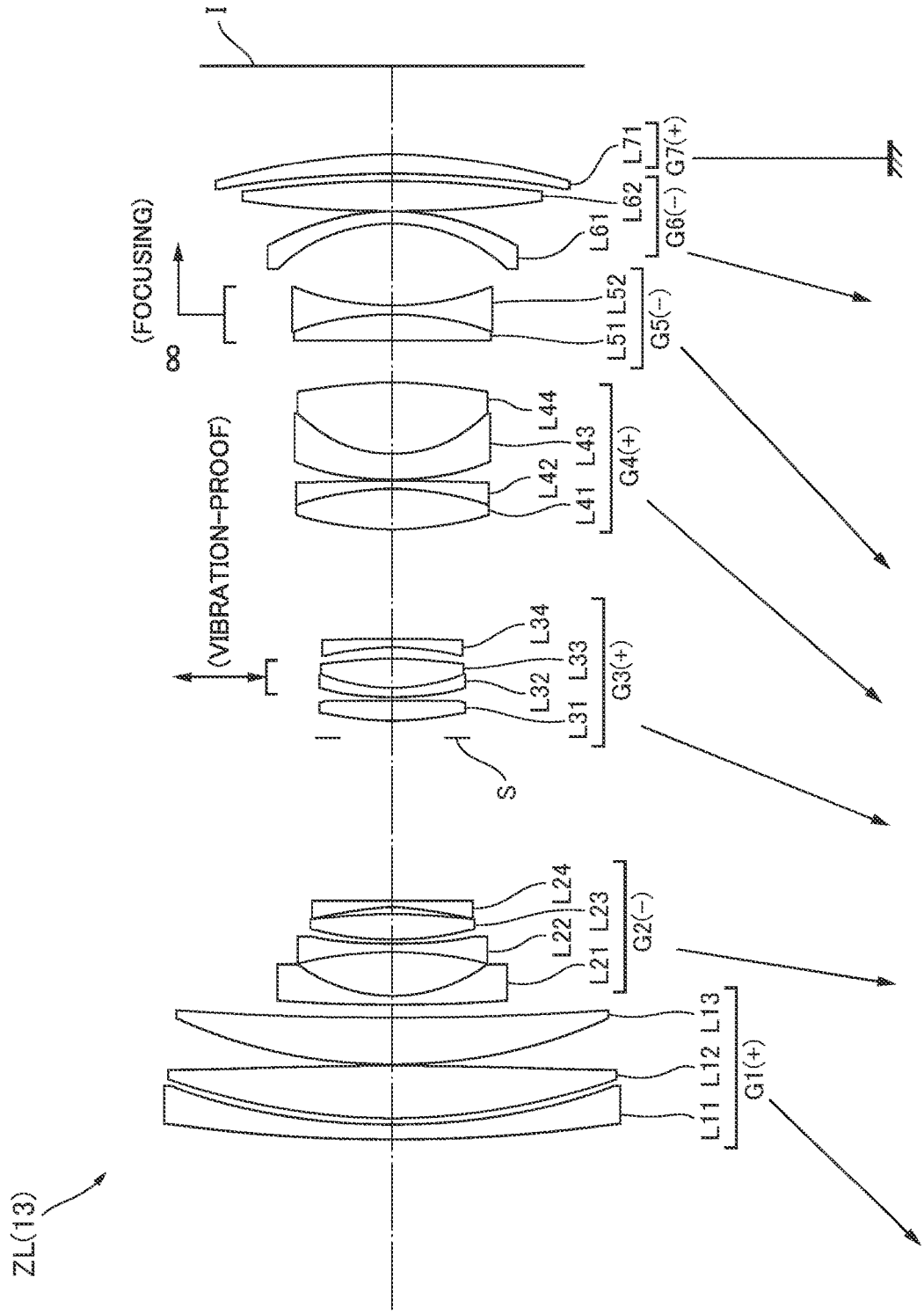

ZL(13)
I
(FOCUSING)
∞
(VIBRATION-PROOF)
S
L11 L12 L13
G1(+)
L21 L22 L23 L24
G2(−)
L31 L32 L33 L34
G3(+)
L41 L42 L43 L44
G4(+)
L51 L52
G5(−)
L61 L62
G6(−)
L71
G7(+)

FIG. 38A

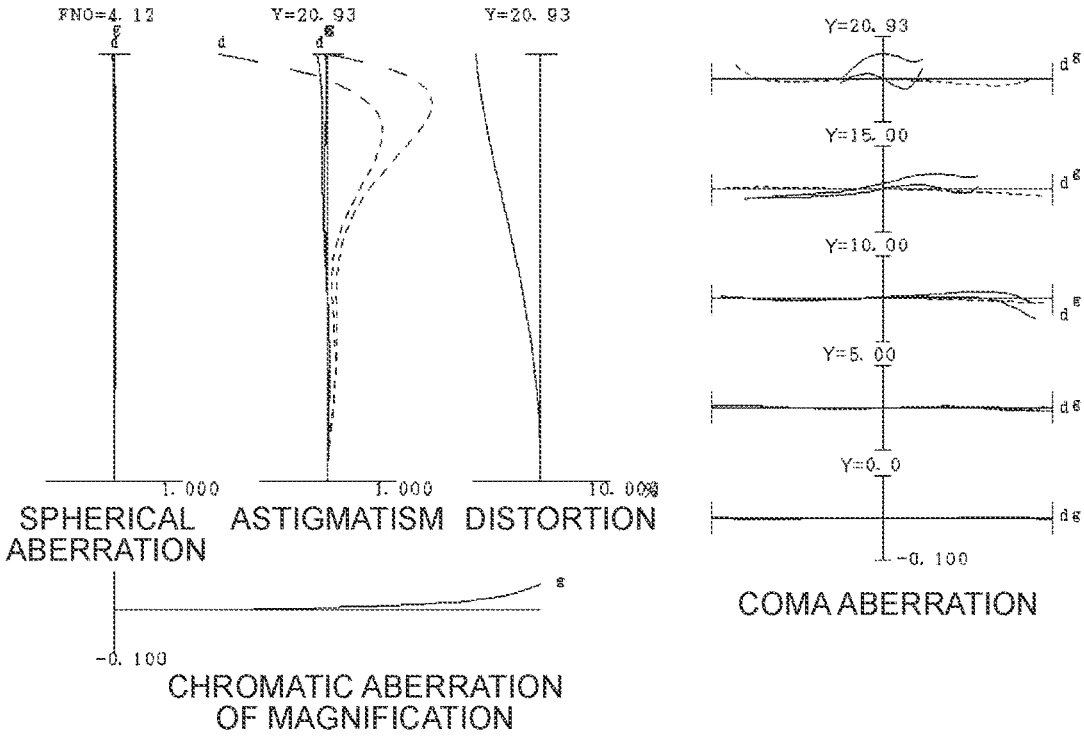


FIG. 38B

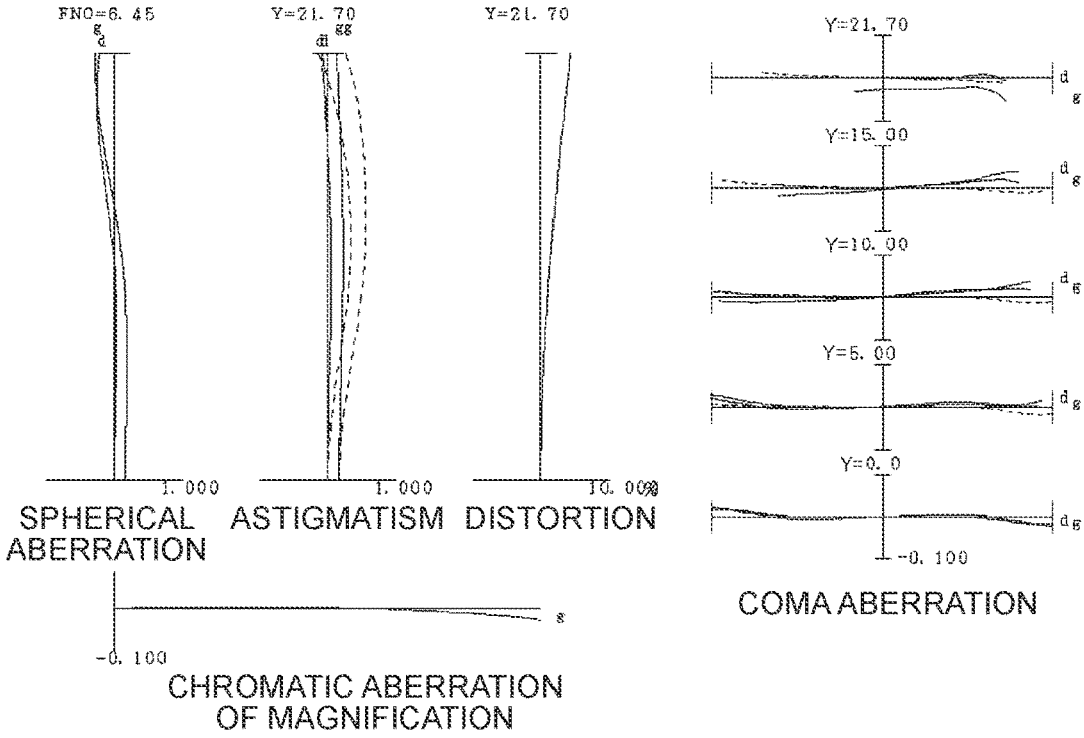
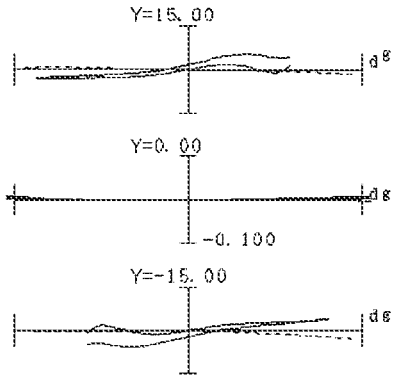
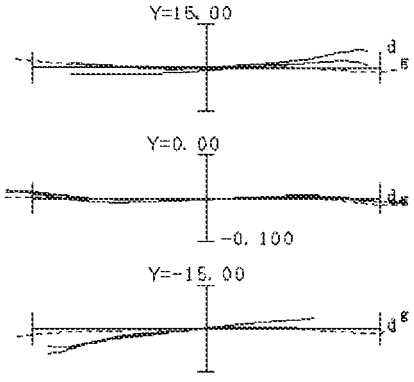


FIG. 39A



COMA ABERRATION

FIG. 39B



COMA ABERRATION

FIG.40

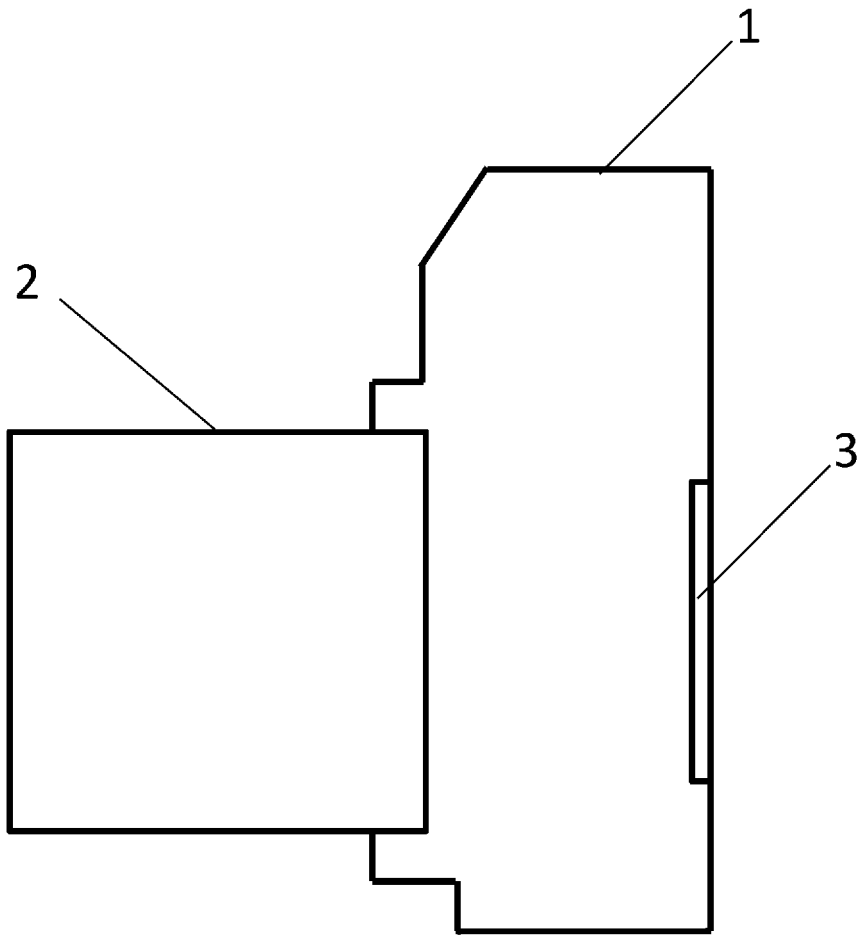
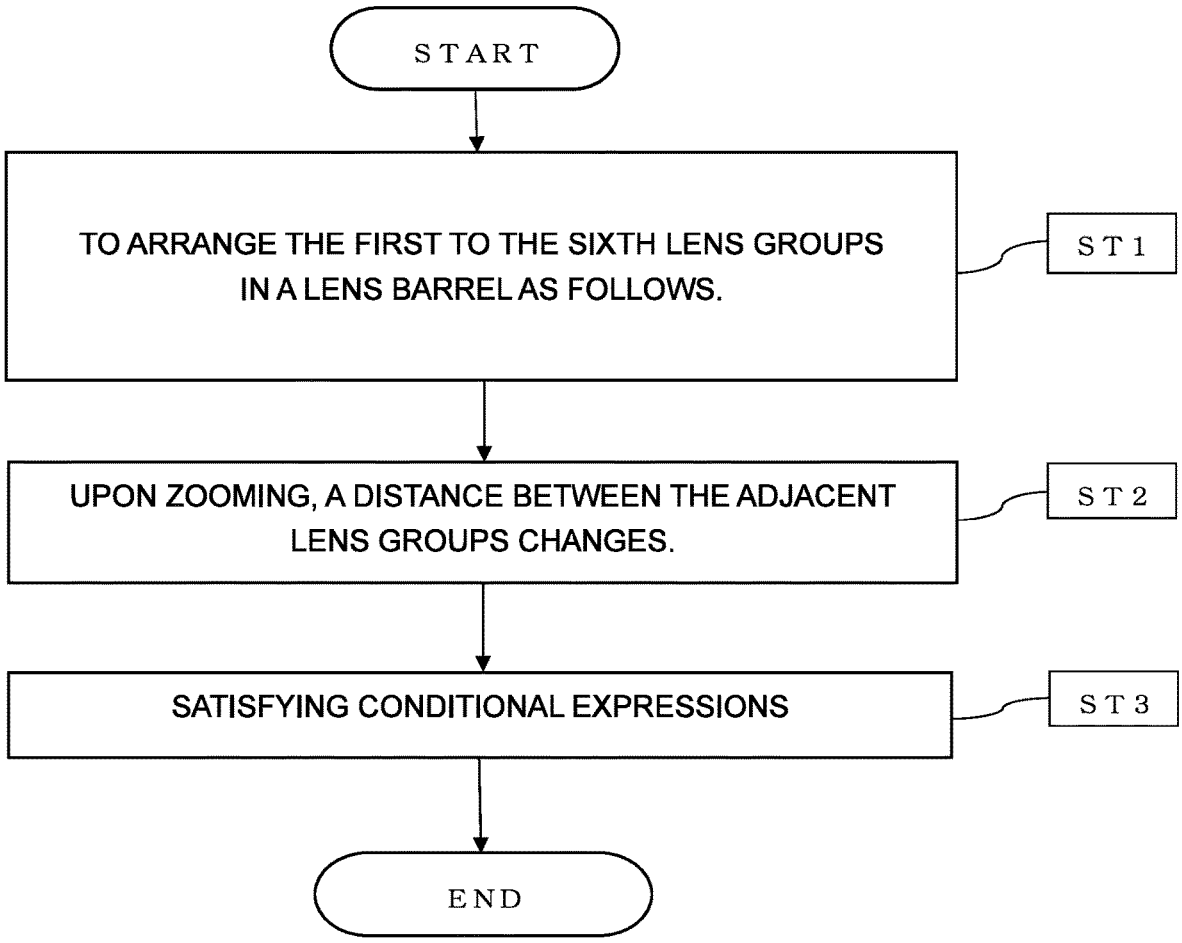


FIG.41


START
TO ARRANGE THE FIRST TO THE SIXTH LENS GROUPS IN A LENS BARREL AS FOLLOWS.
S T 1
UPON ZOOMING, A DISTANCE BETWEEN THE ADJACENT LENS GROUPS CHANGES.
S T 2
SATISFYING CONDITIONAL EXPRESSIONS
S T 3
END

ZOOM OPTICAL SYSTEM, OPTICAL APPARATUS AND METHOD FOR MANUFACTURING THE ZOOM OPTICAL SYSTEM

TECHNICAL FIELD

[0001] The present invention relates to a zoom optical system, an optical apparatus including the same, and a method for manufacturing the zoom optical system.

TECHNICAL BACKGROUND

[0002] Conventionally, zoom optical systems suitable for photographic cameras, electronic still cameras, video cameras and the like have been proposed (for example, see Patent literature 1). If the zooming capability and the angle of view of the zoom optical system are increased, it is difficult to achieve a favorable optical performance, and the zoom optical system tends to increase in size.

PRIOR ARTS LIST

Patent Document

[0003] Patent literature 1: Japanese Laid-Open Patent Publication No. H09-184981(A)

SUMMARY OF THE INVENTION

[0004] A zoom optical system according to a first aspect comprises, in order from an object: a first lens group having a positive refractive power; a second lens group having a negative refractive power; a third lens group having a positive refractive power; a fourth lens group having a positive refractive power; a fifth lens group; and a sixth lens group, wherein upon zooming, a distance between the adjacent lens groups changes, and the zoom optical system satisfies the following conditional expression:

$$1.00<Mv4/Mv3<3.00$$

[0005] where Mv3: an amount of movement of the third lens group upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

[0006] Mv4: an amount of movement of the fourth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

[0007] An optical apparatus according to a second aspect comprises the zoom optical system mounted thereon.

[0008] A method according to a third aspect for manufacturing a zoom optical system that comprises, in order from the object: a first lens group having a positive refractive power; a second lens group having a negative refractive power; a third lens group having a positive refractive power; a fourth lens group having a positive refractive power; a fifth lens group; and a sixth lens group. the method comprises: arranging the lens groups in a lens barrel such that upon zooming, a distance between the adjacent lens groups changes, and satisfying the following conditional expression:

$$1.00<Mv4/Mv3<3.00$$

[0009] where Mv3: an amount of movement of the third lens group upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

[0010] Mv4: an amount of movement of the fourth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a lens configuration diagram of a zoom optical system according to a first example upon focusing on infinity in a wide angle end state;

[0012] FIGS. **2**A and **2**B are various aberration graphs of the zoom optical system according to the first example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0013] FIGS. **3**A and **3**B are coma aberration graphs of the zoom optical system according to the first example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0014] FIG. **4** is a lens configuration diagram of a zoom optical system according to a second example upon focusing on infinity in a wide angle end state;

[0015] FIGS. **5**A and **5**B are various aberration graphs of the zoom optical system according to the second example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0016] FIGS. **6**A and **6**B are coma aberration graphs of the zoom optical system according to the second example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0017] FIG. **7** is a lens configuration diagram of a zoom optical system according to a third example upon focusing on infinity in a wide angle end state;

[0018] FIGS. **8**A and **8**B are various aberration graphs of the zoom optical system according to the third example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0019] FIGS. **9**A and **9**B are coma aberration graphs of the zoom optical system according to the third example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0020] FIG. **10** is a lens configuration diagram of a zoom optical system according to a fourth example upon focusing on infinity in a wide angle end state;

[0021] FIGS. **11**A and **11**B are various aberration graphs of the zoom optical system according to the fourth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0022] FIGS. **12**A and **12**B are coma aberration graphs of the zoom optical system according to the fourth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0023] FIG. **13** is a lens configuration diagram of a zoom optical system according to a fifth example upon focusing on infinity in a wide angle end state;

[0024] FIGS. **14**A and **14**B are various aberration graphs of the zoom optical system according to the fifth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0025] FIGS. **15**A and **15**B are coma aberration graphs of the zoom optical system according to the fifth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0026] FIG. **16** is a lens configuration diagram of a zoom optical system according to a sixth example upon focusing on infinity in a wide angle end state;

[0027] FIGS. **17**A and **17**B are various aberration graphs of the zoom optical system according to the sixth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0028] FIGS. **18**A and **18**B are coma aberration graphs of the zoom optical system according to the sixth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0029] FIG. **19** is a lens configuration diagram of a zoom optical system according to a seventh example upon focusing on infinity in a wide angle end state;

[0030] FIGS. **20**A and **20**B are various aberration graphs of the zoom optical system according to the seventh example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0031] FIGS. **21**A and **21**B are coma aberration graphs of the zoom optical system according to the seventh example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0032] FIG. **22** is a lens configuration diagram of a zoom optical system according to an eighth example upon focusing on infinity in a wide angle end state;

[0033] FIGS. **23**A and **23**B are various aberration graphs of the zoom optical system according to the eighth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0034] FIGS. **24**A and **24**B are coma aberration graphs of the zoom optical system according to the eighth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0035] FIG. **25** is a lens configuration diagram of a zoom optical system according to a ninth example upon focusing on infinity in a wide angle end state;

[0036] FIGS. **26**A and **26**B are various aberration graphs of the zoom optical system according to the ninth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0037] FIGS. **27**A and **27**B are coma aberration graphs of the zoom optical system according to the ninth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0038] FIG. **28** is a lens configuration diagram of a zoom optical system according to a tenth example upon focusing on infinity in a wide angle end state;

[0039] FIGS. **29**A and **29**B are various aberration graphs of the zoom optical system according to the tenth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0040] FIGS. **30**A and **30**B are coma aberration graphs of the zoom optical system according to the tenth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0041] FIG. **31** is a lens configuration diagram of a zoom optical system according to an eleventh example upon focusing on infinity in a wide angle end state;

[0042] FIGS. **32**A and **32**B are various aberration graphs of the zoom optical system according to the eleventh example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0043] FIGS. **33**A and **33**B are coma aberration graphs of the zoom optical system according to the eleventh example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0044] FIG. **34** is a lens configuration diagram of a zoom optical system according to a twelfth example upon focusing on infinity in a wide angle end state;

[0045] FIGS. **35**A and **35**B are various aberration graphs of the zoom optical system according to the twelfth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0046] FIGS. **36**A and **36**B are coma aberration graphs of the zoom optical system according to the twelfth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0047] FIG. **37** is a lens configuration diagram of a zoom optical system according to a thirteenth example upon focusing on infinity in a wide angle end state;

[0048] FIGS. **38**A and **38**B are various aberration graphs of the zoom optical system according to the thirteenth example upon focusing on infinity in the wide-angle end state and a telephoto end state;

[0049] FIGS. **39**A and **39**B are coma aberration graphs of the zoom optical system according to the thirteenth example in the wide-angle end state and the telephoto end state when blur correction is performed;

[0050] FIG. **40** shows a configuration of a camera that comprises a zoom optical system according to this embodiment; and

[0051] FIG. **41** is a flowchart showing a method for manufacturing the zoom optical system according to this embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0052] Hereinafter, a zoom optical system and an optical apparatus according to this embodiment will be described with reference to the drawings. First, a camera (optical apparatus) comprising the zoom optical system according to this embodiment is described with reference to FIG. **40**. As shown in FIG. **40**, the camera **1** is a digital camera that comprises the zoom optical system according to this embodiment as a photographing lens **2**. In the camera **1**, light from an object (photographic object), not shown, is collected by the photographing lens **2**, and reaches an image pickup element **3**. Accordingly, the light from the photographic object is captured by an image pickup element **3**, and is recorded as a photographic object image in a memory, not shown. A photographer can thus take an image of the photographic object through the camera **1**. Note that the camera may be a mirrorless camera, or a single-lens reflex type camera that includes a quick return mirror.

[0053] Next, the zoom optical system (photographing lens) according to this embodiment will be described. As shown in FIG. **1**, a zoom optical system ZL(**1**) that is an example of a zoom optical system (zoom lens) ZL according to this embodiment comprises, in order from an object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5; and a sixth lens group G6. Upon zooming, a distance between the adjacent lens groups changes. Accordingly, variation in astigmatism and spherical aberration upon zooming can be suppressed.

[0054] In the configuration described above, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (1).

$$1.000 < Mv4/Mv3 < 3.00 \quad (1)$$

[0055] where Mv3: an amount of movement of the third lens group G3 upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

[0056] Mv4: an amount of movement of the fourth lens group G4 upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +).

[0057] According to this embodiment, the zoom optical system that has a high zooming ratio and a favorable optical performance, and the optical apparatus that comprises the zoom optical system can be obtained. The zoom optical system ZL according to this embodiment may be a zoom optical system ZL(**2**) shown in FIG. **4**, a zoom optical system ZL(**3**) shown in FIG. **7**, a zoom optical system ZL(**4**) shown in FIG. **10**, or a zoom optical system ZL(**5**) shown in FIG. **13**. The zoom optical system ZL according to this embodiment may be a zoom optical system ZL(**6**) shown in FIG. **16**, a zoom optical system ZL(**7**) shown in FIG. **19**, a zoom optical system ZL(**8**) shown in FIG. **22**, or a zoom optical system ZL(**9**) shown in FIG. **25**. The zoom optical system ZL according to this embodiment may be a zoom optical system ZL(**10**) shown in FIG. **28**, a zoom optical system ZL(**11**) shown in FIG. **31**, or a zoom optical system ZL(**13**) shown in FIG. **37**.

[0058] The conditional expression (1) defines the ratio between the amount of movement of the third lens group G3 and the amount of movement of the fourth lens group G4 upon zooming from the wide angle end state to the telephoto end state. By satisfying the conditional expression (1), variation in spherical aberration and field curves upon zooming can be suppressed.

[0059] If the corresponding value of the conditional expression (1) falls below the lower limit value, it is difficult to suppress variation in field curves upon zooming. By setting the lower limit value of the conditional expression (1) to 1.05, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (1) may be set to 1.10, 1.15, 1.18, 1.20, 1.23, 1.25, 1.28, 1.30, 1.33 and further to 1.35.

[0060] If the corresponding value of the conditional expression (1) exceeds the upper limit value, it is difficult to correct the field curves in the wide angle end state. By setting the upper limit value of the conditional expression (1) to 2.80, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (1) may be set to 2.50, 2.30, 2.00, 1.80, 1.65, 1.62, 1.60, 1.58, 1.55, 1.53, and further to 1.50.

[0061] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (2).

$$0.000 < Mv2/fw < 10.00 \quad (2)$$

[0062] where Mv2: an amount of movement of the second lens group G2 upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +), and

[0063] fw: a focal length of the zoom optical system ZL in the wide angle end state.

[0064] The conditional expression (2) defines the amount of movement of the second lens group G2 upon zooming from the wide angle end state to the telephoto end state. Note that the conditional expression (2) means that the second lens group G2 moves toward the object upon zooming from the wide angle end state to the telephoto end state. By satisfying the conditional expression (2), the spherical aberration in the telephoto end state can be favorably corrected.

[0065] If the corresponding value of the conditional expression (2) falls below the lower limit value, it is difficult to correct the spherical aberration in the telephoto end state. By setting the lower limit value of the conditional expression (2) to 0.05, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (2) may be set to 0.08, 0.10, 0.13, 0.15, 0.18, 0.20, 0.22, and further to 0.24.

[0066] If the corresponding value of the conditional expression (2) exceeds the upper limit value, it is difficult to correct the spherical aberration in the telephoto end state. By setting the upper limit value of the conditional expression (2) to 8.00, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (2) may be set to 5.00, 3.00, 2.50, 2.20, 2.00, 1.80, 1.50, 1.30, 1.10, 0.95, 0.90, 0.85, 0.80, 0.75, and further to 0.70.

[0067] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (3).

$$3.00 < ft/fw < 30.00 \quad (3)$$

[0068] where ft: a focal length of the zoom optical system ZL in the telephoto end state, and

[0069] fw: a focal length of the zoom optical system ZL in the wide angle end state.

[0070] The conditional expression (3) defines the zooming ratio of the zoom optical system ZL. By satisfying the conditional expression (3), the advantageous effects of this embodiment can be exerted to the maximum at a high zooming ratio. By setting the lower limit value of the conditional expression (3) to 3.30, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (3) may be set to 3.50, 4.00, 4.50, 5.00, 6.00, and further to 7.00. By setting the upper limit value of the conditional expression (3) to 25.00, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (3) may be set to 20.00, 15.00, 10.00, 9.00, and further to 8.00.

[0071] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (4).

$$35.0° < \omega w < 75.0° \quad (4)$$

[0072] where ωw: a half angle of view of the zoom optical system ZL in the wide angle end state.

[0073] The conditional expression (4) defines the half angle of view of the zoom optical system ZL in the wide angle end state. By satisfying the conditional expression (4), the field curves can be favorably corrected. By setting the

lower limit value of the conditional expression (4) to 38.0°, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (4) may be set to 40.0°. By setting the upper limit value of the conditional expression (4) to 70.0°, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (4) may be set to 60.0°, 50.0°, and further to 45.0°.

[0074] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (5).

$$2.5°<\omega t<15.0° \quad (5)$$

[0075] where ωt: a half angle of view of the zoom optical system ZL in the telephoto end state.

[0076] The conditional expression (5) defines the half angle of view of the zoom optical system ZL in the telephoto end state. By satisfying the conditional expression (5), the advantageous effects of this embodiment can be exerted to the maximum at a high zooming ratio. By setting the lower limit value of the conditional expression (5) to 4.0°, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (5) may be set to 5.0° and further to 5.5°. By setting the upper limit value of the conditional expression (5) to 13.0°, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (5) may be set to 12.0°, 11.0°, 10.0°, and further to 9.0°.

[0077] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (6).

$$-0.30<fw/f123w<0.60 \quad (6)$$

[0078] where fw: a focal length of the zoom optical system ZL in the wide angle end state, and

[0079] f123w: a combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the wide angle end state.

[0080] The conditional expression (6) defines the ratio between the focal length of the zoom optical system ZL and the combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the wide angle end state. Note that the conditional expression (6) means that the first lens group G1, the second lens group G2 and the third lens group G3 are substantially afocal in the wide angle end state. By satisfying the conditional expression (6), the spherical aberration and field curves in the wide angle end state can be favorably corrected.

[0081] If the corresponding value of the conditional expression (6) falls below the lower limit value, it is difficult to correct the spherical aberration in the wide angle end state. By setting the lower limit value of the conditional expression (6) to −0.28, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (6) may be set to −0.25, −0.20, −0.15, and further to −0.12.

[0082] If the corresponding value of the conditional expression (6) exceeds the upper limit value, it is difficult to correct the spherical aberration in the wide angle end state. By setting the upper limit value of the conditional expression (6) to 0.55, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (6) may be set to 0.50, 0.45, 0.40, 0.35, 0.30, 0.25, 0.20, 0.15, 0.10, and further to 0.05.

[0083] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (7).

$$-1.50<ft/f123t<1.00 \quad (7)$$

[0084] where ft: a focal length of the zoom optical system ZL in the telephoto end state, and

[0085] f123t: a combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the telephoto end state.

[0086] The conditional expression (7) defines the ratio between the focal length of the zoom optical system ZL and the combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the telephoto end state. Note that the conditional expression (7) means that the first lens group G1, the second lens group G2 and the third lens group G3 are substantially afocal in the telephoto end state. By satisfying the conditional expression (7), the spherical aberration and field curves in the telephoto end state can be favorably corrected.

[0087] If the corresponding value of the conditional expression (7) falls below the lower limit value, it is difficult to correct the spherical aberration in the telephoto end state. By setting the lower limit value of the conditional expression (7) to −1.35, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (7) may be set to −1.00, −0.90, and further to −0.80.

[0088] If the corresponding value of the conditional expression (7) exceeds the upper limit value, it is difficult to correct the spherical aberration in the telephoto end state. By setting the upper limit value of the conditional expression (7) to 0.50, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (7) may be set to 0.20, 0.10, −0.10, and further to −0.20.

[0089] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (8).

$$0.20<BFw/fw<0.60 \quad (8)$$

[0090] where BFw: a distance to an image surface from a lens surface of the zoom optical system ZL closest to an image in the wide angle end state, and

[0091] fw: a focal length of the zoom optical system ZL in the wide angle end state.

[0092] The conditional expression (8) defines the ratio between the back focus of the zoom optical system ZL and the focal length of the zoom optical system ZL in the wide angle end state. By satisfying the conditional expression (8), the field curves in the wide angle end state can be efficiently corrected.

[0093] If the corresponding value of the conditional expression (8) falls below the lower limit value, it is difficult to correct the field curves in the wide angle end state. By setting the lower limit value of the conditional expression (8) to 0.25, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects

of this embodiment, the lower limit value of the conditional expression (8) may be set to 0.30, 0.35, 0.37 and further to 0.40.

[0094] If the corresponding value of the conditional expression (8) exceeds the upper limit value, correction of the field curves in the wide angle end state becomes insufficient. By setting the upper limit value of the conditional expression (8) to 0.56, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (8) may be set to 0.54, 0.52, and further to 0.50.

[0095] Preferably, in the zoom optical system ZL according to this embodiment, upon focusing, the fifth lens group G5 moves with respect to an image surface. Accordingly, variation in spherical aberration upon focusing can be suppressed.

[0096] Preferably, in the zoom optical system ZL according to this embodiment, the fifth lens group G5 includes at least one positive lens, and at least one negative lens. Accordingly, variation in field curves upon focusing can be suppressed.

[0097] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (9).

$$1.00<(-f5)/fw<16.00 \quad (9)$$

where f5: a focal length of the fifth lens group G5, and

[0098] fw: a focal length of the zoom optical system ZL in the wide angle end state.

[0099] The conditional expression (9) defines the ratio between the focal length of the fifth lens group G5 and the focal length of the zoom optical system ZL in the wide angle end state. By satisfying the conditional expression (9), the field curves caused upon focusing can be favorably corrected.

[0100] If the corresponding value of the conditional expression (9) falls below the lower limit value, it is difficult to suppress the field curves caused upon focusing. The amount of movement of the fifth lens group G5 upon focusing increases, which in turn increases the size of the lens barrel. By setting the lower limit value of the conditional expression (9) to 1.10, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (9) may be set to 1.20, 1.30, 1.40, and further to 1.45.

[0101] If the corresponding value of the conditional expression (9) exceeds the upper limit value, correction of the field curves upon focusing becomes insufficient. The amount of movement of the fifth lens group G5 upon focusing increases, which in turn increases the size of the lens barrel. By setting the upper limit value of the conditional expression (9) to 15.50, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (9) may be set to 10.00, 8.00, 5.00, 4.00, 3.00, 2.45, 2.38, 2.33, 2.28, 2.25, and further to 2.10.

[0102] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (10).

$$1.000<Mv5/Mv6<3.00 \quad (10)$$

[0103] where Mv5: an amount of movement of the fifth lens group G5 upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +), and

[0104] Mv6: an amount of movement of the sixth lens group G6 upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +).

[0105] The conditional expression (10) defines the ratio between the amount of movement of the fifth lens group G5 and the amount of movement of the sixth lens group G6 upon zooming from the wide angle end state to the telephoto end state. By satisfying the conditional expression (10), the field curves can be favorably corrected.

[0106] If the corresponding value of the conditional expression (10) falls below the lower limit value, it is difficult to suppress the field curves caused in the fifth lens group G5. By setting the lower limit value of the conditional expression (10) to 1.10, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (10) may be set to 1.20, 1.30, and further to 1.40.

[0107] If the corresponding value of the conditional expression (10) exceeds the upper limit value, it is difficult to correct the field curves in the fifth lens group G5. By setting the upper limit value of the conditional expression (10) to 2.50, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (10) may be set to 2.00, 1.80, and further to 1.60.

[0108] Preferably, in the zoom optical system ZL according to this embodiment, upon zooming, the first lens group G1 moves with respect to the image surface. Accordingly, a high zooming ratio can be achieved.

[0109] Preferably, in the zoom optical system ZL according to this embodiment, the first lens group G1 consists of three or more lenses. Accordingly, particularly in the telephoto end state, the spherical aberration can be favorably corrected. Furthermore, a high zooming ratio can be achieved.

[0110] Preferably, the zoom optical system ZL according to this embodiment satisfies the following conditional expression (11).

$$0.30<Mv1/(ft-fw)<0.80 \quad (11)$$

[0111] where Mv1: an amount of movement of the first lens group G1 upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +),

[0112] ft: a focal length of the zoom optical system ZL in the telephoto end state, and

[0113] fw: a focal length of the zoom optical system ZL in the wide angle end state.

[0114] The conditional expression (11) defines the amount of movement of the first lens group G1 with respect to variation in focal length upon zooming from the wide angle end state to the telephoto end state. By satisfying the conditional expression (11), the spherical aberration and field curves in the telephoto end state can be favorably corrected.

[0115] If the corresponding value of the conditional expression (11) falls below the lower limit value, it is

difficult to correct the spherical aberration in the telephoto end state. By setting the lower limit value of the conditional expression (11) to 0.32, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (11) may be set to 0.33, 0.34, and further to 0.35.

[0116] If the corresponding value of the conditional expression (11) exceeds the upper limit value, it is difficult to correct the field curves in the telephoto end state. Furthermore, the diameter of the first lens group G1 increases, which in turn increases the weight of the lens barrel. By setting the upper limit value of the conditional expression (11) to 0.77, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (11) may be set to 0.70, 0.65, 0.58, 0.50, 0.45, and further to 0.40.

[0117] Preferably, in the zoom optical system ZL according to this embodiment, an air lens is provided in the sixth lens group G6, and the zoom optical system ZL satisfies the following conditional expression (12).

$$0.00<(RAr2+RAr1)/(RAr2-RAr1)<2.00 \quad (12)$$

[0118] where RAr1: a radius of curvature of an object-side lens surface of the air lens of the sixth lens group G6, and

[0119] RAr2: a radius of curvature of an image-side lens surface of the air lens of the sixth lens group G6.

[0120] The conditional expression (12) defines the shape factor of the air lens provided in the sixth lens group G6. By satisfying the conditional expression (12), the field curves can be favorably corrected.

[0121] If the corresponding value of the conditional expression (12) falls below the lower limit value, it is difficult to correct the field curves. By setting the lower limit value of the conditional expression (12) to 0.01, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the lower limit value of the conditional expression (12) may be set to 0.10, 0.20, 0.28, 0.30, 0.40, and further to 0.45.

[0122] If the corresponding value of the conditional expression (12) exceeds the upper limit value, it is difficult to correct the field curves. By setting the upper limit value of the conditional expression (12) to 1.90, the advantageous effects of this embodiment can be more secured. To further secure the advantageous effects of this embodiment, the upper limit value of the conditional expression (12) may be set to 1.70, 1.50, 1.20, and further to 1.00.

[0123] Preferably, in the zoom optical system ZL according to this embodiment, upon zooming, at least the first lens group G1, the third lens group G3, the fourth lens group G4, the fifth lens group G5, and the sixth lens group G6 move with respect to an image surface. Accordingly, the variation in magnification of each lens group upon zooming can be increased. Furthermore, the aberrations occurring at the third lens group G3 upon zooming can be corrected by the fourth lens group G4.

[0124] Preferably, in the zoom optical system ZL according to this embodiment, the lens groups moving upon zooming move toward the object upon zooming from the wide angle end state to the telephoto end state. Accordingly, a sufficient zooming ratio satisfying the performance of this embodiment can be secured.

[0125] Subsequently, referring to FIG. **41**, a method of manufacturing the zoom optical system ZL according to this embodiment is generally described. First, in order from an object, a first lens group G1 having a positive refractive power, a second lens group G2 having a negative refractive power, a third lens group G3 having a positive refractive power, a fourth lens group G4 having a positive refractive power, a fifth lens group G5, and a sixth lens group G6, are arranged (step ST1). It is configured such that the distance between the adjacent lens groups changes upon zooming (step ST2). Each lens is arranged in a lens barrel so as to satisfy at least the conditional expression (1) described above (step ST3). According to such a manufacturing method, the zoom optical system that has a high zooming ratio and a favorable optical performance can be manufactured.

EXAMPLES

[0126] Hereinafter, zoom optical systems ZL according to examples of this embodiment will be described with reference to the drawings. FIGS. **1**, **4**, **7**, **10**, **13**, **16**, **19**, **22**, **25**, **28**, **31**, **34** and **37** are sectional views showing configurations and refractive power distributions of the zoom optical systems ZL {ZL(**1**) to ZL(**13**)} according to first to thirteenth examples. The first to the eleventh examples and the thirteenth example are those of present inventions. The twelfth example is a reference example. In each diagram, the movement direction along the optical axis of each lens group that moves upon zooming from the wide angle end state to the telephoto end state is indicated by an arrow. Furthermore, the movement direction of a focusing group upon focusing from the infinity to a short distant object is indicated by an arrow accompanied by characters "FOCUSING". At least a part of the third lens group G3 is assumed as a vibration-proof group. The movement direction upon image blur correction is indicated by an arrow accompanied by characters "VIBRATION-PROOF".

[0127] In these drawings (FIGS. **1**, **4**, **7**, **10**, **13**, **16**, **19**, **22**, **25**, **28**, **31**, **34** and **37**), each lens group is represented by a combination of a symbol G and a numeral, and each lens is represented by a combination of a symbol L and a numeral. In this case, to prevent the number of types and the numbers of symbols and numerals from being large and complicated, the lens groups and the like are represented using combinations of symbols and numerals independently among the examples. Accordingly, even though the same combinations of symbols and numerals are used among the examples, such usage does not mean the same configuration.

[0128] Tables 1 to 13 are hereinafter shown. Among them, Table 1 is a table showing each data item in the first example, Table 2 is that in the second example, Table 3 is that in the third example, Table 4 is that in the fourth example, Table 5 is that in the fifth example, Table 6 is that in the sixth example, Table 7 is that in the seventh example, Table 8 is that in the eighth example, Table 9 is that in the ninth example, Table 10 is that in the tenth example, Table 11 is that in the eleventh example, Table 12 is that in the twelfth example, and Table 13 is that in the thirteenth example. In each example, d-line (wavelength λ=587.6 nm), and g-line (wavelength λ=435.8 nm) are selected as calculation targets of aberration characteristics.

[0129] In tables of [General data], FNO indicates the F-number, ω indicates the half angle of view (the unit is ° (degrees)), and Y indicates the image height. TL indicates a

distance obtained by adding BF to the distance from the lens foremost surface to the lens last surface on the optical axis upon focusing on infinity. BF indicates the air equivalent distance (back focus) from the lens last surface to the image surface I on the optical axis upon focusing on infinity. Note that these values are indicated for zoom states at the wide-angle end (W), a first intermediate focal length (M1), a second intermediate focal length (M2), and the telephoto end (T). f123w indicates the combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the wide angle end state. f123t indicates the combined focal length of the first lens group G1, the second lens group G2 and the third lens group G3 in the telephoto end state.

[0130] In the table of [Lens data], Surface number indicates the order of the optical surface from the object side along the direction in which the ray travels, R indicates the radius of curvature (the surface whose center of curvature resides on the image side is regarded to have a positive value) of each optical surface, D indicates the surface distance from each optical surface to the next optical surface (or the image surface) on the optical axis, nd is the refractive index of the material of the optical member for d-line, and vd indicates the Abbe number of the material of the optical member with reference to d-line. The radius of curvature "∞" indicates a plane or an aperture, and (Aperture stop S) indicates an aperture stop. The description of the air refractive index nd=1.00000 is omitted. In a case where the lens surface is an aspherical surface, the surface number is assigned * symbol, and the field of the radius of curvature R indicates the paraxial radius of curvature.

[0131] In the table of [Aspherical surface data], the shape of the aspherical surface indicated in [Lens data] is indicated by the following expression W. X(y) indicates the distance (sag amount) from the tangent plane at the vertex of the aspherical surface to the position on the aspherical surface at the height y along the optical axis direction. R indicates the radius of curvature (paraxial radius of curvature) of the reference spherical surface. κ indicates the conic constant. Ai indicates the i-order aspherical coefficient. "E-n" indicates "$\times 10^{-n}$". For example, 1.234E-05=1.234×10^{-5}. Note that the second-order aspherical coefficient A2 is zero, and its description is omitted.

$$X(y)=(y^2/R)/\{1+(1-\kappa\times y^2/R^2)^{1/2}\}+A4\times y^4+A6\times y^6+A8\times y^8+A10\times y^{10}+A12\times y^{12} \quad (A)$$

[0132] The table of [Lens group data] shows the first surface (the surface closest to the object) and the focal length of each lens group.

[0133] The table of [Variable distance data] shows the surface distances at surface numbers where the surface distance is "Variable" in the table showing [Lens data]. Here, surface distances in the zoom states at the wide-angle end (W), the first intermediate focal length (M1), the second intermediate focal length (M2) and the telephoto end (T) upon the infinity focus and the short range focus are indicated. In [Variable distance data], f indicates the focal length of the entire lens system, and β indicates the photographing magnification.

[0134] The table of [Conditional expression corresponding value] shows the value corresponding to each conditional expression.

[0135] Hereinafter, among all the data values, "mm" is generally used for the listed focal length f, radius of curvature R, surface distance D, other lengths and the like if not otherwise specified. However, there is no limitation thereto, because the optical system can achieve equivalent optical performances even if being proportionally enlarged or reduced.

[0136] The description of the table so far is common to all the examples. Hereinafter, redundant description is omitted.

First Example

[0137] A first example is described with reference to FIGS. **1** to **3**A and **3**B and Table 1. FIG. **1** is a lens configuration diagram of a zoom optical system according to the first example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**1**) according to the first example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in the directions indicated by arrows in FIG. **1** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move. The sign (+) or (−) assigned to each lens group symbol indicates the refractive power of the corresponding lens group. This similarly applies to all the following examples.

[0138] The first lens group G1 consists of, in order from the object: a negative meniscus lens L11 having a convex surface facing the object; a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0139] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a biconcave negative lens L24.

[0140] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0141] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a negative meniscus lens L42 having a concave surface facing the object; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0142] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0143] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens

L61 and the positive lens L62. The image side surface of the negative meniscus lens L61 is of aspherical shape. The image surface I is disposed on the image side of the sixth lens group G6.

[0144] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0145] The following Table 1 lists values of data on the zoom optical system according to the first example.

TABLE 1

[General Data]

Zooming ratio 7.848
f123w = −217.63848
f123t = −267.32298

	W	M1	M2	T
FNO	4.12109	5.58779	6.39998	6.50002
ω	42.58698	22.66696	11.13686	6.13014
Y	20.50	21.70	21.70	21.70
TL	126.45486	144.98844	168.50373	188.4741

[Lens Data]

Surface Number	R	D	νd	nd
1	185.7354	2.0000	31.27	1.903660
2	75.9813	1.0263		
3	81.5981	6.4204	67.90	1.593190
4	−494.4016	0.1000		
5	59.1320	6.1300	67.90	1.593190
6	390.1369	D1(Variable)		
7	236.0277	1.2500	32.33	1.953750
8	19.0394	5.0675		
9	−46.6700	1.1000	52.33	1.755000
10	68.1612	0.4169		
11	37.1210	3.3840	20.88	1.922860
12	−52.5580	0.5124		
13	−32.9357	1.0000	46.59	1.816000
14	416.8076	D2(Variable)		
15	∞	2.0000		(Aperture Stop S)
16	39.8204	2.5136	35.72	1.902650
17	−292.5261	0.5000		
18	36.7161	1.0000	29.12	2.001000
19	20.9452	3.3404	53.74	1.579570
20	−76.0620	1.4447		
21	−35.5626	1.0000	32.33	1.953750
22	−290.1606	D3(Variable)		
23	37.1374	4.6344	42.73	1.834810
24	−37.1374	1.0000	31.27	1.903660
25	−308.9768	0.1000		
26	31.6449	2.7756	32.33	1.953750
27	15.2741	8.7030	81.49	1.497100
28*	−40.3095	D4(Variable)		
29	1365.4927	3.0634	23.80	1.846660
30	−35.3251	1.0000	40.13	1.851350
31*	32.6144	D5(Variable)		
32	−16.9998	1.4000	42.51	1.820800
33*	−22.5398	0.1000		
34	626.7496	3.5530	37.57	1.683760
35	−77.6296	BF		

TABLE 1-continued

[Aspherical Surface Data]

28th Surface

κ = 1.0000, A4 = 3.13017E−05, A6 = −1.03090E−07
A8 = 6.53525E−10, A10 = −2.57830E−12, A12 = 0.32673E−14

31st Surface

κ = 1.0000, A4 = −6.66636E−06, A6 = 5.10546E−08
A8 = 1.72567E−11, A10 = −2.40595E−12, A12 = 0.98445E−14

33rd Surface

κ = 1.0000, A4 = −1.93366E−06, A6 = −2.05750E−08
A8 = 8.81224E−11, A10 = −2.94021E−13, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	98.9899
G2	7	−16.5057
G3	16	48.48369
G4	23	28.91747
G5	29	−39.0895
G6	32	−15588.34

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72001	49.99999	105.05133	193.99063
D0	∞	∞	∞	∞
D1	1.50000	17.29645	38.92328	54.52847
D2	18.83905	10.91446	4.55495	1.10018
D3	12.23175	6.39417	3.18615	1.47844
D4	5.54311	4.42699	5.70823	2.00068
D5	10.05055	17.00460	18.93085	24.34574
BF	11.75486	22.41624	30.66474	38.48515

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06221	−0.11053	−0.17918	−0.28386
D0	365.9340	397.4004	473.8851	503.9147
D1	1.50000	17.29645	38.92328	54.52847
D2	18.83905	10.91446	4.55495	1.10018
D3	12.23175	6.39417	3.18615	1.47844
D4	6.43705	6.05192	10.02051	11.69839
D5	9.15661	15.37967	14.61857	14.64803
BF	11.78171	22.50112	30.88824	39.04500

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.402
Conditional Expression(2) Mv2/fw = 0.364
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 42.587
Conditional Expression(5) ωt = 6.130
Conditional Expression(6) fw/f123w = −0.114
Conditional Expression(7) ft/f123t = −0.726
Conditional Expression(8) BFw/fw = 0.476
Conditional Expression(9) (−f5)/fw = 1.581
Conditional Expression(10) Mv5/Mv6 = 1.535
Conditional Expression(11) Mv1/(ft − fw) = 0.366
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.931

[0146] FIGS. **2**A and **2**B are various aberration graphs of the zoom optical system according to the first example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **3**A and **3**B are coma aberration graphs of the zoom optical system according to the first example in the wide-angle end state and the telephoto end state when blur correction is performed. In the aberration

graphs in FIGS. **2**A and **2**B, FNO indicates the F-number, and Y indicates the image height. The spherical aberration graph indicates the value of the F-number corresponding to the maximum diameter. The astigmatism graph and the distortion graph each indicate the maximum value of the image height. The coma aberration graph indicates the value of each image height. The coma aberration graphs in FIGS. **3**A and **3**B indicate the value of each image height. In each aberration graph, d indicates d-line (wavelength)=587.6 nm), and g indicates g-line (wavelength λ=435.8 nm). In the astigmatism graph, a solid line indicates a sagittal image surface, and a broken line indicates a meridional image surface. Note that also in the aberration graph in each example described below, symbols similar to those in this example are used, and redundant description is omitted.

[0147] The various aberration graphs show that the zoom optical system according to the first example favorably corrects the various aberrations, and has an excellent imaging performance.

Second Example

[0148] A second example is described with reference to FIGS. **4** to **6**A and **6**B and Table 2. FIG. **4** is a lens configuration diagram of a zoom optical system according to the second example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**2**) according to the second example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; a third lens group G3 that has a positive refractive power and is provided with an aperture stop S; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the third lens group G3 provided with the aperture stop S, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in the directions indicated by arrows in FIG. **4** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0149] The first lens group G1 consists of, in order from the object: a negative meniscus lens L11 having a convex surface facing the object; a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0150] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object.

[0151] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; an aperture stop S; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0152] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41, and a negative meniscus lens L42 having a concave surface facing the object; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0153] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0154] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image side surface of the negative meniscus lens L61 is of aspherical shape. The image surface I is disposed on the image side of the sixth lens group G6.

[0155] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0156] The following Table 2 lists values of data on the zoom optical system according to the second example.

TABLE 2

[General Data]

Zooming ratio 7.848
f123w = −377.733
f123t = −288.19144

	W	M1	M2	T
FNO	4.12000	5.60000	6.20000	6.49999
ω	43.04718	22.53540	10.65017	6.13829
Y	20.91	21.70	21.70	21.70
TL	125.95528	142.51715	167.85323	186.8435

[Lens Data]

Surface Number	R	D	νd	nd
1	188.64525	2.00000	31.27	1.903660
2	77.80524	0.84780		
3	80.41425	6.51915	67.90	1.593190
4	−471.30377	0.10000		
5	62.30684	5.66572	67.90	1.593190
6	358.24871	D1(Variable)		
7	230.01286	1.25000	43.79	1.848500
8	18.45421	5.50336		
9	−40.33983	1.10000	52.34	1.755000
10	79.65336	0.38546		
11	39.14822	3.37749	23.80	1.846660
12	−47.38891	0.46523		
13	−31.94449	1.00000	46.59	1.816000
14	−2729.77760	D2(Variable)		
15	41.64137	2.51154	35.73	1.902650
16	−289.39118	0.40000		
17	∞	0.10000		(Aperture Stop S)
18	38.12143	1.00000	29.12	2.001000
19	21.49924	3.26023	53.74	1.579570
20	−73.20919	1.47119		
21	−34.94662	1.00000	32.33	1.953750
22	−165.99888	D3(Variable)		
23	37.20805	4.18411	42.73	1.834810

TABLE 2-continued

24	−43.17368	1.00003	31.27	1.903660
25	−659.56023	1.54931		
26	28.71779	1.32801	32.33	1.953750
27	14.76801	9.10325	81.49	1.497100
28*	−42.86465	D4 (Variable)		
29	255.99237	3.36761	23.80	1.846660
30	−33.68693	1.00000	40.13	1.851350
31*	31.06431	D5(Variable)		
32	−23.57856	1.40000	45.21	1.794457
33*	−50.21699	0.10000		
34	91.45040	3.78568	29.84	1.800000
35	−197.78095	BF		

[Aspherical Surface Data]

28th Surface

κ = 1.0000, A4 = 2.56920E−05, A6 = −9.38399E−08
A8 = 4.71077E−10, A10 = −1.70196E−12, A12 = 0.00000E+00

31st Surface

κ = 1.0000, A4 = −6.78111E−06, A6 = 6.47335E−08
A8 = −3.28125E−10, A10 = 2.56418E−13, A12 = 0.00000E+00

33rd Surface

κ = 1.0000, A4 = 3.30419E−06, A6 = −1.76274E−09
A8 = 1.66657E−12, A10 = 1.80471E−14, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	102.16195
G2	7	−16.76640
G3	15	47.83089
G4	23	29.71748
G5	29	−41.62356
G6	32	−236.16863

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72031	50.00094	110.00281	194.00483
D0	∞	∞	∞	∞
D1	1.50000	17.23398	38.86323	55.71214
D2	19.13452	10.77832	3.90279	1.10000
D3	12.16022	5.90688	2.59619	1.47832
D4	5.12094	4.32200	5.90770	2.00000
D5	10.00915	17.06861	18.78383	23.80987
BF	11.25528	20.43218	31.02431	35.96804

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06086	−0.10794	−0.18504	−0.27368
D0	374.0451	407.4838	482.1484	513.1582
D1	1.50000	17.23398	38.86323	55.71214
D2	19.13452	10.77832	3.90279	1.10000
D3	12.16022	5.90688	2.59619	1.47832
D4	6.06949	6.02341	10.42761	11.59738
D5	9.06059	15.36720	14.26392	14.21249
BF	11.25529	20.4322	31.02445	35.96847

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.432
Conditional Expression(2) Mv2/fw = 0.270
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 43.047
Conditional Expression(5) ωt = 6.138
Conditional Expression(6) fw/f123w = −0.112
Conditional Expression(7) ft/f123t = −0.783
Conditional Expression(8) BFw/fw = 0.455
Conditional Expression(9) (−f5)/fw = 1.684

TABLE 2-continued

Conditional Expression(10) Mv5/Mv6 = 1.558
Conditional Expression(11) Mv1/(ft − fw) = 0.360
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.291

[0157] FIGS. **5**A and **5**B are various aberration graphs of the zoom optical system according to the second example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **6**A and **6**B are coma aberration graphs of the zoom optical system according to the second example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the second example favorably corrects the various aberrations, and has an excellent imaging performance.

Third Example

[0158] A third example is described with reference to FIGS. **7** to **9**A and **9**B and Table 3. FIG. **7** is a lens configuration diagram of a zoom optical system according to the third example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**3**) according to the third example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in the directions indicated by arrows in FIG. **7** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0159] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0160] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a negative meniscus lens L22 having a convex surface facing the object; a positive meniscus lens L23 having a convex surface facing the object; and a cemented lens composed of a positive meniscus lens L24 having a concave surface facing the object and a negative meniscus lens L25 having a concave surface facing the object. The image side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L25 is of aspherical shape.

[0161] The third lens group G3 consists of a positive meniscus lens L31 having a convex surface facing the object. The image side surface of the positive meniscus lens L31 is of aspherical shape.

[0162] The fourth lens group G4 consists of, in order from the object: a biconvex positive lens L41; a cemented lens composed of a negative meniscus lens L42 having a convex surface facing the object and a positive meniscus lens L43

having a convex surface facing the object; and a cemented lens composed of a biconvex positive lens L44 and a negative meniscus lens L45 having a concave surface facing the object. The image side surface of the negative meniscus lens L45 is of aspherical shape.

[0163] The fifth lens group G5 consists of a cemented lens composed of a positive meniscus lens L51 having a concave surface facing the object and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0164] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a positive meniscus lens L62 having a concave surface facing the object. An air lens is formed between the negative meniscus lens L61 and the positive meniscus lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0165] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the positive meniscus lens L31 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0166] The following Table 3 lists values of data on the zoom optical system according to the third example.

TABLE 3

[General Data]

Zooming ratio 7.850
f123w = −526.69259
f123t = −297.45559

	W	M1	M2	T
FNO	4.12000	5.00001	6.14000	6.50003
ω	41.94830	22.05780	10.36801	5.96172
Y	21.34	21.70	21.70	21.70
TL	118.25612	134.48400	163.70742	182.4804

[Lens Data]

Surface Number	R	D	νd	nd
1	151.3952	2.0000	23.80	1.846660
2	87.2806	5.9280	67.90	1.593190
3	−1349.8590	0.1000		
4	76.7487	4.4238	67.90	1.593190
5	320.3570	D1(Variable)		
6*	395.1403	1.2500	40.66	1.883000
7	17.9444	4.0881		
8	172.0131	1.0000	27.15	1.944421
9	41.2622	0.6317		
10	28.0910	3.7608	20.88	1.922860
11	282.0417	1.6588		
12	−43.9082	1.6452	25.64	1.784720
13	−19.4929	1.1000	43.36	1.839318
14*	−367.3130	D2(Variable)		
15	∞	1.8230		(Aperture Stop S)
16*	25.2025	2.7754	59.33	1.609605
17	116.8971	D3(Variable)		
18	27.7315	3.2255	67.90	1.593190
19	−829.3049	0.7234		
20	31.9256	2.0849	32.32	1.953747
21	14.4283	4.6386	70.32	1.487490
22	87.2035	0.7730		
23	61.3969	5.2420	82.57	1.497820
24	−18.0219	4.1197	37.22	1.882023
25*	−25.6911	D4(Variable)		
26	−1678.9249	3.0141	25.26	1.902000
27	−33.6869	1.0000	40.12	1.851080
28*	40.9152	D5(Variable)		
29	−15.4450	1.2500	46.59	1.816000
30	−29.1017	0.1000		
31	−162.7939	2.9649	29.37	1.950000
32	−61.0034	BF		

[Aspherical Surface Data]

6th Surface

κ = 1.9193, A4 = 5.26888E−06, A6 = −1.61582E−08
A8 = 5.37910E−11, A10 = −9.15512E−14, A12 = 0.00000E+00

14th Surface

κ = 6.0000, A4 = 8.64764E−07, A6 = −1.04249E−08
A8 = −8.45595E−12, A10 = 4.36832E−13, A12 = 0.00000E+00

16th Surface

κ = −0.0411, A4 = −5.82687E−06, A6 = 1.89727E−08
A8 = −3.04157E−10, A10 = 1.94188E−12, A12 = 0.00000E+00

25th Surface

κ = 1.0633, A4 = 1.55522E−05, A6 = −4.60661E−08
A8 = 2.01166E−10, A10 = −8.69226E−13, A12 = 0.00000E+00

28th Surface

κ = 0.0000, A4 = −8.62706E−06, A6 = 9.53672E−08
A8 = −5.21848E−10, A10 = 1.74761E−12, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	111.43064
G2	6	−17.83112
G3	16	52.10796
G4	18	30.96133
G5	26	−50.42308
G6	29	−77.20586

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72028	50.00010	110.02145	194.04302
D0	∞	∞	∞	∞
D1	1.56355	16.96393	42.92481	58.95782
D2	18.90672	9.77960	3.82294	0.50000
D3	10.22026	5.92276	2.94135	1.20000
D4	6.49920	5.70254	5.42574	2.50000
D5	10.45535	15.54950	18.80771	23.47480
BF	9.29011	19.24474	28.46394	34.52694

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06124	−0.12203	−0.24452	−0.38142
D0	373.1327	356.9049	327.6814	308.9084
D1	1.56355	16.96393	42.92481	58.95782
D2	18.90672	9.77960	3.82294	0.50000
D3	10.22026	5.92276	2.94135	1.20000
D4	7.63429	7.89257	11.92314	16.79614
D5	9.32026	13.35947	12.31031	9.17866
BF	9.31670	19.34984	28.88441	35.54307

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.357
Conditional Expression(2) Mv2/fw = 0.276

TABLE 3-continued

Conditional Expression(3) ft/fw = 7.850
Conditional Expression(4) ωw = 41.948
Conditional Expression(5) ωt = 5.962
Conditional Expression(6) fw/f123w = −0.047
Conditional Expression(7) ft/f123t = −0.652
Conditional Expression(8) BFw/fw = 0.376
Conditional Expression(9) (−f5)/fw = 2.040
Conditional Expression(10) Mv5/Mv6 = 1.516
Conditional Expression(11) Mv1/(ft − fw) = 0.379
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 1.435

[0167] FIGS. **8**A and **8**B are various aberration graphs of the zoom optical system according to the third example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **9**A and **9**B are coma aberration graphs of the zoom optical system according to the third example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the third example favorably corrects the various aberrations, and has an excellent imaging performance.

Fourth Example

[0168] A fourth example is described with reference to FIGS. **10** to **12**A and **12**B and Table 4. FIG. **10** is a lens configuration diagram of a zoom optical system according to the fourth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**4**) according to the fourth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **10** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0169] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0170] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a negative meniscus lens L22 having a convex surface facing the object; a biconvex positive lens L23; a cemented lens composed of a positive meniscus lens L24 having a concave surface facing the object and a negative meniscus lens L25 having a concave surface facing the object. The object side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L25 is of aspherical shape.

[0171] The third lens group G3 consists of, in order from the object: a positive meniscus lens L31 having a convex surface facing the object; and a cemented lens composed of a biconvex positive lens L32 and a negative meniscus lens L33 having a concave surface facing the object. The image side surface of the positive meniscus lens L31 is of aspherical shape.

[0172] The fourth lens group G4 consists of, in order from the object: a positive meniscus lens L41 having a convex surface facing the object; a cemented lens composed of a negative meniscus lens L42 having a convex surface facing the object and a positive meniscus lens L43 having a convex surface facing the object; and a cemented lens composed of a positive meniscus lens L44 having a concave surface facing the object and a negative meniscus lens L45 having a concave surface facing the object. The image side surface of the negative meniscus lens L45 is of aspherical shape.

[0173] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0174] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a plano-convex positive lens L62 having a plane facing the image surface I. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0175] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the positive lens L32 and the negative meniscus lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0176] The following Table 4 lists values of data on the zoom optical system according to the fourth example.

TABLE 4

[General Data]

Zooming ratio 7.848
f123w = 102.18699
f123t = −1535.17561

	W	M1	M2	T
FNO	4.12000	5.00001	6.14000	6.50003
ω	41.94830	22.05780	10.36801	5.96172
Y	21.65	21.70	21.70	21.70
TL	122.11284	138.25648	173.12226	195.4602

[Lens Data]

Surface Number	R	D	νd	nd
1	157.9423	2.0000	23.80	1.846660
2	81.8879	5.9036	67.90	1.593190
3	−2013.3747	0.1000		
4	63.5017	4.6636	63.34	1.618000
5	210.8809	D1(Variable)		
6*	318.1018	1.2500	40.66	1.883000
7	16.7008	4.7201		
8	704.9777	1.3500	25.79	1.940573
9	35.9277	0.1354		
10	25.6246	4.3288	20.88	1.922860
11	−84.8316	1.1878		

TABLE 4-continued

12	−26.8353	2.5514	26.72	1.759928
13	−14.0619	1.1000	40.66	1.883000
14*	−120.1155	D2(Variable)		
15	∞	1.7168		(Aperture Stop S)
16*	25.0707	2.5492	56.42	1.650119
17	50.5707	1.8201		
18	2141.2793	3.9646	47.10	1.718816
19	−19.4561	1.2000	29.37	1.950000
20	−40.3974	D3(Variable)		
21	33.1155	2.7430	58.12	1.622989
22	102.1338	0.1000		
23	26.3197	4.3495	29.37	1.950000
24	14.1783	4.4212	70.32	1.487490
25	72.5822	1.6811		
26	−306.2709	4.3812	82.57	1.497820
27	−18.7373	1.2500	37.22	1.882023
28*	−24.4766	D4 (Variable)		
29	119.2349	3.5589	25.92	1.805628
30	−33.6869	1.0000	40.12	1.851080
31*	32.8619	D5 (Variable)		
32	−22.4629	1.2500	40.66	1.883000
33	−43.8572	0.1000		
34	61.5070	4.1976	33.02	1.689260
35	∞	BF		

[Aspherical Surface Data]

6th Surface

κ = 6.0000, A4 = 9.24936E−06, A6 = 4.48621E−09
A8 = −4.48203E−11, A10 = 1.65001E−13, A12 = 0.00000E+00

14th Surface

κ = 5.8635, A4 = −1.80704E−06, A6 = 1.46957E−08
A8 = −7.35664E−11, A10 = −5.50824E−13, A12 = 0.00000E+00

16th Surface

κ = 0.0729, A4 = −4.52720E−06, A6 = 2.52623E−08
A8 = −1.11420E−10, A10 = 1.41519E−13, A12 = 0.00000E+00

28th Surface

κ = 1.0568, A4 = 1.62692E−05, A6 = −9.59061E−09
A8 = −6.35322E−11, A10 = 1.73247E−13, A12 = 0.00000E+00

31st Surface

κ = 1.0365, A4 = −5.49985E−06, A6 = 5.29125E−08
A8 = −9.39998E−11, A10 = 1.17057E−13, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	106.62052
G2	6	−16.22739
G3	16	41.04090
G4	21	40.60874
G5	29	−49.86905
G6	32	−140.23760

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.71999	49.99999	109.99995	193.99988
D0	∞	∞	∞	∞
D1	1.50000	18.53350	35.68354	55.45436
D2	17.47125	8.41504	3.49296	0.78316
D3	8.24386	2.43196	1.40000	1.44036
D4	6.00186	7.14502	5.60349	2.49999
D5	10.03309	14.70182	17.27531	20.33845
BF	9.28884	17.45519	40.09303	45.36996

TABLE 4-continued

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06195	−0.10885	−0.20280	−0.28444
D0	369.2759	403.0933	448.2665	495.9287
D1	1.50000	18.53350	35.68354	55.45436
D2	17.47125	8.41504	3.49296	0.78316
D3	8.24386	2.43196	1.40000	1.44036
D4	7.29302	9.64626	10.19294	12.30143
D5	8.74193	12.20058	12.68586	10.53701
BF	9.31603	17.53910	40.38407	45.94189

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.189
Conditional Expression(2) Mv2/fw = 0.785
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 41.948
Conditional Expression(5) ωt = 5.962
Conditional Expression(6) fw/f123w = 0.242
Conditional Expression(7) ft/f123t = −0.126
Conditional Expression(8) BFw/fw = 0.376
Conditional Expression(9) (−f5)/fw = 2.017
Conditional Expression(10) Mv5/Mv6 = 1.286
Conditional Expression(11) Mv1/(ft − fw) = 0.433
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.168

[0177] FIGS. **11**A and **11**B are various aberration graphs of the zoom optical system according to the fourth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **12**A and **12**B are coma aberration graphs of the zoom optical system according to the fourth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the fourth example favorably corrects the various aberrations, and has an excellent imaging performance.

Fifth Example

[0178] A fifth example is described with reference to FIGS. **13** to **15**A and **15**B and Table 5. FIG. **13** is a lens configuration diagram of a zoom optical system according to the fifth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**5**) according to the fifth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a positive refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **13** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0179] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0180] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object. The object side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L24 is of aspherical shape.

[0181] The third lens group G3 consists of, in order from the object: a positive meniscus lens L31 having a convex surface facing the object; and a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33. The image side surface of the positive meniscus lens L31 is of aspherical shape.

[0182] The fourth lens group G4 consists of, in order from the object: a positive meniscus lens L41 having a convex surface facing the object; a cemented lens composed of a negative meniscus lens L42 having a convex surface facing the object and a biconvex positive lens L43; and a cemented lens composed of a biconvex positive lens L44 and a biconcave negative lens L45. The image side surface of the negative lens L45 is of aspherical shape.

[0183] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0184] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0185] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0186] The following Table 5 lists values of data on the zoom optical system according to the fifth example.

TABLE 5

[General Data]

Zooming ratio 7.848
f123w = 148.33142
f123t = −861.38789

	W	M1	M2	T
FNO	4.12000	5.60000	6.20000	6.49999
ω	42.61146	22.53540	10.65017	6.13829
Y	20.68	21.70	21.70	21.70
TL	122.11284	137.15660	162.89036	188.0553

[Lens Data]

Surface Number	R	D	νd	nd
1	120.0314	2.0000	23.80	1.846660
2	75.2829	6.2421	82.57	1.497820
3	−2364.7242	0.1000		
4	64.4734	5.0111	64.74	1.607834
5	308.6603	D1(Variable)		
6*	221.2774	1.2500	40.66	1.883000
7	17.1630	5.2789		
8	−38.7201	1.1000	33.32	1.903162
9	95.9763	0.1000		
10	40.1060	4.3656	20.88	1.922860
11	−33.5026	0.9492		
12	−22.3899	1.1000	40.67	1.882762
13*	−102.6938	D2(Variable)		
14	∞	0.8341		(Aperture Stop S)
15*	31.3299	2.7567	63.86	1.517039
16	394.2979	1.0000		
17	76.9690	1.1000	25.78	1.906571
18	30.3656	2.9449	45.71	1.623046
19	−89.6818	D3(Variable)		
20	23.7528	3.8095	41.66	1.659437
21	83.0826	2.6863		
22	52.1032	1.8385	32.25	1.954620
23	15.4393	5.4045	70.32	1.487490
24	−39.4485	0.1000		
25	37.0327	3.3978	67.89	1.593103
26	−838.1647	1.2500	43.15	1.810385
27*	82.5521	D4(Variable)		
28	623.8813	3.1385	22.74	1.808090
29	−33.6869	1.0000	41.21	1.836497
30*	32.8807	D5(Variable)		
31	−21.3174	1.2500	27.35	1.663819
32	−31.8044	0.1043		
33	92.9303	3.5471	28.93	1.727721
34	−394.1540	BF		

[Aspherical Surface Data]

6th Surface

κ = 5.7341, A4 = 1.16802E−06, A6 = 2.03518E−09
A8 = 1.81447E−11, A10 = 8.58869E−14, A12 = 0.00000E+00

13th Surface

κ = 3.2914, A4 = −1.11111E−06, A6 = 1.49282E−09
A8 = −3.72110E−11, A10 = 6.45032E−13, A12 = 0.00000E+00

15th Surface

κ = 0.0277, A4 = −8.27654E−06, A6 = 1.77158E−08
A8 = −1.81439E−10, A10 = 1.08193E−12, A12 = 0.00000E+00

27th Surface

κ = 1.9922, A4 = 1.24262E−05, A6 = −1.46784E−08
A8 = 3.73707E−10, A10 = −2.02655E−12, A12 = 0.00000E+00

30th Surface

κ = 1.9072, A4 = −8.91746E−06, A6 = 3.65180E−08
A8 = −5.04265E−10, A10 = 1.78607E−12, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	99.41971
G2	6	−15.74038
G3	15	41.65192
G4	20	37.63548
G5	28	−40.12367
G6	31	7119.59107

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.71999	49.99997	109.99986	193.99963
D0	∞	∞	∞	∞
D1	1.50000	16.60455	37.99399	53.23975
D2	18.36307	10.15010	3.58234	1.66590
D3	13.02129	4.83611	1.75831	1.47831

TABLE 5-continued

D4	6.45265	6.28920	8.47335	2.50000
D5	9.82549	18.17413	19.06777	25.32113
BF	9.29124	17.44341	28.35549	40.19113

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06170	−0.10941	−0.19455	−0.28776
D0	370.3154	405.3261	459.3561	504.3260
D1	1.50000	16.60455	37.99399	53.23975
D2	18.36307	10.15010	3.58234	1.66590
D3	13.02129	4.83611	1.75831	1.47831
D4	7.54590	8.23070	14.07758	12.27653
D5	8.73225	16.23263	13.46354	15.54460
BF	9.31427	17.51651	28.58674	40.69673

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.374
Conditional Expression(2) Mv2/fw = 0.575
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 42.611
Conditional Expression(5) ωt = 6.138
Conditional Expression(6) fw/f123w = 0.167
Conditional Expression(7) ft/f123t = −0.225
Conditional Expression(8) BFw/fw = 0.376
Conditional Expression(9) (−f5)/fw = 1.623
Conditional Expression(10) Mv5/Mv6 = 1.501
Conditional Expression(11) Mv1/(ft − fw) = 0.390
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.490

[0187] FIGS. **14**A and **14**B are various aberration graphs of the zoom optical system according to the fifth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **15**A and **15**B are coma aberration graphs of the zoom optical system according to the fifth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the fifth example favorably corrects the various aberrations, and has an excellent imaging performance.

Sixth Example

[0188] A sixth example is described with reference to FIGS. **16** to **18**A and **18**B and Table 6. FIG. **16** is a lens configuration diagram of a zoom optical system according to the sixth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**6**) according to the sixth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a positive refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **16** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0189] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0190] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object. The image side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L24 is of aspherical shape.

[0191] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0192] The fourth lens group G4 consists of, in order from the object: a biconvex positive lens L41; a cemented lens composed of a negative meniscus lens L42 having a convex surface facing the object and a biconvex positive lens L43; and a cemented lens composed of a positive meniscus lens L44 having a convex surface facing the object and a negative meniscus lens L45 having a concave surface facing the object. The image side surface of the positive lens L41 is of aspherical shape. The image side surface of the negative meniscus lens L45 is of aspherical shape.

[0193] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0194] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0195] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0196] The following Table 6 lists values of data on the zoom optical system according to the sixth example.

TABLE 6

[General Data]

Zooming ratio 7.848
f123w = −136.43292
f123t = −215.16315

	W	M1	M2	T
FNO	4.12001	5.60001	6.20000	6.50003
ω	42.67959	22.59339	10.65052	6.14768
Y	20.58	21.70	21.70	21.70
TL	122.11285	136.30769	162.04178	189.3093

TABLE 6-continued

[Lens Data]

Surface Number	R	D	νd	nd
1	152.2083	2.0000	23.80	1.846660
2	89.2068	6.1884	82.57	1.497820
3	−413.2934	0.1000		
4	63.3220	4.8830	67.90	1.593190
5	253.0230	D1(Variable)		
6*	137.9264	1.2500	40.66	1.882996
7	17.8991	4.7805		
8	−47.3363	1.1000	36.88	1.897432
9	75.2485	0.1000		
10	39.7397	4.1374	21.58	1.918850
11	−39.0575	0.7197		
12	−24.5868	1.1000	47.49	1.802013
13*	−591.6627	D2(Variable)		
14	∞	0.7464		(Aperture Stop S)
15	46.9722	2.7552	48.04	1.768500
16	−93.2395	0.5000		
17	51.8617	1.1000	29.95	1.987022
18	25.2907	2.9021	45.71	1.623046
19	−73.0708	1.4973		
20	−29.7887	1.0273	35.73	1.902641
21	−85.6917	D3(Variable)		
22*	28.7123	3.8190	45.24	1.768369
23	−400.5317	2.3100		
24	68.0478	1.0008	32.32	1.953752
25	17.9627	5.9680	78.66	1.495797
26	−34.0844	0.1000		
27	42.3850	3.2656	67.90	1.593190
28	754.0925	1.2500	44.96	1.790885
29*	73.7905	D4(Variable)		
30	256.5317	3.3327	22.74	1.808090
31	−33.6869	1.0000	40.27	1.839964
32*	28.6240	D5(Variable)		
33	−20.9675	1.2500	27.35	1.663819
34	−40.3074	0.1000		
35	173.3096	4.5044	31.21	1.841022
36	−72.4610	BF		

[Aspherical Surface Data]

6th Surface

κ = 0.0442, A4 = −4.01520E−06, A6 = 2.02052E−08
A8 = −1.03759E−10, A10 = 3.37776E−13, A12 = 0.00000E+00

13th Surface

κ = 1.0000, A4 = −6.36415E−06, A6 = 2.72142E−08
A8 = −2.64695E−10, A10 = 8.53046E−13, A12 = 0.00000E+00

22nd Surface

κ = 1.0000, A4 = −4.72982E−06, A6 = 7.21651E−09
A8 = −1.20147E−10, A10 = 3.75555E−13, A12 = 0.00000E+00

29th Surface

κ = 1.0000, A4 = 1.53597E−05, A6 = −6.12529E−09
A8 = 2.59000E−10, A10 = −2.05818E−12, A12 = 0.00000E+00

32nd Surface

κ = 1.0000, A4 = −5.88848E−06, A6 = 4.28279E−08
A8 = −4.85291E−10, A10 = 2.28998E−12, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	101.19406
G2	6	−16.04849
G3	15	49.36913
G4	22	29.13636
G5	30	−37.13373
G6	33	424.58679

TABLE 6-continued

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72000	50.00001	109.99999	194.00003
D0	∞	∞	∞	∞
D1	1.50000	15.59832	37.20427	54.26539
D2	17.63580	9.34111	2.87140	1.75361
D3	12.53679	5.46459	2.24074	1.47831
D4	6.27123	7.05298	10.12752	2.50000
D5	9.72366	16.01381	16.16341	24.55336
BF	9.65757	18.04908	28.64666	39.97093

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06163	−0.10967	−0.19523	−0.28795
D0	370.3333	405.5358	460.2499	503.0996
D1	1.50000	15.59832	37.20427	54.26539
D2	17.63580	9.34111	2.87140	1.75361
D3	12.53679	5.46459	2.24074	1.47831
D4	7.24735	8.97695	15.97733	12.28747
D5	8.74755	14.08984	10.31360	14.76589
BF	9.68397	18.12250	28.87954	40.47730

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.365
Conditional Expression(2) Mv2/fw = 0.584
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 42.680
Conditional Expression(5) ωt = 6.148
Conditional Expression(6) fw/f123w = −0.181
Conditional Expression(7) ft/f123t = −0.902
Conditional Expression(8) BFw/fw = 0.391
Conditional Expression(9) (−f5)/fw = 1.502
Conditional Expression(10) Mv5/Mv6 = 1.489
Conditional Expression(11) Mv1/(ft − fw) = 0.397
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.623

[0197] FIGS. **17**A and **17**B are various aberration graphs of the zoom optical system according to the sixth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **18**A and **18**B are coma aberration graphs of the zoom optical system according to the sixth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the sixth example favorably corrects the various aberrations, and has an excellent imaging performance.

Seventh Example

[0198] A seventh example is described with reference to FIGS. **19** to **21**A and **21**B and Table 7. FIG. **19** is a lens configuration diagram of a zoom optical system according to the seventh example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**7**) according to the seventh example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative

refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **19** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0199] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0200] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object. The image side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L24 is of aspherical shape.

[0201] The third lens group G3 consists of, in order from the object: a positive meniscus lens L31 having a convex surface facing the object; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a plano-concave negative lens L34 having a plane facing the image surface I. The image side surface of the positive meniscus lens L31 is of aspherical shape.

[0202] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a negative meniscus lens L42 having a concave surface facing the object; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0203] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0204] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a positive meniscus lens L62 having a concave surface facing the object. An air lens is formed between the negative meniscus lens L61 and the positive meniscus lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0205] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0206] The following Table 7 lists values of data on the zoom optical system according to the seventh example.

TABLE 7

[General Data]

Zooming ratio 7.854
f123w = 108.15193
f123t = −1180.72115

	W	M1	M2	T
FNO	4.11505	5.74532	6.36855	6.68279
ω	42.27184	21.88249	10.96245	6.04244
Y	20.89	21.70	21.70	21.70
TL	120.45755	140.80075	169.77272	195.4575

[Lens Data]

Surface Number	R	D	νd	nd
1	164.3654	2.0000	23.80	1.846660
2	73.5155	5.9893	67.90	1.593190
3	−2353.2843	0.1000		
4	72.1741	4.8721	46.03	1.721059
5	384.7599	D1(Variable)		
6*	168.2036	1.5000	40.66	1.883000
7	15.8326	4.3451		
8	−82.6447	1.5000	40.66	1.883000
9	46.2086	0.1000		
10	30.2898	4.8789	23.29	1.872769
11	−32.6789	0.7547		
12	−24.9555	1.5000	40.66	1.883000
13*	−2867.4336	D2(Variable)		
14	∞	1.5000		(Aperture Stop S)
15*	24.1510	3.7103	57.75	1.633994
16	594.5882	1.0000		
17	70.2793	1.5000	35.28	1.801392
18	17.4502	4.9253	46.90	1.702987
19	−111.3896	1.2478		
20	−65.1233	1.5000	44.85	1.743972
21	∞	D3(Variable)		
22	132.6869	3.6334	82.57	1.497820
23	−33.2203	1.5000	23.99	1.871866
24	−81.5274	0.1000		
25	26.2321	1.5000	40.98	1.869660
26	16.8448	7.0033	57.83	1.512954
27*	−36.7178	D4(Variable)		
28	100.0646	3.3139	24.26	1.791180
29	−60.0000	1.5000	40.12	1.851080
30*	35.0435	D5(Variable)		
31	−19.8065	1.5000	40.79	1.877404
32	−36.0179	0.1000		
33	−118.6453	3.6033	27.58	1.755201
34	−51.9780	BF		

[Aspherical Surface Data]

6th Surface

κ = 2.0000, A4 = −3.54713E−06, A6 = 8.39421E−09
A8 = 5.74900E−12, A10 = −2.30186E−14, A12 = 0.00000E+00

13th Surface

κ = 1.0000, A4 = −8.88610E−06, A6 = 8.60054E−10
A8 = 9.35296E−11, A10 = −8.32892E−13, A12 = 0.00000E+00

15th Surface

κ = 1.0000, A4 = −1.25166E−05, A6 = 2.21212E−08
A8 = −2.03902E−10, A10 = 7.07567E−13, A12 = 0.00000E+00

27th Surface

κ = 1.0000, A4 = 2.74577E−05, A6 = −5.57744E−08
A8 = 3.60461E−10, A10 = −1.20456E−12, A12 = 0.00000E+00

30th Surface

κ = 1.0000, A4 = −6.49026E−06, A6 = 5.84808E−08
A8 = −3.26107E−10, A10 = 9.49542E−13, A12 = 0.00000E+00

TABLE 7-continued

[Lens Group Data]

Group	First surface	Focal length
G1	1	99.48878
G2	6	−15.91549
G3	15	36.81358
G4	22	35.33722
G5	28	−59.27007
G6	31	−101.60759

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.70000	50.00001	105.00002	194.00004
D0	∞	∞	∞	∞
D1	1.50000	16.10146	35.16750	51.29406
D2	17.11600	9.70503	4.20600	1.50000
D3	7.94167	4.05851	2.07803	1.50000
D4	8.59873	5.64789	5.81781	1.50000
D5	9.32368	17.20284	18.29639	22.86408
BF	9.30000	21.40756	37.52953	50.12197

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06152	−0.11064	−0.19039	−0.28955
D0	370.9313	400.5881	451.6162	495.9313
D1	1.50000	16.10146	35.16750	51.29406
D2	17.11600	9.70503	4.20600	1.50000
D3	7.94167	4.05851	2.07803	1.50000
D4	10.20432	7.93636	10.79923	11.04504
D5	7.71809	14.91438	13.31498	13.31904
BF	9.32674	21.49415	37.78613	50.71489

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.158
Conditional Expression(2) Mv2/fw = 1.017
Conditional Expression(3) ft/fw = 7.890
Conditional Expression(4) ωw = 42.272
Conditional Expression(5) ωt = 6.042
Conditional Expression(6) fw/f123w = 0.227
Conditional Expression(7) ft/f123t = −0.171
Conditional Expression(8) BFw/fw = 0.375
Conditional Expression(9) (−f5)/fw = 2.392
Conditional Expression(10) Mv5/Mv6 = 1.332
Conditional Expression(11) Mv1/(ft − fw) = 0.439
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 1.872

[0207] FIGS. **20**A and **20**B are various aberration graphs of the zoom optical system according to the seventh example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **21**A and **21**B are coma aberration graphs of the zoom optical system according to the seventh example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the seventh example favorably corrects the various aberrations, and has an excellent imaging performance.

Eighth Example

[0208] An eighth example is described with reference to FIGS. **22** to **24**A and **24**B and Table 8. FIG. **22** is a lens configuration diagram of a zoom optical system according to the eighth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**8**) according to the eighth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **22** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0209] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0210] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object. The image side surface of the negative meniscus lens L21 is of aspherical shape.

[0211] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0212] The fourth lens group G4 consists of, in order from the object: a biconvex positive lens L41; a cemented lens composed of a positive meniscus lens L42 having a concave surface facing the object and a negative meniscus lens L43 having a concave surface facing the object; and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0213] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0214] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0215] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0216] The following Table 8 lists values of data on the zoom optical system according to the eighth example.

TABLE 8

[General Data]

Zooming ratio 7.854
f123w = −440.44611
f123t = −323.78995

	W	M1	M2	T
FNO	4.12083	5.77298	6.33626	6.49162
ω	42.50455	22.44807	11.22387	6.10280
Y	20.61	21.70	21.70	21.70
TL	120.46149	143.33661	170.26168	190.2487

[Lens Data]

Surface Number	R	D	νd	nd
1	119.2532	2.0000	25.26	1.902000
2	75.4740	6.1889	82.57	1.497820
3	−685.9404	0.1000		
4	62.0223	5.1009	67.90	1.593190
5	237.4793	D1(Variable)		
6*	153.6662	1.5000	46.59	1.816000
7	15.5464	4.2474		
8	−40.2333	1.5002	43.79	1.848500
9	79.6309	0.1006		
10	32.2669	3.5980	22.74	1.808090
11	−38.3529	0.7346		
12	−22.0127	1.5000	43.79	1.848500
13	−91.6465	D2(Variable)		
14	∞	1.5000		(Aperture Stop S)
15	44.5290	2.6559	44.85	1.743972
16	−81.3774	0.7000		
17	33.2106	3.4046	30.99	1.940752
18	19.5338	3.9016	59.70	1.508752
19	−65.3422	1.3372		
20	−26.7545	1.5000	29.68	1.730111
21	−101.6153	D3(Variable)		
22	33.1030	6.1030	70.40	1.487502
23	−28.7765	0.1000		
24	−116.0123	3.4598	68.30	1.507497
25	−28.0491	1.5000	32.03	1.910214
26	−237.5876	0.2542		
27	95.5133	3.1295	59.13	1.611115
28*	−51.7400	D4(Variable)		
29	333.8201	3.4464	22.74	1.808090
30	−49.9705	1.5000	44.82	1.743986
31*	31.2247	D5(Variable)		
32	−27.4502	1.5000	66.16	1.531180
33	−59.8926	0.1000		
34	164.9552	2.6581	27.80	1.749763
35	−519.6427	BF		

[Aspherical Surface Data]

6th Surface

κ = 1.0000, A4 = 2.54661E−06, A6 = 1.57681E−08
A8 = −1.62633E−10, A10 = 6.99665E−13, A12 = 0.00000E+00

28th Surface

κ = 1.0000, A4 = 2.83706E−05, A6 = −3.41484E−08
A8 = 2.83345E−10, A10 = −4.50609E−13, A12 = 0.00000E+00

31st Surface

κ = 1.0000, A4 = −4.24770E−06, A6 = 6.21761E−08
A8 = −2.79037E−10, A10 = 4.34156E−13, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	99.94559
G2	6	−15.36108
G3	15	40.04464
G4	22	30.83594
G5	29	−50.14179
G6	32	−238.46610

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.69999	49.99998	104.99995	193.99998
D0	∞	∞	∞	∞
D1	1.50000	18.29170	34.70486	54.28408
D2	15.49680	10.23739	4.55937	1.50000
D3	10.63532	4.97092	1.81398	1.50000
D4	8.89670	4.43383	5.19646	1.51052
D5	9.31183	19.43911	21.83341	25.83333
BF	9.29998	20.64280	36.83273	40.29998

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06119	−0.09874	−0.16976	−0.27724
D0	371.9273	449.0522	522.1271	502.1400
D1	1.50000	18.29170	34.70486	54.28408
D2	15.49680	10.23739	4.55937	1.50000
D3	10.63532	4.97092	1.81398	1.50000
D4	10.32916	6.26965	9.16937	12.36360
D5	7.87937	17.60328	17.86051	14.98025
BF	9.32282	20.70243	37.00914	40.76930

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.295
Conditional Expression(2) Mv2/fw = 0.688
Conditional Expression(3) ft/fw = 7.854
Conditional Expression(4) ωw = 42.505
Conditional Expression(5) ωt = 6.103
Conditional Expression(6) fw/f123w = −0.056
Conditional Expression(7) ft/f123t = −0.599
Conditional Expression(8) BFw/fw = 0.377
Conditional Expression(9) (−f5)/fw = 2.030
Conditional Expression(10) Mv5/Mv6 = 1.533
Conditional Expression(11) Mv1/(ft − fw) = 0.412
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.467

[0217] FIGS. **23**A and **23**B are various aberration graphs of the zoom optical system according to the eighth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **24**A and **24**B are coma aberration graphs of the zoom optical system according to the eighth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the eighth example favorably corrects the various aberrations, and has an excellent imaging performance.

Ninth Example

[0218] A ninth example is described with reference to FIGS. **25** to **27**A and **27**B and Table 9. FIG. **25** is a lens configuration diagram of a zoom optical system according to the ninth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**9**) according to the ninth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second

lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **25** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0219] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a plano-convex positive lens L12 having a plane facing the image surface I; and a positive meniscus lens L13 having a convex surface facing the object.

[0220] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object. The image side surface of the negative meniscus lens L21 is of aspherical shape. The image side surface of the negative meniscus lens L24 is of aspherical shape.

[0221] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a biconvex positive lens L32 and a negative meniscus lens L33 having a concave surface facing the object; and a cemented lens composed of a biconcave negative lens L34 and a positive meniscus lens L35 having a convex surface facing the object. The image side surface of the positive lens L31 is of aspherical shape. The image side surface of the negative lens L34 is of aspherical shape.

[0222] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L41 having a convex surface facing the object and a biconvex positive lens L42; and a biconvex positive lens L43. The image side surface of the positive lens L43 is of aspherical shape.

[0223] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52.

[0224] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0225] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative lens L34 and the positive meniscus lens L35 in the third lens group G3 constitutes a vibration-proof group that has a negative refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0226] The following Table 9 lists values of data on the zoom optical system according to the ninth example.

TABLE 9

[General Data]

Zooming ratio 7.854
f123w = 46.29531
f123t = 1060.13724

	W	M1	M2	T
FNO	4.11505	5.74532	6.36855	6.68279
ω	42.27184	21.88249	10.96245	6.04244
Y	21.03	21.70	21.70	21.70
TL	121.00241	139.79338	169.47903	195.5079

[Lens Data]

Surface Number	R	D	νd	nd
1	215.1564	1.5000	23.80	1.846660
2	73.5337	7.2326	67.90	1.593190
3	∞	0.1000		
4	75.0074	5.6048	40.66	1.883000
5	344.8006	D1(Variable)		
6*	43.3708	1.5000	40.66	1.883000
7	13.2343	4.8088		
8	−47.4291	1.5052	40.66	1.883000
9	43.0037	0.1000		
10	28.6036	4.0197	20.88	1.922860
11	−55.9891	1.1418		
12	−23.0332	1.5000	40.66	1.883000
13*	−75.5957	D2(Variable)		
14	∞	1.5000		(Aperture Stop S)
15*	28.4224	4.3742	52.85	1.598604
16	−48.8993	0.1504		
17	30.2173	5.4129	70.40	1.487490
18	−31.5840	1.5000	21.23	1.903627
19	−130.7132	1.2693		
20*	−107.8541	1.5000	41.09	1.854203
21	30.6579	3.0466	26.18	1.822542
22	165.6444	D3(Variable)		
23	33.3486	1.5007	40.66	1.883000
24	13.1929	6.5567	65.07	1.544771
25	−190.2474	0.7289		
26	37.5609	4.7319	62.98	1.574225
27*	−76.3130	D4(Variable)		
28	80.1779	3.4856	27.58	1.755201
29	−127.8937	1.5007	45.13	1.740338
30	26.5334	D5(Variable)		
31	−26.2026	1.5000	60.35	1.619799
32	−54.6221	0.1000		
33	586.6701	2.5595	28.29	1.738351
34	−391.8753	BF		

[Aspherical Surface Data]

6th Surface

κ = 1.0000, A4 = −6.29772E−06, A6 = −1.23182E−08
A8 = 7.32161E−11, A10 = −3.10876E−13, A12 = 0.00000E+00

13th Surface

κ = 1.0000, A4 = −8.92953E−06, A6 = −3.71644E−08
A8 = 8.09196E−10, A10 = −5.73691E−12, A12 = 0.00000E+00

15th Surface

κ = 1.0000, A4 = −1.00000E−05, A6 = 2.20240E−08
A8 = −1.02146E−10, A10 = 0.00000E+00, A12 = 0.00000E+00

20th Surface

κ = 1.0000, A4 = 3.32815E−06, A6 = 1.66254E−09
A8 = 0.00000E+00, A10 = 0.00000E+00, A12 = 0.00000E+00

TABLE 9-continued

27th Surface

κ = 1.0000, A4 = 1.00000E−05, A6 = −3.83755E−08
A8 = −1.30773E−10, A10 = −1.22891E−12, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	102.37710
G2	6	−14.98474
G3	15	29.62517
G4	23	38.66055
G5	28	−56.76096
G6	31	−113.46417

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.70007	50.00020	105.00052	194.00105
D0	∞	∞	∞	∞
D1	1.50000	18.30775	37.76411	54.05443
D2	18.03389	8.85777	3.84960	1.50000
D3	6.37316	1.62225	1.50000	1.64117
D4	2.43986	3.20131	6.09526	1.50000
D5	9.22531	23.30380	13.81143	17.88006
BF	13.00010	14.07041	36.02852	48.50219

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.08175	−0.12239	−0.19581	−0.27905
D0	272.2777	353.4868	423.8013	497.7725
D1	1.50000	18.30775	37.76411	54.05443
D2	18.03389	8.85777	3.84960	1.50000
D3	6.37316	1.62225	1.50000	1.64117
D4	4.23169	6.02328	12.47110	12.78692
D5	7.43347	20.48182	7.43559	6.59314
BF	13.03743	14.15445	36.24396	48.93981

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.123
Conditional Expression(2) Mv2/fw = 0.889
Conditional Expression(3) ft/fw = 7.887
Conditional Expression(4) ωw = 42.272
Conditional Expression(5) ωt = 6.042
Conditional Expression(6) fw/f123w = 0.534
Conditional Expression(7) ft/f123t = 0.179
Conditional Expression(8) BFw/fw = 0.526
Conditional Expression(9) (−f5)/fw = 2.298
Conditional Expression(10) Mv5/Mv6 = 1.244
Conditional Expression(11) Mv1/(ft − fw) = 0.438
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.830

[0227] FIGS. **26**A and **26**B are various aberration graphs of the zoom optical system according to the ninth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **27**A and **27**B are coma aberration graphs of the zoom optical system according to the ninth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the ninth example favorably corrects the various aberrations, and has an excellent imaging performance.

Tenth Example

[0228] A tenth example is described with reference to FIGS. **28** to **30**A and **30**B and Table 10. FIG. **28** is a lens configuration diagram of a zoom optical system according to the tenth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**10**) according to the tenth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **28** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0229] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a positive meniscus lens L12 having a convex surface facing the object; and a positive meniscus lens L13 having a convex surface facing the object.

[0230] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a biconcave negative lens L24.

[0231] The third lens group G3 consists of, in order from the object: a positive meniscus lens L31 having a convex surface facing the object; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0232] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a negative meniscus lens L42 having a concave surface facing the object; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0233] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0234] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0235] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical

axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0236] The following Table 10 lists values of data on the zoom optical system according to the tenth example.

TABLE 10

[General Data]

Zooming ratio 4.692
f123w = −96.28619
f123t = −88.05735

	W	M1	M2	T
FNO	3.66063	4.51062	5.00831	5.83006
ω	42.43419	22.44766	13.66195	10.17394
Y	20.54	21.70	21.70	21.70
TL	116.50601	138.64669	160.34507	171.5048

[Lens Data]

Surface Number	R	D	νd	nd
1	200.0000	2.0000	23.80	1.846660
2	108.2337	4.6568	70.32	1.487490
3	1133.5711	0.1000		
4	68.7807	5.7029	70.32	1.487490
5	816.5190	D1(Variable)		
6	102.0974	1.2000	46.59	1.816000
7	17.6075	4.9760		
8	−59.3363	1.1000	51.28	1.659368
9	81.2225	0.1000		
10	29.1388	3.9898	23.80	1.846660
11	−107.8110	0.8213		
12	−38.4600	1.0000	46.59	1.816000
13	157.0586	D2(Variable)		
14	∞	2.0000		(Aperture Stop S)
15	41.9442	2.5619	35.72	1.902650
16	552.5411	0.5000		
17	41.0223	0.9000	29.12	2.001000
18	23.0700	4.0200	53.74	1.579570
19	−69.7834	1.5452		
20	−27.7457	1.0000	32.33	1.953750
21	−68.0384	D3(Variable)		
22	33.5256	5.9460	46.59	1.816000
23	−23.9703	1.0000	32.35	1.850260
24	−83.2531	0.1000		
25	30.5301	1.1000	32.35	1.850260
26	14.8810	10.3852	70.32	1.487490
27*	−119.0936	D4(Variable)		
28	81.3890	3.7144	23.80	1.846660
29	−48.0181	1.0000	42.73	1.834810
30*	23.7254	D5(Variable)		
31	−24.5058	1.4000	46.59	1.816000
32	−48.4638	0.1000		
33	142.4943	3.2707	37.57	1.683760
34	−160.0000	BF		

[Aspherical Surface Data]

27th Surface

κ = 1.0000, A4 = 4.39579E−05, A6 = −4.15837E−09
A8 = 6.65149E−10, A10 = 0.00000E+00, A12 = 0.00000E+00

30th Surface

κ = 1.0000, A4 = −2.71688E−06, A6 = 1.79186E−08
A8 = −3.84607E−10, A10 = 0.00000E+00, A12 = 0.00000E+00

TABLE 10-continued

[Lens Group Data]

Group	First surface	Focal length
G1	1	143.63567
G2	6	−20.08403
G3	15	60.03586
G4	22	25.77538
G5	28	−42.36974
G6	31	−151.12346

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72617	50.01050	85.01086	116.00340
D0	∞	∞	∞	∞
D1	1.50000	23.91054	40.85381	47.19819
D2	17.00636	9.20443	4.57783	1.50000
D3	9.28353	3.49337	1.15783	0.30000
D4	2.84012	1.14718	1.49326	3.00287
D5	7.97989	15.46299	17.45245	16.80067
BF	11.70601	19.23807	28.61979	36.51305

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06049	−0.09676	−0.16160	−0.22064
D0	377.2840	455.1463	433.4475	422.2840
D1	1.50000	23.91054	40.85381	47.19819
D2	17.00636	9.20443	4.57783	1.50000
D3	9.28353	3.49337	1.15783	0.30000
D4	3.90798	2.82710	4.74497	7.99347
D5	6.91203	13.78307	14.20074	11.81007
BF	11.72550	19.28822	28.75888	36.77070

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.362
Conditional Expression(2) Mv2/fw = 0.376
Conditional Expression(3) ft/fw = 4.692
Conditional Expression(4) ωw = 42.434
Conditional Expression(5) ωt = 10.174
Conditional Expression(6) fw/f123w = −0.257
Conditional Expression(7) ft/f123t = −1.323
Conditional Expression(8) BFw/fw = 0.473
Conditional Expression(9) (−f5)/fw = 1.714
Conditional Expression(10) Mv5/Mv6 = 1.356
Conditional Expression(11) Mv1/(ft − fw) = 0.603
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.492

[0237] FIGS. **29**A and **29**B are various aberration graphs of the zoom optical system according to the tenth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **30**A and **30**B are coma aberration graphs of the zoom optical system according to the tenth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the tenth example favorably corrects the various aberrations, and has an excellent imaging performance.

Eleventh Example

[0238] An eleventh example is described with reference to FIGS. **31** to **33**A and **33**B and Table 11. FIG. **31** is a lens configuration diagram of a zoom optical system according to the eleventh example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**11**) according to the eleventh example consists of, in order from the object: a first lens group G1 having a positive refractive power; a

second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; a fifth lens group G5 having a negative refractive power; and a sixth lens group G6 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **31** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move.

[0239] The first lens group G1 consists of, in order from the object: a cemented lens composed of a negative meniscus lens L11 having a convex surface facing the object and a positive meniscus lens L12 having a convex surface facing the object; and a positive meniscus lens L13 having a convex surface facing the object.

[0240] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object.

[0241] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0242] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a biconcave negative lens L42; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0243] The fifth lens group G5 consists of a cemented lens composed of a positive meniscus lens L51 having a concave surface facing the object and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0244] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62. The image surface I is disposed on the image side of the sixth lens group G6.

[0245] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0246] The following Table 11 lists values of data on the zoom optical system according to the eleventh example.

TABLE 11

[General Data]

Zooming ratio 3.438
f123w = 2466.12612
f123t = −146.93338

	W	M1	M2	T
FNO	3.65039	4.00020	—	4.50024
ω	43.52469	−22.45389	—	13.66502
Y	21.27	21.70	—	21.70
TL	116.50677	138.27327	—	161.50351

[Lens Data]

Surface Number	R	D	νd	nd
1	200.0000	2.0000	23.80	1.846660
2	112.4996	4.3442	70.32	1.487490
3	642.3642	0.1000		
4	64.3082	6.0839	70.32	1.487490
5	1033.4518	D1(Variable)		
6	91.5822	1.2000	46.59	1.816000
7	16.7434	5.0969		
8	−135.4549	1.1000	50.66	1.670176
9	31.8729	0.1000		
10	24.8799	4.1715	23.80	1.846660
11	−336.6794	1.4309		
12	−32.2164	1.0000	46.59	1.816000
13	−277.8484	D2(Variable)		
14	∞	2.0000		(Aperture Stop S)
15	41.4933	2.7587	43.79	1.848500
16	−1372.9949	0.5000		
17	55.1173	0.9000	34.87	1.847939
18	24.3945	4.0805	56.69	1.586546
19	−76.2325	1.8582		
20	−25.5665	1.0000	34.04	1.847872
21	−37.7016	D3(Variable)		
22	46.4011	6.1027	46.59	1.816000
23	−43.4919	1.0000	26.59	1.847083
24	397.5382	0.1000		
25	28.7499	1.1000	32.35	1.850260
26	18.0000	11.1086	70.32	1.487490
27*	−25.6478	D4(Variable)		
28	−713.8966	3.6653	23.80	1.846660
29	−42.0000	1.0000	45.28	1.796882
30*	31.7158	D5(Variable)		
31	−19.7135	1.4000	62.26	1.536206
32	−42.7591	0.1000		
33	217.1940	3.3164	37.57	1.683760
34	−160.0000	BF		

[Aspherical Surface Data]

27th Surface

κ = 1.0000, A4 = 6.15332E−05, A6 = −2.11407E−07
A8 = 7.47121E−10, A10 = −1.12141E−12, A12 = 0.00000E+00

30th Surface

κ = 1.0000, A4 = −1.68999E−05, A6 = 1.65258E−07
A8 = −4.68439E−10, A10 = 7.74341E−13, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	135.65910
G2	6	−17.91192
G3	15	42.22744
G4	22	26.33888
G5	28	−39.70963
G6	31	−150.62287

TABLE 11-continued

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72587	50.00755	—	85.00361
D0	∞	∞	—	∞
D1	1.50000	24.03902	—	40.38368
D2	13.72037	5.30562	—	1.50000
D3	8.95150	2.76337	—	0.30000
D4	1.97894	1.51459	—	1.15589
D5	10.03133	16.68380	—	19.50588
BF	11.70677	19.34900	—	30.04020

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06066	−0.11959	—	−0.19908
D0	377.2840	355.5169	—	332.2840
D1	1.50000	24.03902	—	40.38368
D2	13.72037	5.30562	—	1.50000
D3	8.95150	2.76337	—	0.30000
D4	2.78309	3.22774	—	4.28073
D5	9.22718	14.97065	—	16.38104
BF	11.72684	19.42705	—	30.25539

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.472
Conditional Expression(2) Mv2/fw = 0.247
Conditional Expression(3) ft/fw = 3.438
Conditional Expression(4) ωw = 43.525
Conditional Expression(5) ωt = 13.665
Conditional Expression(6) fw/f123w = 0.010
Conditional Expression(7) ft/f123t = −0.579
Conditional Expression(8) BFw/fw = 0.473
Conditional Expression(9) (−f5)/fw = 1.616
Conditional Expression(10) Mv5/Mv6 = 1.517
Conditional Expression(11) Mv1/(ft − fw) = 0.746
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = 0.671

[0247] FIGS. **32**A and **32**B are various aberration graphs of the zoom optical system according to the eleventh example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **33**A and **33**B are coma aberration graphs of the zoom optical system according to the eleventh example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the eleventh example favorably corrects the various aberrations, and has an excellent imaging performance.

Twelfth Example

[0248] A twelfth example is described with reference to FIGS. **34** to **36**A and **36**B and Table 12. FIG. **34** is a lens configuration diagram of a zoom optical system according to the twelfth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**12**) according to the twelfth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a positive refractive power; and a fifth lens group G5 having a negative refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4 and the fifth lens group G5 move in directions indicated by arrows in FIG. **34** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the fifth lens group G5 integrally move.

[0249] The first lens group G1 consists of, in order from the object: a negative meniscus lens L11 having a convex surface facing the object; a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0250] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object.

[0251] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a biconcave negative lens L34.

[0252] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a negative meniscus lens L42 having a concave surface facing the object; a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44; and a cemented lens composed of a biconvex positive lens L45 and a biconcave negative lens L46. The image side surface of the positive lens L44 is of aspherical shape. The image side surface of the negative lens L46 is of aspherical shape.

[0253] The fifth lens group G5 consists of, in order from the object: a negative meniscus lens L51 having a concave surface facing the object; and a biconvex positive lens L52. The image side surface of the negative meniscus lens L51 is of aspherical shape. The image surface I is disposed on the image side of the fifth lens group G5.

[0254] In this example, by moving the cemented lens composed of the positive lens L45 and the negative lens L46 in the fourth lens group G4 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. Note that upon zooming with focusing on the short distant object, the cemented lens composed of the positive lens L45 and the negative lens L46, which serves as a focusing group, moves by a different amount of movement, with respect to the cemented lens composed of the positive lens L41 and the negative meniscus lens L42, and the cemented lens composed of the negative meniscus lens L43 and the positive lens L44 in the fourth lens group. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0255] The following Table 12 lists values of data on the zoom optical system according to the twelfth example.

TABLE 12

[General Data]

Zooming ratio 7.848
f123w = −297.77158
f123t = −199.23081

	W	M1	M2	T
FNO	4.12000	5.69956	6.30000	6.50003
ω	42.96973	22.56096	11.03929	6.08825
Y	21.29	21.70	21.70	21.70
TL	129.0507	143.6432	173.1936	191.4323

[Lens Data]

Surface Number	R	D	νd	nd
1	181.0189	2.0855	31.27	1.90366
2	74.7364	0.8982		
3	78.3131	6.0267	67.90	1.59319
4	−878.3490	0.1429		
5	65.0716	4.8340	67.90	1.59319
6	661.4054	D1(Variable)		
7	171.3932	1.1000	35.72	1.90265
8	18.9469	5.2527		
9	−57.6716	1.0000	52.33	1.75500
10	53.7286	0.4722		
11	34.8478	3.1650	20.88	1.92286
12	−81.2943	1.3566		
13	−31.9419	0.9000	46.59	1.81600
14	−487.1030	D2(Variable)		
15	∞	2.0101		(Aperture Stop S)
16	45.9039	2.3316	35.72	1.90265
17	−163.4046	0.5000		
18	33.6170	1.1581	29.12	2.00100
19	19.7670	3.5655	53.74	1.57957
20	−85.9122	1.3700		
21	−41.3606	1.0329	32.33	1.95375
22	1717.1475	D3(Variable)		
23	37.7633	4.9751	42.73	1.83481
24	−38.9447	1.0000	31.27	1.90366
25	−804.1582	0.1000		
26	29.7427	3.0986	32.33	1.95375
27	15.4408	8.8739	81.49	1.49710
28*	−39.9876	D4(Variable)		
29	10338.5730	3.6738	23.80	1.84666
30	−27.6080	1.0000	40.13	1.85135
31*	31.8891	D5(Variable)		
32	−29.8624	1.4000	40.13	1.85135
33*	−63.8559	0.1000		
34	66.4034	4.5715	37.57	1.68376
35	−424.4531	BF		

[Aspherical Surface Data]

28th Surface

κ = 1.0000, A4 = 2.91470E−05, A6 = −1.17772E−07
A8 = 9.21285E−10, A10 = −5.94865E−12, A12 = 0.14842E−13

31st Surface

κ = 1.0000, A4 = −5.83910E−06, A6 = 1.34714E−07
A8 = −1.32747E−09, A10 = 8.60735E−12, A12 = −0.22325E−13

33rd Surface

κ = 1.0000, A4 = 4.26328E−06, A6 = −4.06929E−09
A8 = 4.06528E−11, A10 = −1.22140E−13, A12 = 0.00000E+00

[Lens Group Data]

Group	First surface	Focal length
G1	1	105.7291
G2	7	−16.8196
G3	16	48.27007
G4	23	44.51528
G5	32	−372.043

TABLE 12-continued

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72000	49.99999	104.99993	194.00004
D0	∞	∞	∞	∞
D1	1.73220	15.73205	40.41864	55.46338
D2	20.01315	10.99318	5.91904	1.09143
D3	13.56296	6.37783	3.44003	1.69135
D4	4.09147	4.09147	4.09147	4.09147
D5	9.90112	17.08625	20.02405	21.77273
BF	11.75486	21.36749	31.30545	39.32701

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06144	−0.10950	−0.17803	−0.26497
D0	370.94930	406.35680	476.80640	558.56770
D1	1.73220	15.73205	40.41864	55.46338
D2	20.01315	10.99318	5.91904	1.09143
D3	13.56296	6.37783	3.44003	1.69135
D4	4.89647	5.55202	7.55497	11.91843
D5	9.09612	15.62570	16.56055	13.94577
BF	11.75486	21.36749	31.30545	39.32701

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.614
Conditional Expression(2) Mv2/fw = 0.350
Conditional Expression(3) ft/fw = 7.848
Conditional Expression(4) ωw = 42.970
Conditional Expression(5) ωt = 6.088
Conditional Expression(6) fw/f123w = −0.083
Conditional Expression(7) ft/f123t = −0.974
Conditional Expression(8) BFw/fw = 0.476
Conditional Expression(9) (−f5)/fw = 15.051
Conditional Expression(11) Mv1/(ft − fw) = 0.369

[0256] FIGS. **35**A and **35**B are various aberration graphs of the zoom optical system according to the twelfth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **36**A and **36**B are coma aberration graphs of the zoom optical system according to the twelfth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the twelfth example favorably corrects the various aberrations, and has an excellent imaging performance.

Thirteenth Example

[0257] A thirteenth example is described with reference to FIGS. **37** to **39**A and **39**B and Table 13. FIG. **37** is a lens configuration diagram of a zoom optical system according to the thirteenth example upon focusing on infinity in a wide angle end state. The zoom optical system ZL(**13**) according to the thirteenth example consists of, in order from the object: a first lens group G1 having a positive refractive power; a second lens group G2 having a negative refractive power; an aperture stop S; a third lens group G3 having a positive refractive power; a fourth lens group G4 having a

positive refractive power; a fifth lens group G5 having a negative refractive power; a sixth lens group G6 having a negative refractive power; and a seventh lens group G7 having a positive refractive power. Upon zooming from the wide angle end state to the telephoto end state, the first lens group G1, the second lens group G2, the aperture stop S, the third lens group G3, the fourth lens group G4, the fifth lens group G5 and the sixth lens group G6 move in directions indicated by arrows in FIG. **37** along the optical axis, and the distances between the adjacent lens groups change. Note that upon zooming, the aperture stop S, the third lens group G3 and the sixth lens group G6 integrally move. Upon zooming, the seventh lens group G7 is fixed with respect to the image surface I.

[0258] The first lens group G1 consists of, in order from the object: a negative meniscus lens L11 having a convex surface facing the object; a biconvex positive lens L12; and a positive meniscus lens L13 having a convex surface facing the object.

[0259] The second lens group G2 consists of, in order from the object: a negative meniscus lens L21 having a convex surface facing the object; a biconcave negative lens L22; a biconvex positive lens L23; and a negative meniscus lens L24 having a concave surface facing the object.

[0260] The third lens group G3 consists of, in order from the object: a biconvex positive lens L31; a cemented lens composed of a negative meniscus lens L32 having a convex surface facing the object and a biconvex positive lens L33; and a negative meniscus lens L34 having a concave surface facing the object.

[0261] The fourth lens group G4 consists of, in order from the object: a cemented lens composed of a biconvex positive lens L41 and a negative meniscus lens L42 having a concave surface facing the object; and a cemented lens composed of a negative meniscus lens L43 having a convex surface facing the object and a biconvex positive lens L44. The image side surface of the positive lens L44 is of aspherical shape.

[0262] The fifth lens group G5 consists of a cemented lens composed of a biconvex positive lens L51 and a biconcave negative lens L52. The image side surface of the negative lens L52 is of aspherical shape.

[0263] The sixth lens group G6 consists of, in order from the object: a negative meniscus lens L61 having a concave surface facing the object; and a biconvex positive lens L62. An air lens is formed between the negative meniscus lens L61 and the positive lens L62.

[0264] The seventh lens group G7 consists of a positive meniscus lens L71 having a concave surface facing the object. An image surface I is disposed on the image side of the seventh lens group G7.

[0265] In this example, by moving the fifth lens group G5 toward the image surface I, focusing from a far distant object to a short distant object (from an infinity object to a finite distance object) is achieved. In this example, the cemented lens composed of the negative meniscus lens L32 and the positive lens L33 in the third lens group G3 constitutes a vibration-proof group that has a positive refractive power and is movable in a direction perpendicular to the optical axis, and corrects the displacement of the imaging position due to camera shake and the like (an image blur on the image surface I).

[0266] The following Table 13 lists values of data on the zoom optical system according to the thirteenth example.

TABLE 13

[General Data]

Zooming ratio 7.852
f123w = −242.5247
f123t = −265.90409

	W	M1	M2	T
FNO	4.12000	5.69956	6.30000	6.50003
ω	42.96973	22.56096	11.03929	6.08825
Y	20.93	21.70	21.70	21.70
TL	127.52968	144.84356	169.66796	191.04949

[Lens Data]

Surface Number	R	D	νd	nd
1	183.1489	1.7000	31.27	1.90366
2	76.2993	0.8845		
3	78.7954	6.1936	67.90	1.59319
4	−594.6799	0.1000		
5	61.9988	5.6077	67.90	1.59319
6	371.0839	D1(Variable)		
7	190.1957	1.1000	35.72	1.90265
8	19.1266	5.1112		
9	−52.1202	1.0000	52.33	1.75500
10	58.1840	0.5132		
11	36.9591	3.1252	20.88	1.92286
12	−69.4993	0.6909		
13	−34.0835	0.9000	46.59	1.81600
14	−15713.5710	D2(Variable)		
15	∞	2.0000		(Aperture Stop S)
16	40.7989	2.3289	35.72	1.90265
17	−299.8253	0.5000		
18	38.9427	1.0000	29.12	2.00100
19	21.5486	3.5304	53.74	1.57957
20	−63.7114	1.3676		
21	−35.4002	1.0000	32.33	1.95375
22	−265.5862	D3(Variable)		
23	37.7375	4.7476	42.73	1.83481
24	−37.5607	1.0000	31.27	1.90366
25	−325.9958	0.1000		
26	31.4406	3.1004	32.33	1.95375
27	15.3849	8.5803	81.49	1.49710
28*	−42.3410	D4(Variable)		
29	572.4423	3.1728	23.80	1.84666
30	−34.5910	1.0000	40.13	1.85135
31*	31.5461	D5(Variable)		
32	−19.9700	1.4000	40.13	1.85135
33*	−28.8707	0.1000		
34	136.4370	3.5760	37.57	1.68376
35	−114.7970	D6(Variable)		
36	−118.5432	2.3370	63.88	1.51680
37	−70.3002	BF		

[Aspherical Surface Data]

28th Surface

κ = 1.0000, A4 = 3.78774E−05, A6 = −4.14498E−07
A8 = 6.80734E−09, A10 = −6.10728E−11, A12 = 0.20806E−12

31st Surface

κ = 1.0000, A4 = −1.36815E−05, A6 = 2.49099E−07
A8 = −3.33308E−09, A10 = 2.73107E−11, A12 = −0.88099E−13

33rd Surface

κ = 1.0000, A4 = 1.98989E−06, A6 = −1.03153E−08
A8 = 4.34935E−11, A10 = −1.04756E−13, A12 = 0.00000E+00

TABLE 13-continued

[Lens Group Data]

Group	First surface	Focal length
G1	1	103.06116
G2	7	−17.00821
G3	16	49.18043
G4	23	29.23287
G5	29	−39.13048
G6	32	−1300.48544
G7	36	328.82617

[Variable Distance Data]

	W Infinity	M1 Infinity	M2 Infinity	T Infinity
f	24.72000	50.00000	104.99999	194.09403
D0	∞	∞	∞	∞
D1	1.50000	17.37231	39.57659	56.44287
D2	19.29037	11.03703	4.56142	1.16368
D3	12.96315	6.13632	3.05308	1.47831
D4	4.87593	4.10851	5.56324	1.90252
D5	9.73283	17.32708	18.95559	24.19108
D6	0.80000	10.49492	19.59067	27.50369
BF	10.60000	10.59999	10.59998	10.59995

	W Short-distance	M1 Short-distance	M2 Short-distance	T Short-distance
β	−0.06123	−0.10885	−0.17758	−0.28031
D0	372.47030	405.15640	480.33200	508.95050
D1	1.50000	17.37231	39.57659	56.44287
D2	19.29037	11.03703	4.56142	1.16368
D3	12.96315	6.13632	3.05308	1.47831
D4	5.77372	5.76435	9.89727	11.65975
D5	8.83504	15.67124	14.62157	14.43385
D6	0.80000	10.49492	19.59067	27.50369
BF	11.75486	21.36749	31.30545	39.32701

[Conditional expression corresponding value]

Conditional Expression(1) Mv4/Mv3 = 1.430
Conditional Expression(2) Mv2/fw = 0.347
Conditional Expression(3) ft/fw = 7.852
Conditional Expression(4) ωw = 42.970
Conditional Expression(5) ωt = 6.088
Conditional Expression(6) fw/f123w = −0.102
Conditional Expression(7) ft/f123t = −0.730
Conditional Expression(8) BFw/fw = 0.429
Conditional Expression(9) (−f5)/fw = 1.583
Conditional Expression(10) Mv5/Mv6 = 1.541
Conditional Expression(11) Mv1/(ft − fw) = 0.375
Conditional Expression(12) (RAr2 + RAr1)/(RAr2 − RAr1) = −0.651

[0267] FIGS. **38**A and **38**B are various aberration graphs of the zoom optical system according to the thirteenth example upon focusing on infinity in the wide-angle end state and the telephoto end state. FIGS. **39**A, and **39**B are coma aberration graphs of the zoom optical system according to the thirteenth example in the wide-angle end state and the telephoto end state when blur correction is performed. The various aberration graphs show that the zoom optical system according to the thirteenth example favorably corrects the various aberrations, and has an excellent imaging performance.

[0268] According to each example, the zoom optical system with various aberrations, such as the spherical aberration, being favorably corrected, can be achieved.

[0269] Here, each of the examples described above indicates a specific example of the invention of the present application. The invention of the present application is not limited thereto.

[0270] Note that the following details can be appropriately adopted in a range without degrading the optical performance of the zoom optical system according to this embodiment.

[0271] As numerical examples of the zoom optical system, systems having the five-, six- and seven-element group configurations have been described. However, the present application is not limited thereto. A zoom optical system having another group configuration (for example, a four- or eight-element group configuration) may be configured. Specifically, a configuration may be adopted where a lens or a lens group is added on the most-object side or the most-image side of the zoom optical system. Note that the lens group indicates a portion that includes at least one lens separated by air distances changing during zooming.

[0272] The lens surface may be formed to be a spherical surface or a plane, or formed to be an aspherical surface. A case where lens surfaces are spherical surfaces or planes is preferable because the case facilitates processing, assembly and adjustment of lenses, and can prevent degradation of optical performances due to errors in the processing, assembly and adjustment. Furthermore, it is preferable because degradation of drawing performances is small even in case the image surface deviates.

[0273] In a case where the lens surface is an aspherical surface, the aspherical surface may be any of an aspherical surface made by a grinding process, a glass mold aspherical surface made by forming glass into an aspherical shape with a mold, and a composite type aspherical surface made by forming a resin on a surface of glass into an aspherical shape. The lens surface may be a diffractive surface. The lens may be a gradient-index lens (GRIN lens), or a plastic lens.

[0274] To reduce flares and ghosts and achieve a high optical performance having a high contrast, an antireflection film having a high transmissivity over a wide wavelength region may be applied to each lens surface. Accordingly, flares and ghosts can be reduced, and high optical performances having a high contrast can be achieved.

EXPLANATION OF NUMERALS AND CHARACTERS

G1 First lens group	G2 Second lens group
G3 Third lens group	G4 Fourth lens group
G5 Fifth lens group	G6 Sixth lens group
G7 Seventh lens group	
I Image surface	S Aperture stop

1. A zoom optical system, comprising, in order from an object: a first lens group having a positive refractive power; a second lens group having a negative refractive power; a third lens group having a positive refractive power; a fourth lens group having a positive refractive power; a fifth lens group; and a sixth lens group, wherein

upon zooming, a distance between the adjacent lens groups changes, and

the zoom optical system satisfies the following conditional expression:

$$1.00<Mv4/Mv3<3.00$$

where Mv3: an amount of movement of the third lens group upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

Mv4: an amount of movement of the fourth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

2. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$0.00<Mv2/fw<10.00$$

where Mv2: an amount of movement of the second lens group upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +), and

fw: a focal length of the zoom optical system in the wide angle end state.

3. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$3.00<ft/fw<30.00$$

where ft: a focal length of the zoom optical system in the telephoto end state, and

fw: a focal length of the zoom optical system in the wide angle end state.

4. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$35.0°<\omega w<75.0°$$

where ωw: a half angle of view of the zoom optical system in the wide angle end state.

5. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$2.5°<\omega t<15.0°$$

where ωt: a half angle of view of the zoom optical system in the telephoto end state.

6. The zoom optical system according claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$-0.30<fw/f123w<0.60$$

where fw: a focal length of the zoom optical system in the wide angle end state, and

f123w: a combined focal length of the first lens group, the second lens group and the third lens group in the wide angle end state.

7. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$-1.50<ft/f123t<1.00$$

where ft: a focal length of the zoom optical system in the telephoto end state, and

f123t: a combined focal length of the first lens group, the second lens group and the third lens group in the telephoto end state.

8. The zoom optical system according to claim **1**, wherein the zoom optical system satisfies the following conditional expression:

$$0.20<BFw/fw<0.60$$

where BFw: a distance to an image surface from a lens surface of the zoom optical system closest to an image in the wide angle end state, and

fw: a focal length of the zoom optical system in the wide angle end state.

9. The zoom optical system according to claim **1**, wherein upon focusing, the fifth lens group moves with respect to an image surface.

10. The zoom optical system according to claim **1**, wherein the fifth lens group includes at least one positive lens, and at least one negative lens.

11. The zoom optical system according to claim **1**, wherein

the zoom optical system satisfies the following conditional expression:

$$1.00<(-f5)/fw<16.00$$

where f5: a focal length of the fifth lens group, and

fw: a focal length of the zoom optical system in the wide angle end state.

12. The zoom optical system according to claim **1**, wherein

the zoom optical system satisfies the following conditional expression:

$$1.00<Mv5/Mv6<3.00$$

where Mv5: an amount of movement of the fifth lens group upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +), and

Mv6: an amount of movement of the sixth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

13. The zoom optical system according to claim **1**, wherein upon zooming, the first lens group moves with respect to an image surface.

14. The zoom optical system according to claim **1**, wherein the first lens group consists of three or more lenses.

15. The zoom optical system according to claim **1**, wherein

the zoom optical system satisfies the following conditional expression:

$$0.30<Mv1/(ft-fw)<0.80$$

where Mv1: an amount of movement of the first lens group upon zooming from the wide angle end state to the telephoto end state (a sign of the amount of movement toward an object is shown as +),

ft: a focal length of the zoom optical system in the telephoto end state, and

fw: a focal length of the zoom optical system in the wide angle end state.

16. The zoom optical system according to claim **1**, wherein

an air lens is provided in the sixth lens group, and

the zoom optical system satisfies the following conditional expression:

$$0.00<(RAr2+RAr1)/(RAr2-RAr1)<2.00$$

where RAr1: a radius of curvature of an object-side lens surface of the air lens of the sixth lens group, and

RAr2: a radius of curvature of an image-side lens surface of the air lens of the sixth lens group.

17. The zoom optical system according to claim **1**, wherein upon zooming, at least the first lens group, the third lens group, the fourth lens group, the fifth lens group, and the sixth lens group move with respect to an image surface.

18. The zoom optical system according to claim **1**, wherein the lens groups moving upon zooming move toward the object upon zooming from the wide angle end state to the telephoto end state.

19. An optical apparatus, comprising the zoom optical system according to claim **1** mounted thereon.

20. A method for manufacturing a zoom optical system comprising, in order from the object: a first lens group having a positive refractive power; a second lens group having a negative refractive power; a third lens group having a positive refractive power; a fourth lens group having a positive refractive power; a fifth lens group; and a sixth lens group, the method comprising:

arranging the lens groups in a lens barrel such that:

upon zooming, a distance between the adjacent lens groups changes, and

satisfying the following conditional expression:

$$1.00<Mv4/Mv3<3.00$$

where Mv3: an amount of movement of the third lens group upon zooming from a wide angle end state to a telephoto end state (a sign of the amount of movement toward an object is shown as +), and

Mv4: an amount of movement of the fourth lens group upon zooming from the wide angle end state to the telephoto end state (the sign of the amount of movement toward an object is shown as +).

* * * * *